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PHYSICS

Controlled transport of Cold Atoms in a Hollow-Core Fiber

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Abstract

This work presents the design, implementation and characterization of an Optical Dipole Trap (ODT) for capturing cold ^{87}Rb atoms from a Magneto-Optical Trap (MOT) and optical molasses setup.
The ODT has been realized by coupling 1064 nm laser light into a Hollow-Core Photonic Crystal Fiber (HCPCF). Due to the inherently multimodal nature of these fibers, light injection is highly sensitive to misalignment. Monitoring the near-field at the fiber output was found to be the most effective alignment procedure. For this purpose, a near-field imaging system was built to optimize the light injection, achieving a coupling efficiency of 96% with an LP-like fundamental mode profile in test fibers.
Subsequent characterization of the in-vacuum fiber revealed mode degradation and reduced transmission efficiency, likely due to mechanical stress from the vacuum seals.
The temperature of the optical molasses was minimized by systematically tuning the cooling beam detuning and intensity, and the compensation coil currents used to cancel external magnetic fields. A minimum temperature of $T = (9.6 \pm 0.1)\ \mu$K was achieved.
Finally, the loading efficiency from the molasses into the ODT was optimized. After the MOT loading phase, the compensation coils were used to shift the atomic cloud's position, increasing the spatial overlap with the ODT beam. A loading efficiency of $(0.160 \pm 0.005)\%$ was achieved. This result is of the same order of magnitude as the expected value of $(0.41 \pm 0.04)\%$, calculated using the non-shallow trap model proposed in this thesis. The obtained efficiency is sufficient for the subsequent realization of an optical conveyor belt to effectively load the atoms into the fiber's core.
This work establishes a reliable experimental starting point for introducing the second, counter-propagating 1064 nm beam, necessary to realize the optical conveyor belt.

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Introduction

The work presented in this thesis was conducted in the cold atoms laboratory of Professors Francesco Minardi and Marco Prevedelli at the Department of Physics and Astronomy "A. Righi" of the University of Bologna. The primary objective of the experiment is to study a quantum system consisting of a cold rubidium-87 (^{87}Rb) gas loaded into the core of a Hollow-Core Photonic Crystal Fiber (HCPCF). By confining atoms directly within their hollow core, HCPCFs enable tight spatial confinement and strong atom-light interactions over macroscopic distances, establishing a highly versatile platform for advanced quantum technologies. Potential applications for the fiber-atoms system include the realization of highly sensitive, miniaturized quantum sensors (such as atomic magnetometers and interferometers), as well as the realization of quantum memories, which constitute the fundamental building blocks for future quantum networks.

Loading cold atoms into an HCPCF requires an efficient intermediate trapping and transfer mechanism. As a starting point, the source of these cold atoms is provided by a pre-existing setup utilizing a Magneto-Optical Trap (MOT) followed by an optical molasses phase, preparing a cold, sub-Doppler atomic sample. To efficiently load the atoms into the HCPCF, an optical conveyor belt is required. This system is realized by using two counter-propagating Optical Dipole Trap (ODT) beams that interfere to create a standing wave. Tuning the relative frequency of the two branches allows the standing wave to become dynamic, enabling the controlled transport of the cold atoms. Furthermore, the standing wave provides the necessary axial confinement inside the optical fiber's core. This thesis details the implementation and optimization of the first 1064 nm ODT beam, which represents the starting point for the implementation of the complete transfer mechanism.

The thesis is structured as follows:

Chapter 1 establishes the theoretical framework underpinning the experiment. It begins with the physics of optical waveguides, deriving the Linearly Polarized (LP) modes for standard step-index fibers in the weakly guiding approximation. It then introduces HCPCFs, contrasting the Photonic Band Gap (PBG) and Inhibited Coupling (IC) guiding mechanisms. Finally, it reviews the semiclassical theory of the optical dipole force, deriving the conservative trapping potential and scattering rates for multi-level alkali atoms.

Chapter 2 details the experimental apparatus and the analytical methodologies. Following an overview of the vacuum chamber, the MOT, and the laser frequency stabilization systems, it describes the implementation of the 1064 nm ODT setup. This includes the spatial characterization of the ODT laser beam, the design of the optical injection line, and the development of a custom horizontal microscope to monitor the near-field mode to optimize the coupling into the HCPCF. It also describes two theoretical models to calculate a reference ODT capture efficiency in shallow and non-shallow trap regimes. Finally, this chapter outlines the experimental and analytical procedure employed for obtaining the results summarized in the last chapter.

Chapter 3 presents the experimental results and their discussion. It first reports the systematic minimization of the optical molasses temperature. Subsequently, it details the tuning of the compensation magnetic fields required to maximize the spatial overlap between the atomic cloud and the ODT beam, ultimately optimizing the capture efficiency and comparing the experimental results with the theoretical predictions.

0.0.1 Hollow-Core Fibers and Cold Atoms

The integration of laser-cooled neutral atoms with hollow-core optical fibers enables enhanced light–matter interactions compared with conventional free-space geometries. The advantages of this approach arise from the limitations imposed by diffraction in tightly focused optical fields.

In a free-space optical dipole trap, atoms are confined near the focus of a laser beam. The atom–light interaction is determined by two competing length scales. A small beam waist w_0 provides strong transverse confinement and increases the local intensity, enhancing the coupling per atom. However, diffraction limits the distance over which such confinement is maintained. This characteristic length is given by the Rayleigh range:

$$z_R = \frac{\pi w_0^2}{\lambda}. \tag{1}$$

Therefore, reducing w_0 to increase the transverse confinement also decreases z_R proportionally to w_0^2, limiting the achievable interaction length. This diffraction-induced trade-off prevents the simultaneous optimization of transverse confinement and longitudinal interaction length in free-space atom optics.

Hollow-core fibers overcome this limitation by confining atoms within a guided optical mode [1]. Unlike a freely propagating beam, the guided mode is an eigenmode of the waveguide and its transverse profile is maintained by the cladding microstructure. As a result, strong transverse confinement can be preserved over macroscopic propagation distances, from centimeters to meters, without being limited by the Rayleigh range. The interaction length is therefore decoupled from diffraction, while the confined core mode maintains a high atom–light coupling per unit length.

Applications. The combination of strong confinement, long interaction lengths, and efficient coupling between atoms and guided modes enables applications in both quantum technologies and fundamental studies. In metrology, these properties improve the performance of compact quantum sensors, including atomic magnetometers and matter-wave interferometers, by increasing interrogation times and enhancing light–matter coupling [2]. In quantum optics, the quasi-one-dimensional geometry provided by hollow-core fibers enables the investigation of collective emission phenomena, such as subradiance and superradiance [3], and provides a platform for efficient quantum memories and strongly correlated many-body systems [1].

Chapter 1

Theoretical Background

This chapter provides an overview of the theoretical background underlying the main topics addressed in this thesis.
The first part introduces hollow-core fibers and describes their light-guiding mechanisms. This is followed by an overview of atomic trapping techniques, starting from the magneto-optical trap (MOT) and the optical dipole trap, outlining the fundamental principles that enable the cooling and confinement of neutral atoms.
Building on these concepts, more advanced manipulation techniques are then discussed. Specifically, the concept of the optical conveyor belt is introduced, demonstrating how moving optical lattices allow for the controlled transport of atoms. Finally, the theoretical principles of electromagnetically induced transparency (EIT) cooling are presented, highlighting its potential for achieving sub-Doppler temperatures inside the fiber core.

1.1 Hollow-Core Photonic Crystal Fibers

1.1.1 From Conventional to Hollow-Core Optical Fibers

A conventional optical fiber guides light through Total Internal Reflection (TIR). In a step-index fiber, a solid dielectric core of refractive index n_{core} is surrounded by a cladding of lower index n_{clad}, with $n_{\text{core}} > n_{\text{clad}}$. When light propagating inside the core meets the core–cladding interface at an angle steeper than the critical angle $\theta_c = \arcsin(n_{\text{clad}}/n_{\text{core}})$, it is fully reflected and remains trapped within the core over macroscopic distances. Standard telecommunications fibers, based on a silica core ($n \approx 1.45$) slightly doped to increase its index with respect to the cladding, are a well-known example of this guiding mechanism and form the basis of modern optical networks.
A hollow, air- or gas-filled core, where $n_{\text{core}} \approx 1 < n_{\text{clad}} \approx 1.45$, therefore cannot guide by TIR, since the index hierarchy is reversed. However, it would allow light to propagate through a medium far less dispersive, less nonlinear, and less absorptive than silica, while

simultaneously opening the core to atoms, gases, and particles for which a solid core is a physical barrier. The central challenge of hollow-core fiber (HCF) technology is therefore to find a confinement mechanism that does not rely on an index hierarchy between core and cladding.

Hollow-core guidance was first attempted in the 1980s and 1990s using metal-coated waveguides and simple glass capillaries, but these suffered from high losses due to the lack of an efficient low-loss cladding. This limitation was overcome with the invention of the solid-core photonic crystal fiber (PCF) in 1996, whose concept was extended to hollow cores in 1999 with the first hollow-core photonic bandgap fibers (HC-PBGFs), where light confinement is achieved through the photonic bandgap of a periodically microstructured cladding [4].

Alternative designs were therefore developed, including Kagome-lattice hollow-core fibers, introduced in 2002. In these structures, light confinement does not rely on a bandgap but on suppressed coupling between core and cladding modes (see Section 1.1.2). Further improvements were achieved around 2010 with the introduction of negative-curvature (hypocycloid) core boundaries, which significantly reduce confinement losses. This approach led to simplified antiresonant fiber (ARF) designs and, more recently, to nested antiresonant nodeless fibers (NANFs).

The main milestones in this development are summarized in Fig. 1.1. The evolution of these structures can be understood through two main trends: the progressive simplification of the cladding geometry and the transition between different light-confinement mechanisms.

1.1.2 Photonic Bandgap and Inhibited Coupling Guiding Mechanisms

The theoretical description of HCPCFs draws on concepts developed in condensed matter physics. In a crystalline solid, the periodic potential experienced by electrons gives rise to allowed energy bands separated by forbidden gaps in which no electronic states can exist. In the optical case, the periodic electronic potential is replaced with a periodic dielectric modulation: the cladding of an HCPCF is a two-dimensional photonic crystal, and the hollow core introduces a structural defect, much like a vacancy or impurity in a semiconductor. Solving the electromagnetic wave equation in the presence of this periodic structure yields eigenmodes organized in bands, with the possibility of forbidden spectral regions in which the cladding does not support propagating modes.

Mathematically, this is cast as an eigenvalue problem for the mode amplitudes, which take the form $E \propto e^{i\beta z}$ along the fiber axis z, where β is the longitudinal propagation constant. The set of allowed solutions defines the Density of Photonic States (DOPS), conventionally expressed in terms of the effective refractive index $n_{\text{eff}} = \beta/k_0$, where

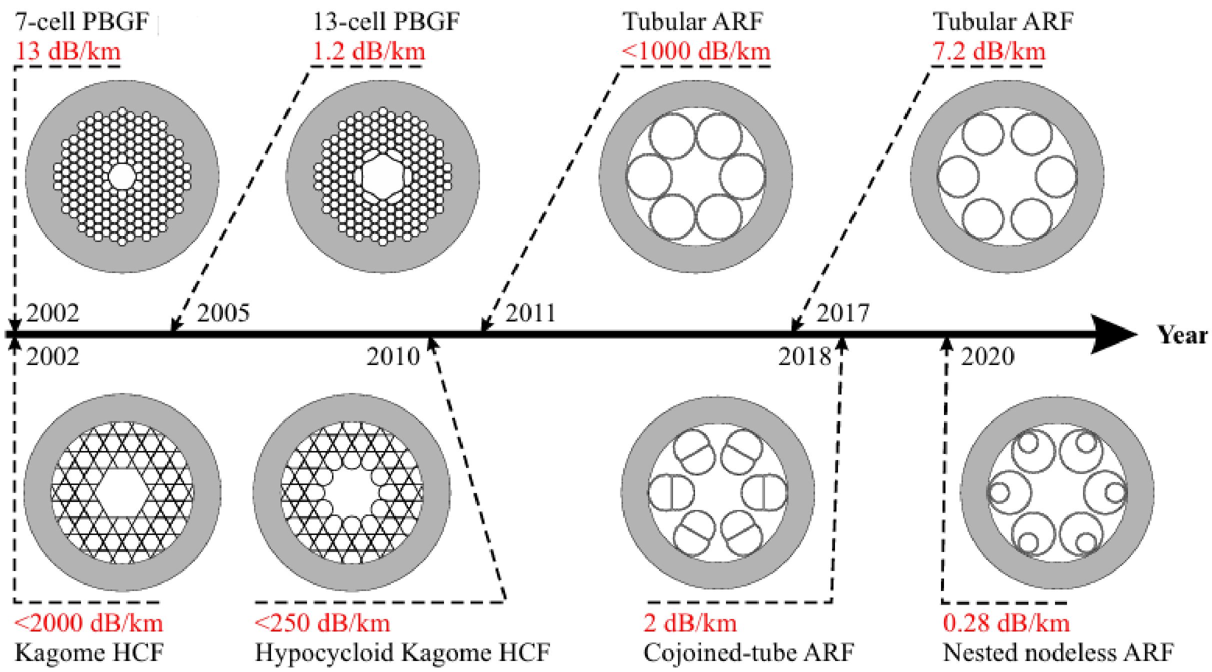


Figure 1.1: *Timeline illustrating the structural evolution of hollow-core optical fibers. The progression highlights the transition from early photonic bandgap fibers (PBGF) to advanced antiresonant architectures (ARF), including Kagome, cojoined-tube, and nested nodeless designs. Attenuation values are given for the wavelength of 1550 nm. Credits [4].*

$k_0 = \omega/c$ is the vacuum wavenumber [5]. A diagram of the DOPS in the (n_{eff}, ω) plane plays, for photons, the same role that a band structure diagram plays for electrons: it identifies which modes can exist at a given frequency and effective index.

Based on the structure of the DOPS, HCPCFs are divided into two distinct guiding regimes: Photonic Band Gap (PBG) and Inhibited Coupling (IC), schematically compared in Fig. 1.2 together with conventional TIR guidance.

Photonic Band Gap (PBG) guidance. In PBG fibers, the periodic cladding is engineered so that for certain ranges of (ω, n_{eff}) the DOPS is zero: no propagating cladding modes exist, and these forbidden regions are the photonic bandgaps. Guidance is achieved by introducing a hollow-core defect whose localized optical mode falls within one of these gaps: the core mode cannot leak into the surrounding structure because there are no available cladding states to couple into at that effective index [7]. PBG fibers deliver extremely low confinement losses, but their transmission is restricted to the narrow spectral bandwidth of the bandgap itself. For applications requiring the simultaneous guidance of widely separated wavelengths, for instance, multiple atomic transitions co-propagating in the same fiber, this narrowness becomes a limitation.

Inhibited Coupling (IC) guidance. Unlike photonic bandgap fibers, IC fibers do not require a forbidden spectral region in the cladding; instead, the core mode coexists with a

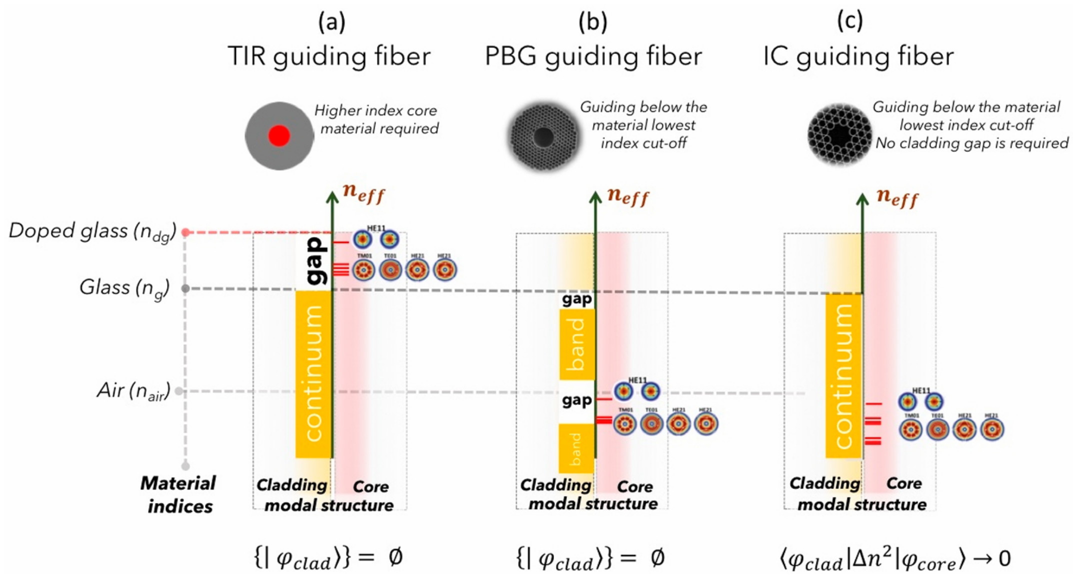


Figure 1.2: *Schematic comparison of the internal reflection, PBG and IC guiding mechanisms. The effective index n_{eff} is plotted on the vertical axis, with the relevant index levels marked by horizontal dashed lines. The graphs display the cladding photonic bands versus n_{eff}, highlighting the core guidance regimes through continua (orange rectangles) and bandgaps (white rectangles). For TIR and PBG fibers, core modes are guided within the gaps. In contrast, IC fibers exhibit no bandgaps, and guidance occurs alongside the cladding mode continuum. Credits [6].*

continuum of cladding modes of similar effective refractive index n_{eff}, and confinement is achieved by suppressing the coupling between them. By treating the silica microstructure as a local dielectric perturbation $\Delta n^2(x, y)$ introduced into a uniform vacuum background, the interaction between a core mode $|\varphi_{\text{core}}\rangle$ and a cladding mode $|\varphi_{\text{clad}}\rangle$ can be described by the overlap integral [6]

$$\langle\varphi_{\text{clad}}|\Delta n^2|\varphi_{\text{core}}\rangle = \iint_{\text{cross-section}} \varphi^*_{\text{clad}}(x, y)\, \Delta n^2(x, y)\, \varphi_{\text{core}}(x, y)\, dx\, dy. \tag{1.1}$$

In this expression, the term Δn^2 restricts the interaction to the physical regions where the silica glass is present. Whenever the fiber geometry is engineered to suppress this integral, the core mode uncouples from the cladding and remains tightly confined.

This behavior is closely related to a Bound State in the Continuum (BIC), a concept introduced by von Neumann and Wigner in 1929 [8] to describe a localized state that remains confined due to symmetry or destructive interference despite its energy lying within a continuous spectrum of radiative states.

Two mechanisms can drive the integral in Eq. 1.1 toward zero. The first is a *reduced spatial overlap*: if the core and cladding fields occupy largely disjoint regions of the transverse plane, the integrand is small everywhere and so is the integral. The second is a *transverse phase mismatch*, which is best seen by decomposing each mode into azimuthal harmonics,

$\varphi(r,\theta) = \sum_m A_m R_m(r) e^{im\theta}$. Substituting into Eq. 1.1 separates the integral into a radial part and an azimuthal part,

$$\langle \varphi_{\mathrm{clad}} | \Delta n^2 | \varphi_{\mathrm{core}} \rangle \propto \int_{r_1}^{r_2} R_{\mathrm{clad}}(r)\, R^*_{\mathrm{core}}(r)\, r\, dr \;\times\; \int_0^{2\pi} e^{i(m_{\mathrm{clad}} - m_{\mathrm{core}})\theta}\, d\theta, \tag{1.2}$$

where m_{clad} and m_{core} are the azimuthal indices of the two modes. The azimuthal integral vanishes whenever $m_{\mathrm{clad}} \neq m_{\mathrm{core}}$: modes with different azimuthal order are orthogonal, and their coupling is suppressed by symmetry regardless of their spatial overlap [6]. [1] By engineering the cladding geometry so that the cladding modes carry a high azimuthal index m while the fundamental core mode has $m \approx 0$, the phase-mismatch term in Eq. 1.2 strongly suppresses the coupling over a broad spectral range.

Because IC guidance is based on local mode interactions rather than on a global photonic bandgap, IC fibers can provide broadband transmission compared with PBG fibers, whose bandwidth is limited by the spectral range of the bandgap [6].

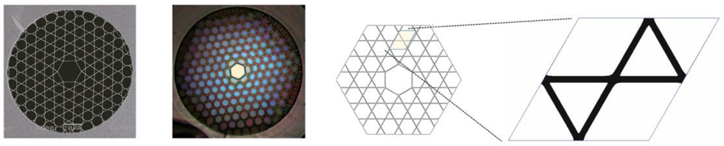
1.1.3 Kagome-Lattice Hollow-Core Fibers

The Kagome-lattice fiber is a primary realization of the Inhibited Coupling guiding mechanism [6]. Its cladding is a trihexagonal tiling, formed by a network of thin silica capillaries that create a periodic pattern of large hexagonal and smaller triangular interstitial air holes (see Fig. 1.3a).

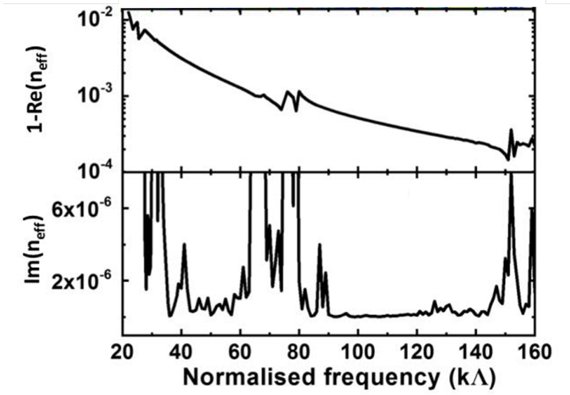
Confinement in this regime is achieved through the two suppression strategies identified in Section 1.1.2. The thin silica struts support cladding modes with a highly oscillatory transverse phase, characterized by a high azimuthal index number m, while the fundamental core mode features a slowly varying, nearly Gaussian profile with $m \approx 0$. The strong mismatch in m makes the azimuthal integral in Eq. 1.2 vanish, so that the core mode cannot couple into the surrounding glass network despite the spectral overlap with the cladding continuum.

This coupling suppression is not, however, absolute across the entire spectrum. The silica struts behave as thin dielectric slabs, and at the wavelengths for which a slab becomes resonant its transverse phase matches that of the core mode leading to hybridization. These discrete resonances produce the sharp loss peaks visible in Fig. 1.3b. Their location follows from the transverse resonance condition of a dielectric slab of thickness t and index n_g surrounded by air. Requiring the optical round-trip phase inside the slab to be an integer multiple of 2π, and accounting for the phase accumulated at the air–glass

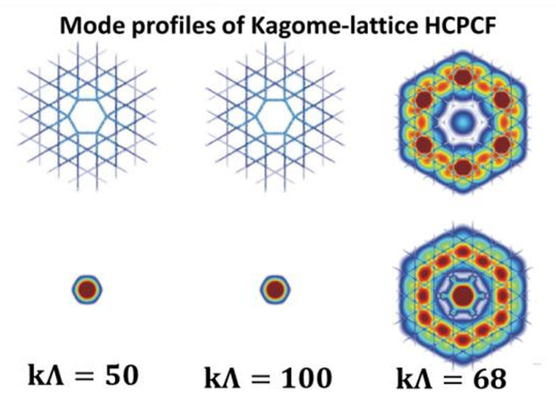
[1] In an idealized axisymmetric model, the azimuthal integral vanishes for modes with different azimuthal indices. In a real Kagome fiber, the discrete cladding geometry breaks perfect cylindrical symmetry, so the suppression of coupling is approximate and depends on the detailed modal overlap and phase matching.



(a)



(b)



(c)

Figure 1.3: *Overview of guidance in a hypocycloid-core Kagome fiber.* ***(a)*** *Scanning electron microscopy micrographs and structural unit cell.* ***(b)*** *Real and imaginary parts of the effective index of the fundamental core mode as a function of the normalized frequency* $k\Lambda$*:* $Im(n_{eff})$*, proportional to the propagation loss, remains low except at a series of sharp resonance peaks.* ***(c)*** *Transverse intensity profile of the same mode at three representative frequencies: tight confinement at* $k\Lambda = 50$ *and* $k\Lambda = 100$*, and loss of confinement at the resonance* $k\Lambda = 68$*. Credits [6].*

interfaces at grazing incidence (the regime relevant to a hollow core), i.e.

$$2\left(\frac{2\pi}{\lambda}\sqrt{n_g^2-1}\right)t = 2\pi j, \tag{1.3}$$

yields the resonance condition [6]

$$\lambda_j = \frac{2t\sqrt{n_g^2-1}}{j}, \qquad j = 1, 2, 3, \ldots, \tag{1.4}$$

where j is the radial mode order of the strut. Equivalently, in terms of the normalized frequency $k\Lambda = (2\pi/\lambda)\Lambda$,

$$k\Lambda = \frac{j\,\pi\,\Lambda}{t\sqrt{n_g^2-1}}. \tag{1.5}$$

At these wavelengths the strut is resonant and transmits rather than reflects the transverse field, so the core mode leaks into the cladding and the confinement breaks down, as shown at $k\Lambda = 68$ in Fig. 1.3c. Between consecutive resonances the slab is antiresonant and acts as a high reflector: the m-mismatch is restored and the light is tightly localized in the core (e.g. at $k\Lambda = 50$ or 100). Because the resonance wavelengths scale linearly with

t, fabricating fibers with nanoscale glass webs pushes the resonances toward shorter wavelengths, thereby opening ultra-broadband transmission windows in the spectral region of interest.

To reduce confinement loss the core boundary can be engineered with a hypocycloidal (or negative-curvature) profile by incorporating a series of inward-pointing glass arcs (as depicted in Fig. 1.1). This geometry increases the phase mismatch and reduces the overlap between the core mode and the surrounding silica [6].

Higher order modes Similar to standard hollow capillaries and conventional fibers, the core of a Kagome IC fiber supports multiple transverse modes, denoted by the $LP_{m,n}$ notation where m and n represent the azimuthal and radial mode numbers.

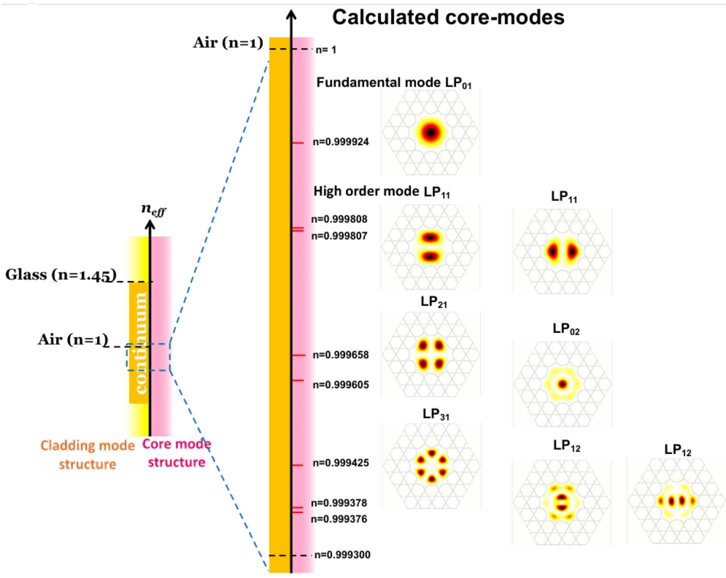


Figure 1.4: *Calculated modal structure of an IC-HCPCF, mapping the fundamental* LP_{01} *mode and higher-order modes against the continuous cladding modal spectrum.*

Fig. 1.4 illustrates this modal hierarchy. Higher-order modes (such as LP_{11} or LP_{02}) exist but experience much higher leakage. Their electromagnetic fields extend radially closer to the silica boundaries, which increases the spatial overlap with the cladding modes, and their non-zero azimuthal index (like $m = 1$ for the LP_{11} mode) reduces the phase mismatch. Thus, while the fiber inherently supports a multimode continuum, if properly designed it can behave as a single-mode waveguide over macroscopic distances due to the attenuation of higher-order modes [6].

1.2 Electromagnetic Trapping of Neutral Atoms

1.2.1 The Magneto-Optical Trap

The Magneto-Optical Trap (MOT) is a technique used to simultaneously confine and cool neutral atoms. It relies on the combination of a position-dependent restoring force and a velocity-dependent damping force [9].

Spatial confinement is produced by a quadrupole magnetic field, generated by a pair of anti-Helmholtz coils. The field is zero at the center of the trap and grows linearly along each direction, with a gradient that induces a position-dependent Zeeman shift of the atomic energy levels. Three pairs of counter-propagating laser beams, red-detuned from the atomic resonance, are sent along the three orthogonal axes. Within each pair, the two beams carry opposite circular polarizations, σ^+ and σ^-, selected so that the beam pushing toward the center is the one brought into resonance by the local Zeeman shift. As an atom drifts away from the origin, the Zeeman shift brings it closer to resonance with the inward-propagating beam, producing a restoring radiation pressure that pushes the atom back toward the center [9]. A schematic representation of the MOT is shown in Fig. 1.5

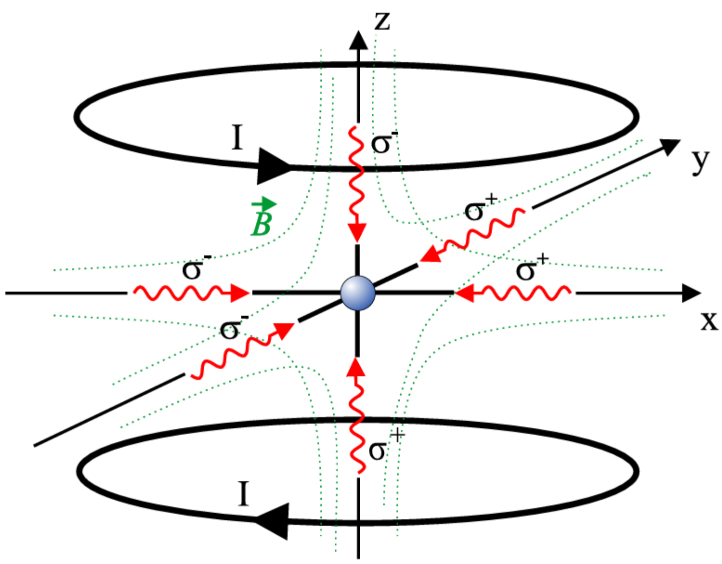


Figure 1.5: *Schematic representation of the Magneto-Optical Trap (MOT) configuration, illustrating the counter-propagating laser beams with alternating σ^+/σ^- polarizations and the anti-Helmholtz coils used to generate the magnetic quadrupole field. From [10]*

The cooling mechanism operating inside the MOT is known as optical molasses [11]. It is generated by the same three pairs of counter-propagating beams used for confinement, when their action is considered in the absence of the magnetic field gradient. The beams are slightly red-detuned from the atomic resonance: when an atom moves, the laser beam opposing its trajectory is Doppler-shifted closer to resonance. The atom therefore absorbs more photons from the opposing beam than from the co-propagating one, resulting in a velocity-dependent friction force that viscously damps the atomic motion.

During this continuous molasses cooling, a competing heating mechanism arises from the spontaneous emission of photons. After absorbing a photon, the atom returns to its ground state by spontaneously re-emitting a photon in a random direction. Each emission imparts a small momentum kick (recoil) to the atom; this random walk in momentum space leads to a diffusion of the atomic velocities and heats the atomic cloud. The thermal equilibrium established between the viscous damping of the molasses (cooling) and the random recoils from spontaneous emission (heating) defines the Doppler temperature limit [11]:

$$T_D = \frac{\hbar\Gamma}{2k_B} \tag{1.6}$$

where $\hbar$ is the reduced Planck constant, Γ is the natural linewidth of the atomic transition, and k_B is the Boltzmann constant. This expression corresponds to the minimum achievable under Doppler cooling and is obtained for an optimal detuning $\Delta = -\Gamma/2$. For ^{87}Rb atoms on the D2 line, it corresponds to a temperature of approximately 146 μK.

1.2.2 Polarization Gradient Cooling

While the MOT cools atoms down to the Doppler limit, further temperature reduction requires Polarization Gradient Cooling (PGC), commonly referred to as Sisyphus cooling. Two main configurations of PGC exist, depending on the relative polarizations of the counter-propagating beams: the lin$\perp$lin geometry, in which the two beams have orthogonal linear polarizations, and the $\sigma^+\sigma^-$ geometry, in which the beams have opposite circular polarizations. The two configurations give rise to distinct polarization patterns and to correspondingly distinct cooling dynamics. In the following, the discussion is restricted to the lin$\perp$lin case, which provides the most direct illustration of the Sisyphus mechanism [12].

The spatial interference of two counter-propagating beams with orthogonal linear polarizations generates a standing wave with a spatially varying polarization state. Over a distance of $\lambda/2$, the local polarization cycles through linear, circular, and orthogonal linear states. For atoms with a degenerate ground state, the light shift (AC Stark shift) depends on the local polarization and on the specific Zeeman sublevel m_F. As an atom moves through this optical field, it climbs a potential hill created by the spatially varying light shift, converting its kinetic energy into potential energy [12].

Upon reaching the peak of the potential hill, the local polarization drives an optical pumping transition that transfers the atom into a lower-energy sublevel. The spontaneously emitted photon carries away the excess energy, leaving the atom at the minimum of a new potential valley. A schematic representation of this process is depicted in Fig. 1.6. This continuous cycle of ascending potential gradients and undergoing subsequent optical pumping dissipates the atom's kinetic energy, resulting in temperatures significantly below the Doppler limit.

The lowest temperatures achievable through PGC are limited by the recoil energy associated with the emission of a single photon. Defining the recoil energy as $E_R = \hbar^2 k^2/(2m)$, with k the photon wave vector and m the atomic mass, the definition for the corresponding recoil temperature is

$$T_R = \frac{\hbar^2 k^2}{2mk_B} = \frac{E_R}{k_B}. \tag{1.7}$$

Temperatures of the order of a few T_R represent the practical lower bound for PGC. At this scale, the thermal kinetic energy of the atom becomes comparable to the energy of a single photon recoil, and the Sisyphus cooling cycle can no longer extract energy efficiently without re-heating the atom through the spontaneous emission process itself. Reaching temperatures below this threshold requires sub-recoil cooling techniques.

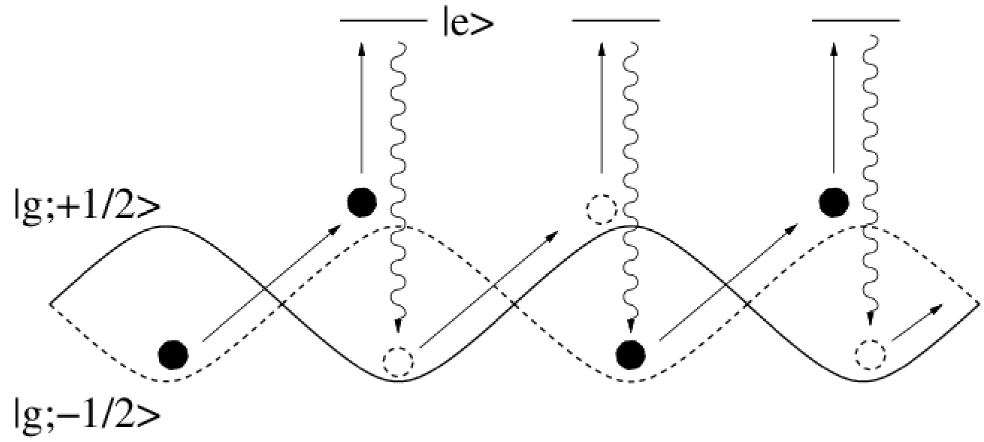


Figure 1.6: *Schematic representation of Polarization Gradient Cooling (Sisyphus cooling). The diagram illustrates the position-dependent light shifts of the ground-state Zeeman sublevels and the optical pumping process that dissipates the atom's kinetic energy.*

1.2.3 The Optical Dipole Trap

In contrast to the dissipative radiation-pressure forces that underlie the MOT, an optical dipole trap (ODT) confines neutral atoms in a nearly conservative potential produced by far-off-resonant light. The mechanism relies on the interaction between the intensity gradient of the light field and the atomic dipole moment that the field itself induces [13]. Because the optical dipole interaction is weak compared with the radiative forces used in a MOT, typical trap depths lie in the millikelvin range or below; atoms must therefore be pre-cooled (through the mechanisms described above) before they can be transferred into an ODT.

1.2.3.1 Oscillator model and semiclassical treatment

The essential features of the optical dipole force can be derived by treating the atom as a classical damped oscillator driven by a classical oscillating electric field. The laser field $\mathbf{E}(\mathbf{r},t) = \hat{\mathbf{e}}\,\tilde{E}(\mathbf{r})\,e^{-i\omega t} + \text{c.c.}$ induces a dipole moment $\mathbf{p}(\mathbf{r},t) = \hat{\mathbf{e}}\,\tilde{p}(\mathbf{r})\,e^{-i\omega t} + \text{c.c.}$ whose amplitude is proportional to the field amplitude through the complex atomic polarizability $\alpha(\omega)$,

$$\tilde{p} = \alpha(\omega)\,\tilde{E}. \tag{1.8}$$

The interaction potential of the induced dipole in the driving field is

$$U_{\text{dip}}(\mathbf{r}) = -\frac{1}{2}\,\langle \mathbf{p}\cdot\mathbf{E}\rangle = -\frac{1}{2\epsilon_0 c}\,\text{Re}(\alpha)\,I(\mathbf{r}), \tag{1.9}$$

where the angular brackets denote time averaging over the optical period, $I = 2\epsilon_0 c|\tilde{E}|^2$ is the field intensity, and the factor 1/2 accounts for the dipole being induced rather than permanent [13]. The potential is proportional to the real (dispersive) part of α, which describes the in-phase component of the dipole oscillation. The resulting dipole force

$$\mathbf{F}_{\text{dip}}(\mathbf{r}) = -\nabla U_{\text{dip}}(\mathbf{r}) = \frac{1}{2\epsilon_0 c}\,\text{Re}(\alpha)\,\nabla I(\mathbf{r}) \tag{1.10}$$

is conservative and proportional to the intensity gradient.
Concurrently, the driven oscillator absorbs power from the field and re-emits it as dipole radiation. The absorbed power

$$P_{\text{abs}}(\mathbf{r}) = \langle \dot{\mathbf{p}}\cdot\mathbf{E}\rangle = \frac{\omega}{\epsilon_0 c}\,\text{Im}(\alpha)\,I(\mathbf{r}) \tag{1.11}$$

is proportional to the imaginary (absorptive) part of α, describing the out-of-phase component of the dipole. Interpreting the absorption as a stream of photons of energy $\hbar\omega$, the corresponding spontaneous scattering rate is [13]

$$\Gamma_{\text{sc}}(\mathbf{r}) = \frac{P_{\text{abs}}}{\hbar\omega} = \frac{1}{\hbar\epsilon_0 c}\,\text{Im}(\alpha)\,I(\mathbf{r}). \tag{1.12}$$

To obtain an explicit expression for $\alpha(\omega)$, the atom is first modeled in Lorentz's picture as an electron of mass m_e and charge $-e$ elastically bound to the core with eigenfrequency ω_0, damped by radiative energy loss. Integration of the equation of motion $\ddot{x} + \Gamma_\omega \dot{x} + \omega_0^2 x = -eE(t)/m_e$ yields [13]

$$\alpha = \frac{e^2}{m_e}\,\frac{1}{\omega_0^2 - \omega^2 - i\omega\Gamma_\omega}, \tag{1.13}$$

where $\Gamma_\omega = e^2\omega^2/(6\pi\epsilon_0 m_e c^3)$ is the classical damping rate from Larmor's formula. In the semiclassical treatment the same functional form is recovered, the only modification being that the damping rate is no longer given by Larmor's formula but is determined by the quantum-mechanical dipole matrix element between ground and excited state [13]:

$$\Gamma = \frac{\omega_0^3}{3\pi\epsilon_0\hbar c^3}\,|\langle e|\hat{\mu}|g\rangle|^2. \tag{1.14}$$

For the D lines of the alkali atoms Na, K, Rb, and Cs, the classical result of Eq. 1.13 agrees with the true decay rate to within a few percent. Expressing the polarizability in terms of this quantum-mechanical Γ gives [13]

$$\alpha(\omega) = 6\pi\epsilon_0 c^3\,\frac{\Gamma/\omega_0^2}{\omega_0^2 - \omega^2 - i(\omega^3/\omega_0^2)\Gamma}. \tag{1.15}$$

Substituting Eq. 1.15 into Eqs. 1.9 and 1.12 yields the general expressions [13]

$$U_{\text{dip}}(\mathbf{r}) = -\frac{3\pi c^2}{2\omega_0^3}\left(\frac{\Gamma}{\omega_0-\omega} + \frac{\Gamma}{\omega_0+\omega}\right) I(\mathbf{r}), \tag{1.16}$$

$$\Gamma_{\text{sc}}(\mathbf{r}) = \frac{3\pi c^2}{2\hbar\omega_0^3}\left(\frac{\omega}{\omega_0}\right)^3\left(\frac{\Gamma}{\omega_0-\omega} + \frac{\Gamma}{\omega_0+\omega}\right)^2 I(\mathbf{r}). \tag{1.17}$$

These expressions are valid for any driving frequency and contain two resonant contributions: the one at $\omega = \omega_0$ and the counter-rotating term resonant at $\omega = -\omega_0$. In most trapping experiments the laser is tuned close to ω_0 with detuning $\Delta \equiv \omega - \omega_0$ satisfying $|\Delta| \ll \omega_0$; the counter-rotating term can then be neglected in the rotating-wave approximation (RWA), and one may set $\omega/\omega_0 \simeq 1$. Equations 1.16–1.17 then reduce to [13]

$$U_{\text{dip}}(\mathbf{r}) = \frac{3\pi c^2}{2\omega_0^3}\,\frac{\Gamma}{\Delta}\,I(\mathbf{r}), \tag{1.18}$$

$$\Gamma_{\text{sc}}(\mathbf{r}) = \frac{3\pi c^2}{2\hbar\omega_0^3}\left(\frac{\Gamma}{\Delta}\right)^2 I(\mathbf{r}). \tag{1.19}$$

1.2.3.2 Quantum-mechanical picture

The same potential arises in the quantum description as an energy-level shift induced by the off-resonant field. The effect of far-detuned laser light can be treated as a perturbation of second order in the electric field, linear in the intensity. Within the dressed-atom approach, the unperturbed states of the combined atom–photon system are considered: in the ground state the atom has zero internal energy and the field carries $n\hbar\omega$, so $E_i = n\hbar\omega$; after absorption of one photon the atom is in the excited state with internal energy $\hbar\omega_0$ and the field carries $(n-1)\hbar\omega$, giving $E_j = \hbar\omega_0 + (n-1)\hbar\omega = -\hbar\Delta + n\hbar\omega$. The energy denominator is therefore $E_i - E_j = \hbar\Delta$. For low saturation, the second-order shift of the ground state due to the coupling with the excited state $|e\rangle$ through the dipole operator $\hat{\mu}$ reads [13]

$$\Delta E = \frac{|\langle e|\hat{\mu}|g\rangle|^2}{\hbar\Delta}\,|\tilde{E}|^2 = \frac{3\pi c^2}{2\omega_0^3}\,\frac{\Gamma}{\Delta}\,I(\mathbf{r}), \tag{1.20}$$

where the dipole matrix element has been replaced by Γ via Eq. 1.14. This perturbative result coincides with the semiclassical dipole potential of Eq. 1.18 [13]: the AC Stark shift of the ground state is the dipole potential, while the excited state experiences the opposite shift. Since in the low-saturation regime the atom resides predominantly in the ground state, the light-shifted ground state acts as the effective conservative potential for the atomic motion. A spatially varying intensity, such as the Gaussian profile of a focused beam, produces a position-dependent shift and hence a trapping potential, as illustrated in Fig. 1.7.

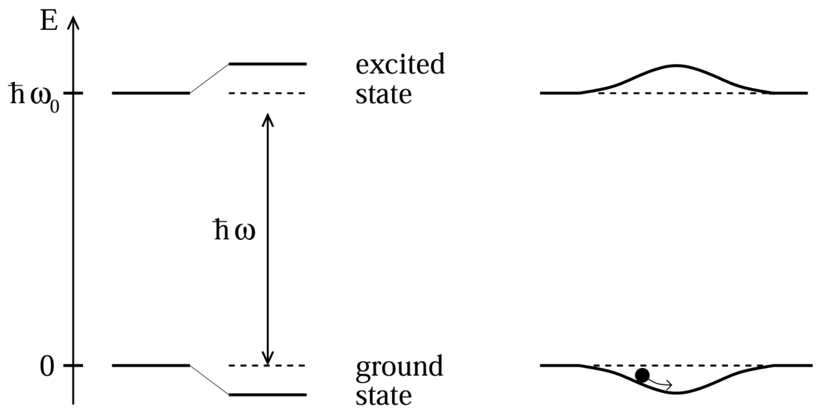


Figure 1.7: *Light shifts for a two-level atom. Left: red-detuned light ($\Delta < 0$) shifts the ground state down and the excited state up by equal amounts. Right: a spatially inhomogeneous field such as a Gaussian laser beam produces a ground-state potential well in which an atom can be trapped.*

The sign of the detuning determines the trapping geometry. For a red-detuned field ($\Delta < 0$) the ground state is shifted downward, yielding an attractive potential that confines atoms near intensity maxima; for a blue-detuned field ($\Delta > 0$) the shift is positive and atoms are pushed toward intensity minima, where spontaneous scattering is also reduced.

Combining Eqs. 1.18–1.19 gives the relation

$$\hbar\Gamma_{\rm sc} = \frac{\Gamma}{\Delta}\,U_{\rm dip}, \tag{1.21}$$

a direct consequence of the link between the absorptive and dispersive response of the oscillator [13]. Since the dipole potential scales as I/Δ while the scattering rate scales as I/Δ^2, a deep potential can be maintained at comparatively low scattering, and thus low heating and decoherence, by using high intensity together with large detuning.

1.2.3.3 Multi-level atoms and the alkali D-lines

Real alkali atoms possess a fine and hyperfine structure that are neglected in the two-level model. The state-dependent dipole potential is then obtained from second-order time-independent perturbation theory. For a non-degenerate ground state $|g_i\rangle$, the energy shift is [13]

$$\Delta E_i = \sum_{j\neq i} \frac{|\langle e_j|\hat{\mu}|g_i\rangle|^2}{E_i - E_j} |\tilde{E}|^2, \tag{1.22}$$

where the sum runs over all electronically excited states $|e_j\rangle$. By the Wigner–Eckart theorem, each dipole matrix element factorizes into a reduced matrix element $\|\mu\|$, depending only on the orbital wavefunctions, and a real transition coefficient c_{ij} encoding the angular-momentum geometry and the laser polarization, $\mu_{ij} = c_{ij}\|\mu\|$. Since $\|\mu\|$ is directly related to Γ through Eq. 1.14, the ground-state shift becomes [13]:

$$\Delta E_i = \frac{3\pi c^2 \Gamma}{2\omega_0^3}\, I(\mathbf{r}) \sum_j \frac{c_{ij}^2}{\Delta_{ij}}, \tag{1.23}$$

with Δ_{ij} the detuning from the $i \to j$ transition. The dipole potential is therefore a sum over all coupled excited states, weighted by their line-strength factors c_{ij}^2 and inverse detunings.

For alkali metals the dominant contributions come from the D_1 ($nS_{1/2} \to nP_{1/2}$) and D_2 ($nS_{1/2} \to nP_{3/2}$) fine-structure lines [13]; higher-lying transitions lie tens of THz away and contribute negligibly in the near-infrared trapping regime used in this work. Under the assumption that the optical detuning is large compared with the excited-state hyperfine splitting, so that the resolved hyperfine components of each fine-structure line can be replaced by their weighted center, the line-strength sum rules allow Eq. 1.23 to be evaluated in closed form for a ground state $|F, m_F\rangle$ [13]:

$$U_{\mathrm{dip}}(\mathbf{r}) = \frac{\pi c^2 \Gamma}{2\omega_0^3} \left(\frac{2 + \mathcal{P}\, g_F m_F}{\Delta_{2,F}} + \frac{1 - \mathcal{P}\, g_F m_F}{\Delta_{1,F}} \right) I(\mathbf{r}), \tag{1.24}$$

where g_F is the Landé factor, $\mathcal{P}$ describes the laser polarization ($\mathcal{P} = 0$ for linear, $\mathcal{P} = \pm 1$ for $\sigma^{\pm}$), and $\Delta_{2,F}$, $\Delta_{1,F}$ are the detunings from the centers of the D_2 and D_1 transitions, respectively (Fig. 1.8).

The structure of Eq. 1.24 follows from the line-strength factors of the two transitions. For linear polarization, symmetry requires both electronic ground states $m_J = \pm 1/2$ to be shifted equally; after coupling to the nuclear spin, all magnetic sublevels of a given F remain degenerate, with line-strength factors $2/3$ for the D_2 line and $1/3$ for the D_1 line [13]. These factors appear as the constants 2 and 1 in the numerators of Eq. 1.24 (the common factor $1/3$ being absorbed into the prefactor). For circular polarization, the m_J degeneracy is lifted: the line-strength factors become $(2 \pm g_F m_F)/3$ for D_2 and

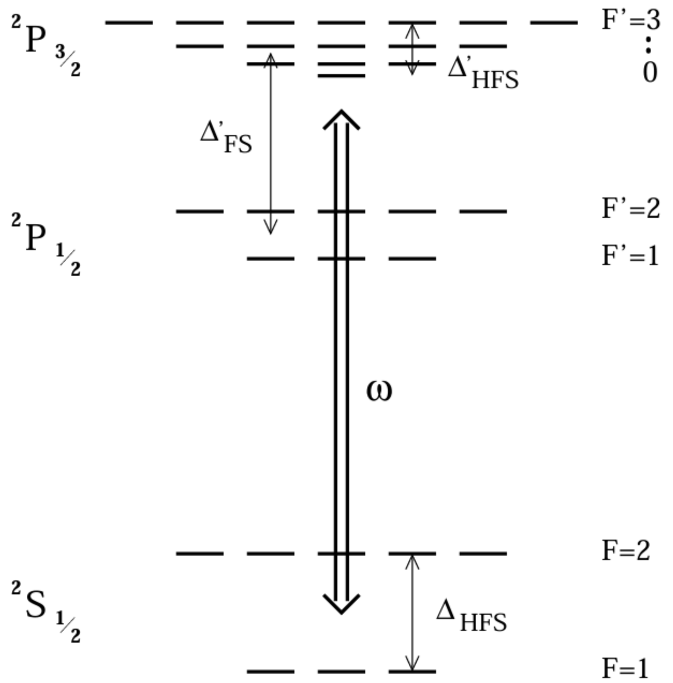


Figure 1.8: *Level scheme of an alkali atom (nuclear spin $I = 3/2$, as in ^{87}Rb) relevant to the dipole-potential calculation. The fine-structure doublet D_1 ($nS_{1/2} \rightarrow nP_{1/2}$) and D_2 ($nS_{1/2} \rightarrow nP_{3/2}$) couples the ground manifold $nS_{1/2}$ to the excited $nP_{1/2}$ and $nP_{3/2}$ manifolds; $\Delta_{1,F}$ and $\Delta_{2,F}$ denote the detunings of the trapping laser from the centers of the two lines. After [13].*

$(1 \mp g_F m_F)/3$ for D_1, where the substitution $m_J \to \frac{1}{2} g_F m_F$ (with $g_J = 2$ for alkali atoms) follows from the analogy with the linear Zeeman effect in weak magnetic fields. This m_F-dependent contribution is the vector light shift, which acts as a fictitious magnetic field along the propagation axis and lifts the degeneracy of the magnetic sublevels. For this reason, linearly polarized light is generally preferred when homogeneous, state-insensitive confinement is required [13].

For the photon scattering rate, the same line-strength factors enter as for the dipole potential, since absorption and light shifts are governed by the same transition matrix elements. For linear polarization one obtains [13]

$$\Gamma_{\rm sc}(\mathbf{r}) = \frac{\pi c^2 \Gamma^2}{2\hbar\omega_0^3} \left(\frac{2}{\Delta_{2,F}^2} + \frac{1}{\Delta_{1,F}^2} \right) I(\mathbf{r}). \tag{1.25}$$

The general expression of Eq. 1.24 reduces to the two-level result of Eq. 1.18 in two limiting regimes. When the detuning is much larger than the fine-structure splitting ($|\Delta| \gg |\Delta_{\rm FS}|$, with $\Delta_{\rm FS} = \omega_{D_2} - \omega_{D_1}$), the two detunings become nearly equal, $\Delta_{1,F} \simeq \Delta_{2,F} \simeq \Delta$. For linear polarization ($\mathcal{P} = 0$) the numerators add to $2 + 1 = 3$, the total oscillator strength of the D-doublet; absorbing this factor into an effective linewidth $\Gamma_{\rm eff}$, Eq. 1.24 takes the two-level form of Eq. 1.18, with Δ measured from the weighted center of the doublet. In the opposite limit, where the laser is tuned close to one line ($|\Delta_{i,F}| \ll |\Delta_{j,F}|$), only the nearest transition contributes appreciably and the sum collapses to a single term, again yielding the two-level form. In both limits, linear polarization suppresses the vector light shift, so the resulting potential is independent of m_F and is identical in form to the two-level dipole potential. If the detuning additionally exceeds the ground-state hyperfine splitting ($|\Delta| \gg \Delta_{HFS}$), all sub-states of the electronic ground manifold are shifted by the same amount and the potential becomes completely independent of both m_F and F. This justifies treating the atom as an effective two-level system throughout this work, where far-detuned, linearly polarized beams are used in all trapping configurations.

1.3 Optical Conveyor Belt

The optical dipole trap discussed in the previous section provides a static conservative potential in which atoms are confined near the focus of a single laser beam. Transporting the trapped cloud over macroscopic distances requires a different architecture, in which the position of the potential minima can be controlled dynamically. The optical conveyor belt fulfills this requirement by replacing the single beam with two counter-propagating, phase-coherent beams: their interference generates a standing wave whose nodes and antinodes can be set into motion by imposing a controlled frequency difference between the two arms. This section derives the geometry of the standing-wave potential, the kinematics of the moving lattice, and the acceleration limits that bound adiabatic transport.

1.3.1 Interference of Counter-Propagating Gaussian Beams and the Stationary Lattice

The fundamental mechanism of the optical conveyor belt relies on the periodic intensity pattern generated by the interference of two counter-propagating laser beams [10]. The resulting standing wave acts as a one-dimensional optical lattice, imposing a tight spatial confinement on atoms via the optical dipole force.

Assuming two coherent Gaussian beams in the fundamental transverse mode propagating along the z-axis, the respective electric fields can be modeled as follows. The first beam, propagating towards $+z$ with angular frequency ω_1, is described by:

$$E_1(z, \rho, t) = E_0 \frac{w_0}{w(z)} \exp\left(-\frac{\rho^2}{w^2(z)}\right) \exp\left(i(kz - \omega_1 t + \phi(z))\right) \tag{1.26}$$

where E_0 denotes the peak field amplitude, $k = 2\pi/\lambda$ the optical wavenumber, $\phi(z)$ the Gouy phase shift, and $\rho = \sqrt{x^2 + y^2}$ the radial coordinate. The beam radius $w(z)$ evolves along the propagation axis according to:

$$w^2(z) = w_0^2 \left(1 + \frac{z^2}{z_R^2}\right), \tag{1.27}$$

with w_0 the waist at the focal plane and $z_R = \pi w_0^2/\lambda$ the Rayleigh length. The second field, counter-propagating towards $-z$ at frequency ω_2, is given by:

$$E_2(z, \rho, t) = E_0 \frac{w_0}{w(z)} \exp\left(-\frac{\rho^2}{w^2(z)}\right) \exp\left(i(-kz - \omega_2 t - \phi(z))\right). \tag{1.28}$$

To determine the local intensity profile for degenerate frequencies ($\omega_1 = \omega_2 = \omega$), the coherent superposition $E_{tot} = E_1 + E_2$ is evaluated. Restricting the analysis to the spatial region localized around the focal point ($z \ll z_R$), the variation of the beam waist and the Gouy phase shift become negligible. The superimposed electric field simplifies to:

$$E_{tot}(z, \rho, t) \approx E_0 \frac{w_0}{w(z)} \exp\left(-\frac{\rho^2}{w^2(z)}\right) e^{-i\omega t} \left[e^{ikz} + e^{-ikz}\right]. \tag{1.29}$$

Applying Euler's relation, the spatial interference manifests as a real cosine envelope $2\cos(kz)$. The effective optical intensity $I(z, \rho)$ experienced by the atoms is proportional to the time-averaged squared amplitude of the field, $\langle |E_{tot}|^2 \rangle$, which yields a stationary standing wave:

$$I(z, \rho) = I_{\max} \frac{w_0^2}{w^2(z)} \exp\left(-\frac{2\rho^2}{w^2(z)}\right) \cos^2(kz). \tag{1.30}$$

The peak intensity $I_{\max} = 4P/(\pi w_0^2)$ follows from the total optical power P distributed across the two beams. Each beam carries power $P/2$ and has a peak intensity $2(P/2)/(\pi w_0^2) = P/(\pi w_0^2)$; the additional factor of 4 in $I_{\max}$ originates from the constructive interference at the antinodes, where the two field amplitudes add in phase.

In the far-detuned regime, where the laser frequency is significantly shifted from any atomic resonance, the resulting dipole force is conservative. The optical potential energy $U(\rho, z)$ is therefore directly proportional to the local intensity distribution:

$$U(\rho, z) = U_0 \frac{w_0^2}{w^2(z)} \exp\left(-\frac{2\rho^2}{w^2(z)}\right) \cos^2(kz), \tag{1.31}$$

where U_0 defines the maximum depth of the individual potential wells, with $U_0 < 0$ for a red-detuned, attractive trap.

The harmonic spatial modulation introduced by the $\cos^2(kz)$ term imposes a longitudinal periodicity, organizing the optical potential into a one-dimensional lattice of trapping sites with a lattice constant of $\lambda/2$.

Within a potential well, atomic motion is restricted to a small region around the trap minimum at $z = 0$ and $\rho = 0$, allowing the harmonic approximation. Applying a second-order Taylor expansion to Eq. (1.31) using $\cos^2(kz) \approx 1 - (kz)^2$ and $\exp(-2\rho^2/w_0^2) \approx 1 - 2\rho^2/w_0^2$, the potential takes the quadratic form

$$U(\rho, z) \approx U_0 \left(1 - \frac{2\rho^2}{w_0^2}\right)(1 - k^2 z^2) \approx U_0 - U_0 k^2 z^2 - U_0 \frac{2\rho^2}{w_0^2}. \tag{1.32}$$

Equating these leading-order restoring terms to the classical energies of a harmonic oscillator, $\frac{1}{2}\kappa z^2$ and $\frac{1}{2}\kappa \rho^2$, yields the effective spring constants $\kappa_z = 2|U_0|k^2$ and $\kappa_{rad} = 4|U_0|/w_0^2$. The characteristic axial and radial oscillation frequencies therefore read:

$$\Omega_z = \sqrt{\frac{2|U_0|k^2}{m}}, \qquad \Omega_{rad} = \sqrt{\frac{4|U_0|}{m w_0^2}}. \tag{1.33}$$

These characteristic frequencies establish the dynamical timescales of the system and set the bound within which external parameters can be varied while preserving adiabaticity.

1.3.2 Geometry of the Trapping Potential

The ratio of the two oscillation frequencies exhibits a geometric dependence, independent of the overall trap depth U_0:

$$\frac{\Omega_z}{\Omega_{rad}} = \frac{kw_0}{\sqrt{2}} = \frac{\sqrt{2}\,\pi w_0}{\lambda}. \tag{1.34}$$

In typical experimental configurations the beam waist exceeds the optical wavelength ($w_0 \gg \lambda$), so that the longitudinal confinement dominates over the transverse confinement ($\Omega_z \gg \Omega_{rad}$). The standing wave then imposes quasi-two-dimensional potential wells with a strongly anisotropic, disc-shaped geometry: along the z-axis the atomic motion is localized to a fraction of λ, while the transverse motion retains a comparatively broad spatial extent. This anisotropy suppresses tunneling between adjacent lattice sites while leaving the transverse dynamics only weakly confined. A schematic representation of the resulting disc-shaped potential wells is shown in Fig. 1.9.

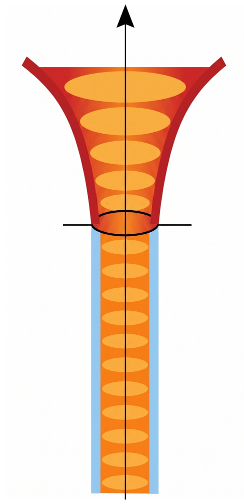


Figure 1.9: *The strongly anisotropic, disc-shaped potential wells of the one-dimensional optical lattice.*

1.3.3 Kinematics of the Moving Standing Wave

Translating the stationary lattice into an active transport potential requires inducing a macroscopic translation of the interference pattern. This is achieved by imposing a controlled frequency difference $\Delta\nu$ between the two counter-propagating laser modes. The dynamic behavior can be derived by applying the detuning either symmetrically across both beams or asymmetrically to a single beam. In both configurations, the coherent superposition of the fields is resolved using the sum-to-product identity $\cos A + \cos B = 2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$.

Symmetric detuning. Consider the case in which the detuning is distributed symmetrically, shifting the optical frequencies to $\nu_1 = \nu + \Delta\nu/2$ and $\nu_2 = \nu - \Delta\nu/2$. The corresponding angular frequencies become $\omega_1 = \omega + \pi\Delta\nu$ and $\omega_2 = \omega - \pi\Delta\nu$. Evaluating the scalar electric fields near the focal region (dropping wavefront curvature and Gouy phase), the two beams are expressed as real functions:

$$E_1(z,t) = E_0(z,\rho)\cos(kz - \omega t - \pi\Delta\nu t), \tag{1.35}$$

$$E_2(z,t) = E_0(z,\rho)\cos(-kz - \omega t + \pi\Delta\nu t) = E_0(z,\rho)\cos(kz + \omega t - \pi\Delta\nu t). \tag{1.36}$$

Superposing the two fields, $E_{tot} = E_1 + E_2$, and mapping the arguments to $A = kz - \omega t - \pi\Delta\nu t$ and $B = kz + \omega t - \pi\Delta\nu t$, the sum-to-product identity yields

$$E_{tot}(z,t) = 2E_0(z,\rho)\cos\left(kz - \pi\Delta\nu t\right)\cos(-\omega t). \tag{1.37}$$

Because the atomic motion does not follow the rapidly oscillating optical carrier, the interaction can be averaged over one optical cycle, reducing the $\cos^2(-\omega t)$ term to a constant factor of $1/2$. The effective intensity, and by extension the conservative trapping potential $U(z,\rho,t)$, is governed by the slowly varying spatial envelope. The static potential thus evolves into a moving periodic lattice:

$$U(z,\rho,t) = U_0\frac{w_0^2}{w^2(z)}\exp\left(-\frac{2\rho^2}{w^2(z)}\right)\cos^2(\pi\Delta\nu t - kz). \tag{1.38}$$

Setting $\Delta\nu = 0$ recovers the stationary lattice structure.

Single-beam detuning. In laboratory implementations it is generally simpler to introduce the frequency offset into a single arm. Defining the unshifted frequency as $\nu_1 = \nu$ and the detuned frequency as $\nu_2 = \nu + \Delta\nu$ (yielding angular frequencies $\omega_1 = \omega$ and $\omega_2 = \omega + 2\pi\Delta\nu$), the interacting fields are

$$E_1(z,t) = E_0(z,\rho)\cos(kz - \omega t), \tag{1.39}$$

$$E_2(z,t) = E_0(z,\rho)\cos(-kz - (\omega + 2\pi\Delta\nu)t) = E_0(z,\rho)\cos(kz + \omega t + 2\pi\Delta\nu t). \tag{1.40}$$

Applying the sum-to-product identity with $A = kz - \omega t$ and $B = kz + \omega t + 2\pi\Delta\nu t$ factors the superposition:

$$E_{tot}(z,t) = 2E_0(z,\rho)\cos\left(kz + \pi\Delta\nu t\right)\cos\left(\omega t + \pi\Delta\nu t\right). \tag{1.41}$$

Upon time-averaging the fast optical component $\cos^2(\omega t + \pi\Delta\nu t) \to 1/2$, the resulting effective potential is

$$U(z,\rho,t) = U_0\frac{w_0^2}{w^2(z)}\exp\left(-\frac{2\rho^2}{w^2(z)}\right)\cos^2(kz + \pi\Delta\nu t). \tag{1.42}$$

This expression coincides, up to an overall sign in the time argument of the envelope, with Eq. (1.38), confirming that an asymmetric detuning imposed on a single beam produces the same effect as a symmetrically distributed detuning.

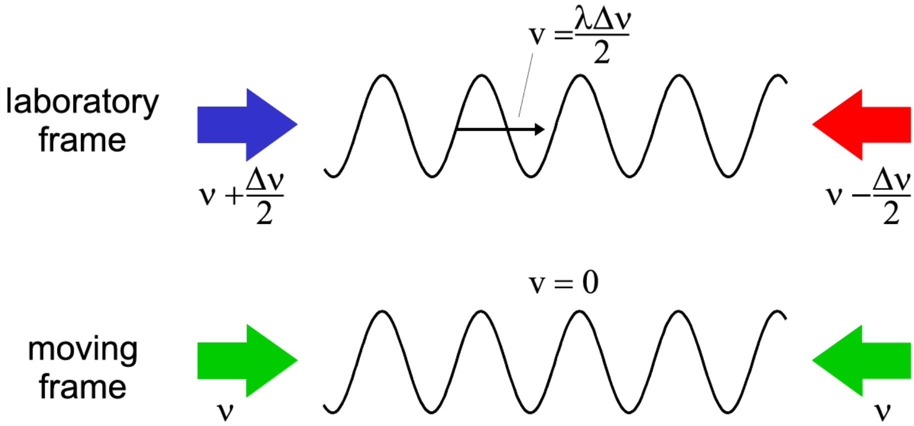


Figure 1.10: *Kinematics of the optical conveyor belt: a mutual detuning $\Delta\nu$ produces a moving standing wave. In the moving reference frame both beams are doppler shifted by the same amount [10].*

The macroscopic velocity of the interference pattern can be understood by considering a moving reference frame. In a coordinate system translating along the z-axis with velocity v, the Doppler effect causes a red shift of the co-propagating beam and a blue shift of the counter-propagating beam. There exists a specific reference frame in which these Doppler shifts exactly compensate for the frequency difference $\Delta\nu$ imposed in the laboratory frame. In this system, the two beams have identical frequencies, and the standing wave appears stationary [10].

The velocity can also be derived directly in the laboratory frame by tracking a point of constant intensity, i.e. by setting the phase argument of the moving envelope to a constant value, $kz \pm \pi\Delta\nu t = \text{const}$. Taking the time derivative yields $k\, dz/dt \pm \pi\Delta\nu = 0$. Substituting $k = 2\pi/\lambda$ and identifying the transport velocity as $v = dz/dt$ gives

$$v = \frac{\lambda\Delta\nu}{2}. \tag{1.43}$$

1.3.4 Trap-Depth Scaling and Adiabatic Compression

The dimensional properties of the trap scale dynamically when optical parameters are varied, for instance during adiabatic compression in which the beam waist is tightened towards w_0. Under constant optical power P, the trap depth obeys $U_0 \propto I_{\max} \propto P/w_0^2$. Combining this scaling with the results of Eq. (1.33) gives the behavior of the trapping frequencies:

$$\Omega_z \propto \sqrt{U_0} \propto \frac{1}{w_0}, \qquad \Omega_{rad} \propto \frac{\sqrt{U_0}}{w_0} \propto \frac{1}{w_0^2}. \tag{1.44}$$

The quadratic dependence of Ω_{rad}, compared to the linear dependence of Ω_z, induces a deformation during focusing: the anisotropy ratio Ω_z/Ω_{rad} scales directly with w_0, so the radial confinement tightens faster than the axial confinement and the aspect ratio of the disc-shaped wells changes during compression.

Adiabatic temperature scaling. Provided that the variation of the trap depth remains slow compared to the inverse characteristic frequencies, the action $J \propto E/\Omega$ is an adiabatic invariant of the harmonic motion [14]. Modeling the mean thermal energy via the equipartition theorem ($E \propto k_B T$), this invariance implies that along each independent degree of freedom

$$\frac{T}{\Omega} = \text{const.} \tag{1.45}$$

Because Ω_z and Ω_{rad} scale at different rates with w_0, an initially isotropic thermal distribution evolves into an anisotropic one: adiabatic compression increases the transverse temperature relative to the axial temperature.

Lamb–Dicke parameter. The axial behavior of the potential also affects the applicability of advanced laser-cooling mechanisms. The ground-state spatial width of the atomic wavepacket along the z-axis is

$$x_0 = \sqrt{\frac{\hbar}{2m\Omega_z}}, \tag{1.46}$$

and the parameter describing the coupling of this motional state to a resonant cooling field of wavenumber $k_c = 2\pi/\lambda_c$ is the Lamb–Dicke parameter

$$\eta = k_c\, x_0 = k_c\sqrt{\frac{\hbar}{2m\Omega_z}}. \tag{1.47}$$

The Lamb–Dicke regime provides the basis for the EIT-cooling treatment developed in Section 1.4.

1.3.5 Dynamics in the Accelerated Potential

Transporting a trapped atom between two spatial positions requires a profile of continuous acceleration followed by symmetric deceleration. This is achieved by applying a linear temporal ramp to the beat frequency $\Delta\nu(t)$.

Transforming the Hamiltonian into the non-inertial frame of the accelerating lattice, an atom subject to an acceleration a experiences a fictitious force $-ma$. The addition of this

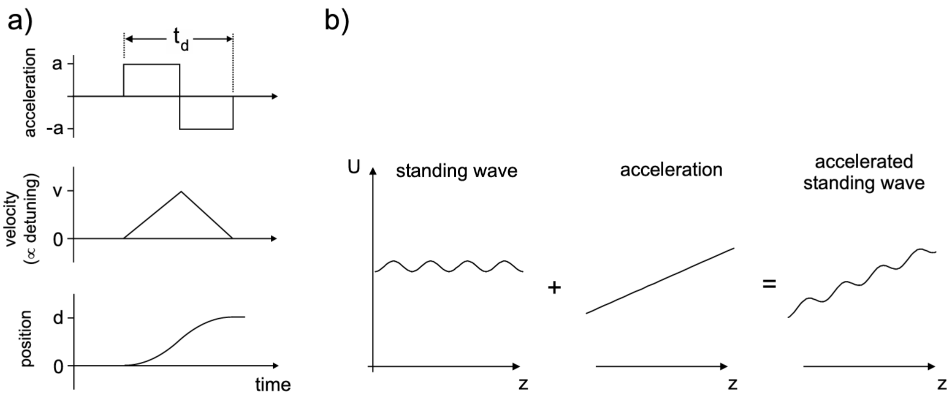


Figure 1.11: *In the frame co-moving with the accelerated lattice, the periodic potential is superimposed with a linear gradient, resulting in a tilted washboard landscape [10].*

linear term along the optical axis modifies the effective potential:

$$U_{\text{eff}}(z') = U_0 \cos^2(kz') + maz', \tag{1.48}$$

where z' is the coordinate in the co-moving frame and $U_0 < 0$ denotes a red-detuned trap. The linear term breaks the symmetry of the periodic profile, lowering the potential barrier in the direction of the fictitious force. Stationary minima of U_{eff} exist provided the equation $dU_{\text{eff}}/dz' = 0$ admits real solutions corresponding to local minima. Differentiating yields

$$\frac{dU_{\text{eff}}(z')}{dz'} = -kU_0 \sin(2kz') + ma = 0 \quad \Longrightarrow \quad ma = kU_0 \sin(2kz'). \tag{1.49}$$

Since $|\sin(2kz')| \leq 1$, real solutions require $|ma| \leq |kU_0|$. As $|a|$ approaches this limit, the local minimum and the adjacent maximum merge into a single inflection point, causing the trapping barrier to vanish completely and the atom to escape. The threshold acceleration for a particle with zero kinetic energy is therefore:

$$a_0 = \frac{k|U_0|}{m}. \tag{1.50}$$

This theoretical bound assumes a particle at absolute zero temperature. For a thermal cloud, the effective potential barrier $\Delta U = U_{\max} - U_{\min}$ decreases with acceleration, causing atoms with non-zero kinetic energy to escape at values significantly below a_0. Furthermore, the local trap frequency approaches zero ($\Omega \to 0$) as $a \to a_0$. These factors impose stringent constraints on the acceleration profiles in optical conveyor belt operations to minimize thermal loss and maintain the adiabatic confinement of the atomic wavepacket.

1.4 Electromagnetically Induced Transparency Cooling

1.4.1 Introduction to EIT Cooling

The preparation of trapped atoms in their motional ground state is a fundamental prerequisite for high-fidelity quantum information processing and the realization of strongly coupled atom-photon systems. Resolved-sideband cooling is effective for this task, but it imposes the restriction of the strong-confinement condition: the natural linewidth of the atomic transition γ must be much smaller than the harmonic trap frequency ν ($\gamma \ll \nu$) [15]. This requirement is difficult to fulfill in shallow or low-frequency traps, where the sidebands are not spectrally resolved. Electromagnetically Induced Transparency (EIT) cooling overcomes this limitation by exploiting quantum interference in a multi-level atomic structure [15, 16]. It achieves ground-state cooling without requiring strong confinement; the relevance of this feature becomes clear when the cooling stage is integrated into a sequence that involves transport and tight spatial confinement.

In the experimental protocol considered in this work, atoms are first cooled in a magneto-optical trap, transferred to an optical dipole trap, and then transported into the hollow core of a photonic crystal fiber by means of an optical conveyor belt. The transport itself is a source of heating, the transverse compression required to match the fiber mode increases the trap frequencies and raises the transverse temperature. By the time the atoms reach the fiber core, their temperature is significantly higher, and re-cooling is required before any precision experiment can be performed. In this section, all frequencies and detunings are expressed as angular frequencies in units of rad/s, and the corresponding linear frequencies in Hz are obtained by dividing by 2π.

1.4.2 The Three-Level Λ System and Coherent Population Trapping

The theoretical framework of EIT cooling is based on a three-level atomic system in a Λ configuration (Fig. 1.12). This consists of two stable or metastable ground states, $|g_1\rangle$ and $|g_2\rangle$, both optically coupled to a common, short-lived excited state $|e\rangle$ with spontaneous decay rate γ [15, 16].

The transition $|g_1\rangle \leftrightarrow |e\rangle$ is driven by a weak probe laser with Rabi frequency Ω_p, satisfying $\Omega_p \ll \Omega_c$, while the transition $|g_2\rangle \leftrightarrow |e\rangle$ is driven by a strong coupling laser with Rabi frequency Ω_c. Under the rotating wave approximation, and assuming the two lasers are tuned to a two-photon resonance with a common detuning Δ, the internal Hamiltonian

H_0 and the interaction potential V_0 in the rotating frame are [16]:

$$H_0 = -\hbar\Delta\left(|g_1\rangle\langle g_1| + |g_2\rangle\langle g_2|\right) \tag{1.51}$$

$$V_0 = \frac{\hbar}{2}\left(\Omega_p|e\rangle\langle g_1| + \Omega_c|e\rangle\langle g_2| + \text{h.c.}\right) \tag{1.52}$$

The incoherent spontaneous emission from the excited state is described by the Liouvillian operator $\mathcal{K}$ [16]:

$$\mathcal{K}\rho = -\frac{\gamma}{2}\left(|e\rangle\langle e|\rho + \rho|e\rangle\langle e|\right) + \sum_{j=1,2}\gamma_j|g_j\rangle\langle e|\rho|e\rangle\langle g_j| \tag{1.53}$$

where γ_1 and γ_2 are the partial decay rates into $|g_1\rangle$ and $|g_2\rangle$, with $\gamma_1 + \gamma_2 = \gamma$ [16]. Physically, $\mathcal{K}$ encodes the fact that population in $|e\rangle$ decays at total rate γ, redistributed between $|g_1\rangle$ and $|g_2\rangle$ according to the branching ratios γ_1/γ and γ_2/γ.

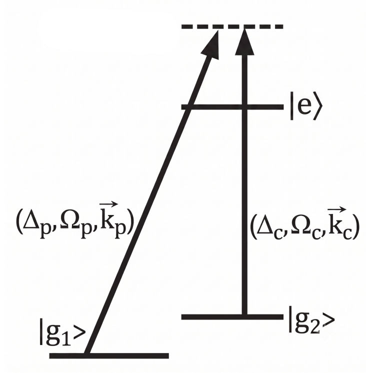


Figure 1.12: *The three-level Λ system used for EIT cooling, consisting of two ground states ($|g_1\rangle$, $|g_2\rangle$) coupled to a common excited state ($|e\rangle$) by a probe and a coupling laser.*

The essential physical mechanism underlying EIT is the emergence of a dark state: a coherent superposition of the two ground states that is decoupled from the excited state by destructive quantum interference between the two optical excitation pathways. At two-photon resonance, this dark state takes the form [16]:

$$|\Psi_D\rangle = \frac{1}{\Omega}\left(\Omega_c|g_1\rangle - \Omega_p|g_2\rangle\right) \tag{1.54}$$

where $\Omega = \sqrt{\Omega_p^2 + \Omega_c^2}$ is the total effective Rabi frequency [16]. An atom optically pumped into $|\Psi_D\rangle$ ceases to absorb light, a phenomenon known as Coherent Population Trapping (CPT) [17].

The decoupling of $|\Psi_D\rangle$ from $|e\rangle$ is verified by direct evaluation of the coupling matrix element $\langle e|V_0|\Psi_D\rangle$. Using Eq. (1.52) and the coefficients of $|\Psi_D\rangle$ in Eq. (1.54), only the $\Omega_p|e\rangle\langle g_1|$ and $\Omega_c|e\rangle\langle g_2|$ terms of V_0 contribute, giving

$$\langle e|V_0|\Psi_D\rangle = \frac{\hbar}{2\Omega}\Big(\Omega_p\Omega_c - \Omega_c\Omega_p\Big) = 0. \tag{1.55}$$

The excitation amplitude contributed by the $|g_1\rangle \to |e\rangle$ pathway, proportional to Ω_p times the Ω_c/Ω weight of $|g_1\rangle$ in $|\Psi_D\rangle$, is cancelled by the amplitude contributed by the $|g_2\rangle \to |e\rangle$ pathway, proportional to Ω_c times the $-\Omega_p/\Omega$ weight of $|g_2\rangle$.

1.4.3 Dressed States and the Fano Absorption Spectrum

The cooling mechanism is analyzed in the dressed-state picture [16]. When the coupling laser interacts with the atom, the unperturbed atomic states are replaced by hybridized energy levels, the eigenstates of the coupled atom–laser Hamiltonian.

In the driven Λ system, the dark state $|\Psi_D\rangle$ is an eigenstate of the full internal Hamiltonian $H_0 + V_0$. The three-level system thus reduces to an effective two-level system spanned by $|e\rangle$ and the orthogonal bright ground state

$$|\psi_C\rangle = \frac{1}{\Omega}\left(\Omega_p|g_1\rangle + \Omega_c|g_2\rangle\right). \tag{1.56}$$

The state $|\psi_C\rangle$ is termed *bright* because, unlike $|\Psi_D\rangle$, it couples to $|e\rangle$ via V_0. Restricting $H_0 + V_0$ to this two-dimensional subspace yields the effective interaction matrix

$$H_{\text{2-level}} = \hbar \begin{pmatrix} 0 & \Omega/2 \\ \Omega/2 & -\Delta \end{pmatrix} \quad \text{in the basis } \{|e\rangle,\, |\psi_C\rangle\}. \tag{1.57}$$

Diagonalizing Eq. (1.57) gives the dressed energies

$$E_\pm = -\frac{\hbar\Delta}{2} \pm \frac{\hbar}{2}\sqrt{\Delta^2 + \Omega^2}, \tag{1.58}$$

with the corresponding orthogonal eigenvectors

$$|\psi_+\rangle = \cos\theta\,|e\rangle + \sin\theta\,|\psi_C\rangle, \qquad |\psi_-\rangle = \sin\theta\,|e\rangle - \cos\theta\,|\psi_C\rangle, \tag{1.59}$$

where the mixing angle θ satisfies $\tan\theta = (\sqrt{\Delta^2 + \Omega^2} - \Delta)/\Omega$ [16].

The dressed-state structure is probed via a pump–probe scheme: the coupling laser dresses the atomic states, while the weak cooling laser probes the resulting spectrum. In EIT cooling, the coupling laser operates far from single-photon resonance, $|\Delta| \gg \Omega$. In this limit the mixing angle θ is small, and the dressed states separate into a predominantly excited-like state and a predominantly ground-like state. Since spontaneous emission occurs from the $|e\rangle$ component, the linewidth of each branch is determined by its excited-state admixture. The state $|\psi_+\rangle$ is excited-state-like and retains approximately the bare atomic linewidth ($\gamma_+ \approx \gamma$). The state $|\psi_-\rangle$ is ground-state-like, with a suppressed linewidth $\gamma_- \approx \gamma\,\Omega^2/(4\Delta^2) \ll \gamma$.

As the probe laser scans across the system, the excitation amplitudes to the broad background ($|\psi_+\rangle$) and to the narrow resonance ($|\psi_-\rangle$) interfere. This interference generates an asymmetric Fano-like absorption profile. For a blue-detuned coupling laser ($\Delta > 0$), the spectrum exhibits three features: a zero-absorption point at the two-photon resonance,

corresponding to the dark state $|\Psi_D\rangle$; a broad absorption resonance centered near zero probe detuning, and a narrow absorption resonance.
The coupling laser induces an AC Stark shift δ that displaces the narrow resonance from the dark-state zero. The exact expression is

$$\delta = \frac{\sqrt{\Delta^2 + \Omega_c^2} - |\Delta|}{2}, \tag{1.60}$$

which in the limit $\Delta \gg \Omega_c$ reduces to $\delta \approx \Omega_c^2/(4\Delta)$. The experimental control of this AC Stark shift is the foundation of the EIT cooling protocol.

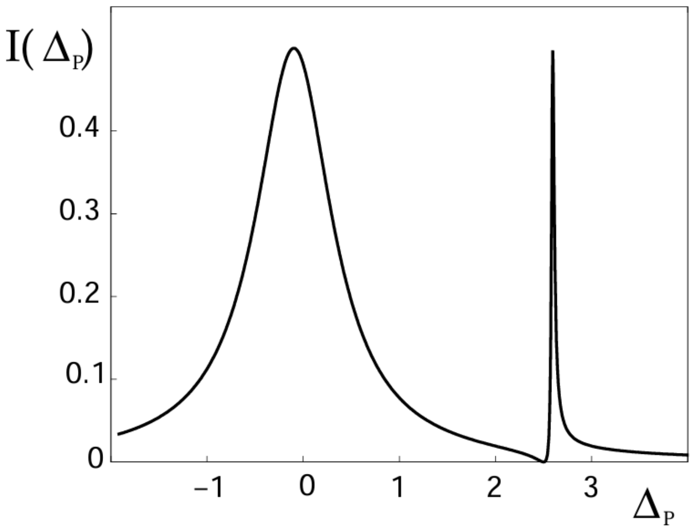


Figure 1.13: *Absorption spectrum of the weak probe laser as a function of its detuning, in the presence of the strong coupling laser. The spectrum exhibits the characteristic asymmetric Fano lineshape of EIT: a zero-absorption point at the two-photon (dark-state) resonance, a broad resonance of width $\gamma_+ \approx \gamma$ associated with $|\psi_+\rangle$, and a narrow, AC-Stark-shifted resonance of width $\gamma_- \ll \gamma$ associated with $|\psi_-\rangle$. Credits: [15]*

1.4.4 Coupling to the Motional Degrees of Freedom

For an atom confined in a harmonic trapping potential of frequency ν, the external center-of-mass motion is quantized into vibrational Fock states $|n\rangle$, with mechanical Hamiltonian $H_{\text{mec}} = \hbar\nu(a^\dagger a + 1/2)$. EIT cooling maps the engineered asymmetric Fano spectrum onto these motional sidebands [15].
The dynamics are evaluated in the Lamb–Dicke regime, where the spatial extent of the atomic wavepacket is much smaller than the optical wavelength (characterized by $\eta \ll 1$, see 1.47). In this two-photon process, the relevant Lamb-Dicke parameter is defined by the effective wavevector $\Delta\vec{k} = \vec{k}_p - \vec{k}_c$, such that $\eta = |\Delta\vec{k}|x_0$. A crucial geometrical requirement for EIT cooling is therefore $\Delta\vec{k} \neq 0$: if the lasers are copropagating, there is no net momentum transfer on the atom. For this reason, the beams must be aligned at

an angle to guarantee a non-zero projection of $\Delta\vec{k}$ along the cooling axis. In this limit, the ideal cooling conditions are achieved by tuning the AC Stark shift to match the trap frequency [15]:

$$\delta \simeq \nu. \tag{1.61}$$

Substituting the approximate expression $\delta \approx \Omega_c^2/(4\Delta)$ yields $\Omega_c^2 \approx 4\Delta\nu$. The exact condition, obtained from Eq. (1.60) without approximation, is

$$\Omega_c^2 = 4\nu(\nu + \Delta), \tag{1.62}$$

which reduces to the approximate form in the limit $\nu \ll \Delta$ typical of EIT cooling. Satisfying this resonance condition produces two simultaneous effects (also shown in Fig. 1.14):

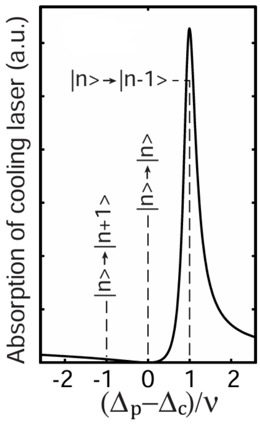


Figure 1.14: *Absorption of the cooling laser as a function of the two-photon detuning, in units of the trap frequency ν. The Fano peak, engineered to sit at $+1$, drives the red-sideband transition $|n\rangle \rightarrow |n-1\rangle$ responsible for cooling, while the carrier ($|n\rangle \rightarrow |n\rangle$) and blue-sideband ($|n\rangle \rightarrow |n+1\rangle$) transitions, marked at 0 and -1, experience comparatively little absorption. Credits: [15]*

- **Suppression of heating (carrier cancellation).** The zero of the Fano profile is aligned with the carrier transition ($|g_1, n\rangle \rightarrow |e, n\rangle$) [15]. An atom in the dark state therefore cannot scatter photons unless it changes its motional state. This suppresses the dominant source of heating that affects standard cooling schemes.

- **Enhancement of cooling (red-sideband excitation).** The narrow dressed-state resonance $|\psi_-\rangle$ is shifted onto the red-sideband transition ($|g_1, n\rangle \rightarrow |e, n-1\rangle$) [15], enhancing the absorption cross-section for the removal of a vibrational quantum. Conversely, the blue sideband ($|g_1, n\rangle \rightarrow |e, n+1\rangle$), which would add a phonon, falls in a spectral region of minimal excitation probability.

After resonant excitation on the red sideband, $|g_1, n\rangle \rightarrow |e, n-1\rangle$, the atom decays at rate γ. For a net phonon to be removed, the decay must predominantly return the atom to $|g_i, n-1\rangle$; that is, it must occur on the carrier transition of the excited manifold, $|e, n-1\rangle \rightarrow |g_i, n-1\rangle$, leaving the vibrational quantum number unchanged, rather than to $|g_i, n\rangle$, which would restore the phonon just removed. This is guaranteed in the Lamb–Dicke regime: the recoil imparted by the spontaneously emitted photon changes n by an amount controlled by η, so that sideband emission ($\Delta n = \pm 1$) is suppressed relative to carrier emission ($\Delta n = 0$) by a factor of order η^2. The cooling cycle thus closes: absorption removes one phonon on the red sideband, the subsequent decay leaves the vibrational state untouched, and the net effect of many such cycles is the removal of motional quanta. The full scheme of EIT cooling, combining the level structure with the corresponding absorption spectrum and the subsequent decay dynamics, is shown in Fig. 1.15.

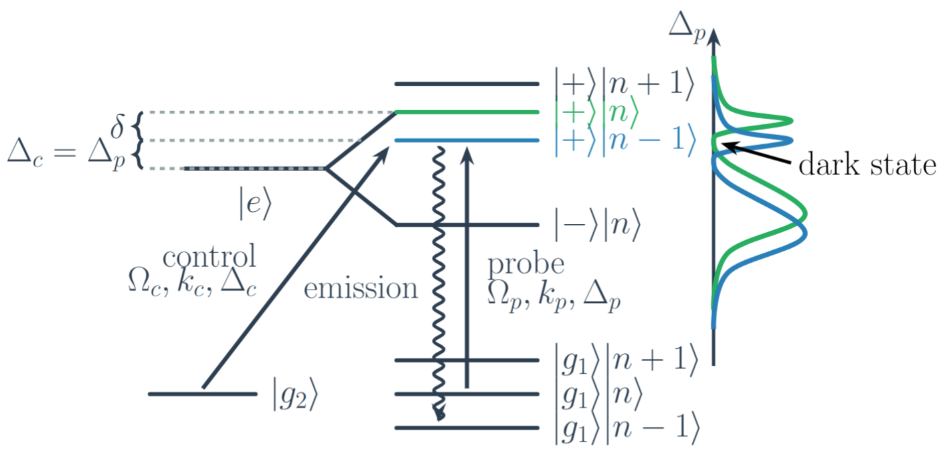


Figure 1.15: *Full picture of the EIT cooling scheme, combining the level structure with the corresponding absorption spectrum. The carrier transition (green, $|g_1, n\rangle \rightarrow |e, n\rangle$) and the red-sideband transition (blue, $|g_1, n\rangle \rightarrow |e, n-1\rangle$) absorption profiles feature a dark-state zero and a narrow, AC-Stark-shifted Fano resonance. The efficiency of the scheme relies on the tuning $\delta \simeq \nu$ (Eq. (1.61)) so that the Fano resonance peak of the red sideband (blue) overlaps with the zero-absorption point of the carrier (green). Consequently, cooling is resonantly enhanced exactly where carrier heating is completely suppressed.*

1.4.5 Cooling Dynamics and the Theoretical Limit

Provided that the cooling laser is weak ($\Omega_p \ll \Omega_c$) and the narrow resonance is not saturated, the motional dynamics can be described by coarse-graining the evolution over the internal-state relaxation time. The populations $P(n)$ of the vibrational states evolve according to a classical rate equation [15, 16]:

$$\frac{d}{dt}P(n) = \eta^2 \left[A_- \left((n+1)P(n+1) - nP(n)\right) + A_+ \left(nP(n-1) - (n+1)P(n)\right)\right] \quad (1.63)$$

where A_- and A_+ are the effective excitation rates of the red and blue sidebands, respectively. Incorporating the full quantum interference around the dark resonance, these rates take the explicit form

$$A_{\pm} = \frac{1}{4}\left(\frac{\Omega_p\Omega_c}{\Omega}\right)^2 \frac{\gamma\nu^2}{[\Omega_c^2/4 - \nu(\nu \mp \Delta)]^2 + \gamma^2\nu^2/4}. \tag{1.64}$$

The evolution of the mean phonon number $\langle n\rangle = \sum_n nP(n)$ follows [15, 16]:

$$\frac{d\langle n\rangle}{dt} = -\eta^2(A_- - A_+)\langle n\rangle + \eta^2 A_+ = -W\langle n\rangle + \eta^2 A_+, \tag{1.65}$$

where $W = \eta^2(A_- - A_+)$ is the net cooling rate. Under the optimal condition $\delta \simeq \nu$, the maximum cooling rate scales as

$$W_{\max} \sim \eta^2 \frac{\Omega_p^2\Omega_c^2}{\Omega^2\gamma}, \tag{1.66}$$

which confirms that fast cooling is achieved for large Rabi frequencies [16].

The steady-state mean vibrational quantum number $\langle n_S\rangle$ is given by the ratio of the heating rate to the net cooling rate:

$$\langle n_S\rangle = \frac{A_+}{A_- - A_+} = \frac{\gamma^2\nu^2 + 4[\Omega_c^2/4 - \nu(\nu+\Delta)]^2}{4\Delta\nu|\Omega_c^2 - 4\nu^2|}. \tag{1.67}$$

The absolute minimum is achieved when the AC Stark shift matches the trap frequency, i.e. when Eq. (1.62) is satisfied, yielding the fundamental theoretical cooling limit [15, 16]:

$$\langle n_S\rangle_{\min} = \left(\frac{\gamma}{4\Delta}\right)^2. \tag{1.68}$$

By choosing a sufficiently large coupling-laser detuning $\Delta \gg \gamma$, $\langle n_S\rangle$ can in principle be made arbitrarily small. However maintaining the resonance condition $\delta \simeq \nu$ at larger Δ requires $\Omega_c^2 \propto \Delta\nu$ to grow accordingly, so faster cooling and lower final occupations demand correspondingly higher coupling-laser intensities.

1.4.6 Theoretical Considerations for in-fiber EIT Cooling

The implementation of EIT cooling inside a tightly focused optical dipole trap, such as the one used to load atoms into a hollow-core photonic crystal fiber (HCPCF), requires revisiting the assumptions of the free-space case in the presence of an intense, spatially varying trapping field. The dominant perturbation is the differential AC Stark shift between the ground and excited states of the cooling cycle, induced by the trapping laser itself.

Differential AC Stark shift. For a far-red-detuned trapping field of angular frequency ω_t and local intensity $I(\mathbf{r})$, the scalar light shift of an atomic level $|i\rangle$ of static polarizability $\alpha_i(\omega_t)$ is

$$U_i(\mathbf{r}) = -\frac{\alpha_i(\omega_t)}{2\varepsilon_0 c}\, I(\mathbf{r}), \tag{1.69}$$

so that the single-photon resonance of the $|g_1\rangle \leftrightarrow |e\rangle$ transition is displaced by the differential shift

$$\Delta_{\mathrm{AC}}(\mathbf{r}) = \frac{U_g(\mathbf{r}) - U_e(\mathbf{r})}{\hbar} = -\frac{\alpha_g(\omega_t) - \alpha_e(\omega_t)}{2\varepsilon_0 c\,\hbar}\, I(\mathbf{r}). \tag{1.70}$$

At the trapping wavelength used for in-fiber loading [2], the excited state of the cooling cycle has polarizability of opposite sign to the ground manifold, so $\alpha_g - \alpha_e$ is large and $|\Delta_{\mathrm{AC}}(0)|$ typically exceeds the bare linewidth. The cooling and probe lasers must therefore be retuned by $\Delta_{\mathrm{AC}}(0)$ to maintain the cooling condition.

Common-mode protection of the dark state. The two hyperfine ground states $|g_1\rangle, |g_2\rangle$ entering the Λ system share essentially the same scalar polarizability, so their light shifts are equal and the relative ground-state splitting is preserved. The dark state $|\Psi_D\rangle$ of Eq. (1.54) is built only from $|g_1\rangle, |g_2\rangle$; its absolute energy is shifted by the common-mode U_g, but its composition and the two-photon resonance condition are unchanged. The Fano transparency zero is thus structurally protected against the trapping field.

The red-sideband transition $|g_1, n\rangle \to |e, n-1\rangle$ is a single-photon transition, sensitive to the absolute shift of $|e\rangle$. Its resonance frequency acquires the displacement $\Delta_{\mathrm{AC}}(0)$, so the optimal cooling condition of Eq. (1.61) generalizes to

$$\delta \;\simeq\; \nu - \Delta_{\mathrm{AC}}(0), \tag{1.71}$$

i.e., the coupling-laser intensity must be tuned so that the AC Stark shift $\delta \simeq \Omega_c^2/(4\Delta)$ of the narrow dressed-state resonance compensates both the trap frequency ν *and* the trapping-induced shift $\Delta_{\mathrm{AC}}(0)$.

Inhomogeneous broadening. The previous treatment gives the correct tuning for an atom strictly on the trap axis. Real atoms, however, oscillate transversely in the dipole potential and sample a range of local intensities. Since the differential AC Stark shift Δ_{AC} is proportional to the local trapping intensity, this thermal motion translates into a spread of the single-photon resonance frequencies sampled by the atomic ensemble.

To quantify this effect, we consider the transverse intensity profile of the fundamental Gaussian mode, which is given by $I(r) = I_0 e^{-2r^2/w_0^2}$, where w_0 is the beam waist. The confining optical dipole potential is directly proportional to this intensity, and can be

[2] For the particular excited state and trapping wavelength considered here and explained in the next chapter, the polarizability has the opposite sign to that of the ground manifold.

written as $U(r) = -U_0 e^{-2r^2/w_0^2}$, with U_0 being the maximum trap depth at the axis ($r = 0$). For cold atoms confined near the trap axis ($r \ll w_0$), we can expand the exponential to first order, $e^{-2r^2/w_0^2} \approx 1 - 2r^2/w_0^2$, yielding the harmonic approximation of the potential:

$$U(r) \approx -U_0 \left(1 - \frac{2r^2}{w_0^2}\right) = -U_0 + \frac{2U_0 r^2}{w_0^2}. \tag{1.72}$$

The potential energy relative to the trap minimum is thus $\Delta U(r) = 2U_0 r^2/w_0^2$. The radial extent of the atomic motion is bounded by the available thermal energy. By equating the local harmonic potential energy to the characteristic one-dimensional thermal energy $k_B T$, i.e.,

$$\frac{2U_0 r_{\text{th}}^2}{w_0^2} = k_B T, \tag{1.73}$$

we can solve for r_{th} to find the effective thermal radius of the atomic cloud:

$$r_{\text{th}} = \frac{w_0}{\sqrt{2}} \sqrt{\frac{k_B T}{U_0}}. \tag{1.74}$$

Beyond this radius, the kinetic energy is generally insufficient to climb the potential barrier. As expected, hotter atoms or shallower traps (larger $k_B T/U_0$) push r_{th} outward into the wings of the Gaussian profile, while colder atoms in deep traps stay confined near the axis where the intensity is nearly uniform.

Because the differential light shift follows the same Gaussian profile as the potential, $\Delta_{\text{AC}}(r) = \Delta_{\text{AC}}(0)e^{-2r^2/w_0^2}$, its spatial variation near the axis can similarly be expanded as:

$$\Delta_{\text{AC}}(r) \approx \Delta_{\text{AC}}(0) \left(1 - \frac{2r^2}{w_0^2}\right). \tag{1.75}$$

The effective spread of the single-photon resonance sampled by the thermal cloud, which we denote as δ_{in}, is given by the difference between the shift on-axis and the shift evaluated at the thermal radius:

$$\delta_{\text{in}} \approx |\Delta_{\text{AC}}(0)| - |\Delta_{\text{AC}}(r_{\text{th}})| = |\Delta_{\text{AC}}(0)| \frac{2r_{\text{th}}^2}{w_0^2}. \tag{1.76}$$

Substituting the expression for r_{th}^2 derived above into this relation, we obtain:

$$\delta_{\text{in}} \approx |\Delta_{\text{AC}}(0)| \frac{2}{w_0^2} \left(\frac{w_0^2 k_B T}{2U_0}\right) = |\Delta_{\text{AC}}(0)| \frac{k_B T}{U_0}. \tag{1.77}$$

When δ_{in} becomes comparable to the trap frequency ν, the narrow red-sideband absorption line is smeared out over a spectral range that overlaps with the carrier transition. This inhomogeneous broadening increases the steady-state occupation with respect to the ideal free-space limit. A quantitative estimate requires averaging the cooling and heating rates over the thermal spatial distribution, which is beyond the scope of this analysis.

Chapter 2

Materials and Methods

This chapter describes the experimental apparatus used in this work. First, a brief overview of the implementation of the Magneto-Optical Trap (MOT) for rubidium-87 atoms (^{87}Rb) is presented (this setup was already available prior to the present work). The chapter then focuses on the work carried out during this thesis, namely the implementation of the Optical Dipole Trap setup employed to transfer and trap atoms from the MOT. A particular focus is put on the coupling between the ODT laser source and the Hollow-Core Photonic Crystal Fiber (HCPCF), that delivers the ODT light in the vacuum chamber.

2.1 Experimental Setup

This section describes the apparatus used to produce, cool, and trap a sample of ^{87}Rb atoms. The atomic sample is prepared via a standard Magneto-Optical Trap (MOT) followed by an optical molasses phase, providing the cold atomic cloud required to load the Optical Dipole Trap (ODT). Spatial confinement and initial cooling rely on three pairs of counter-propagating, orthogonally polarized (σ^+/σ^-), red-detuned laser beams intersecting a quadrupole magnetic field. Following the MOT stage, the magnetic gradient is switched off to perform sub-Doppler cooling in an optical molasses (see 1.2).

2.1.1 Setup Description

The core of the apparatus is an Ultra-High Vacuum (UHV) chamber (Fig. 2.1) maintained at a baseline pressure of $\sim 10^{-10}$ mbar by an ion/getter combination pump (NEXTorr Z200). A heated ampoule of pure ^{87}Rb supplies the atomic vapor, raising the pressure to $\sim 10^{-8}$ mbar during trap loading.
Optical access is provided by eight windows: two large CF100 viewports along the z-axis and four CF40 viewports at 45°, all AR-coated for 780 nm to optimize MOT beam

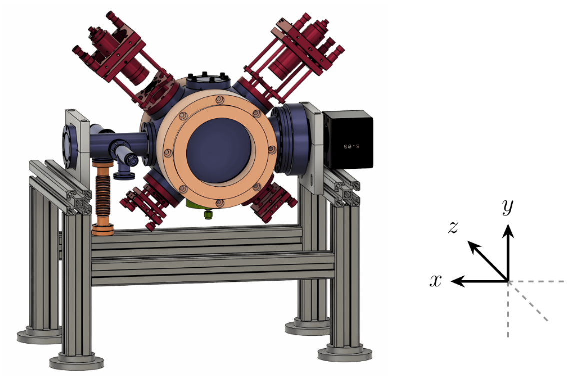


Figure 2.1: *CAD rendering of the vacuum chamber and surrounding experimental setup.*

transmission. Two uncoated CF40 viewports provide access for the second ODT beam and the ion pump. Due to reflections from the ion pump on the x-axis, imaging is performed through the z-axis viewports.

The MOT quadrupole magnetic field is generated by anti-Helmholtz coils (110 mm inner diameter) producing a measured axial gradient of $1.33\,\mathrm{G\,cm^{-1}A^{-1}}$ (and a corresponding radial gradient of half this value with opposite sign) [18]. Three mutually orthogonal pairs of Helmholtz shim coils cancel stray fields and are used to optimize the MOT-ODT spatial overlap, with calibration factors [13]:

$$B_z \approx 1.4\text{ mG/mA}, \qquad B_x \approx 1.3\text{ mG/mA}, \qquad B_y \approx 1.5\text{ mG/mA}.$$

A Hollow-Core Photonic Crystal Fiber (HCPCF) is introduced from the bottom of the chamber via a custom mount, positioning its tip $\sim 10\,\mathrm{mm}$ below the trap center. The optical layout is fiber-coupled using polarization-maintaining (PANDA) fibers [10]. The MOT beams are delivered via three fiber collimators and retro-reflected with quarter-wave plates to maintain the correct polarization sequence.

2.1.2 Laser System and Frequency Stabilization

Cooling operates on the ^{87}Rb $5^2S_{1/2} \to 5^2P_{3/2}$ (D2) transition near 780 nm (Fig. 2.2). The cooling laser drives the closed $F = 2 \to F' = 3$ transition, while a repumper laser on the $F = 1 \to F' = 2$ transition recovers atoms that off-resonantly decay into the $F = 1$ dark state. Both lasers are offset-locked to a common master laser [19, 20].

2.1.2.1 Master laser

The master reference is a 780 nm DFB laser diode stabilized via frequency-modulated saturation spectroscopy in a heated ^{85}Rb vapor cell (Fig. **??**). It is locked to the ^{85}Rb $(F = 3 \to F' = 3) \times (F = 3 \to F' = 4)$ crossover resonance, providing a stable absolute

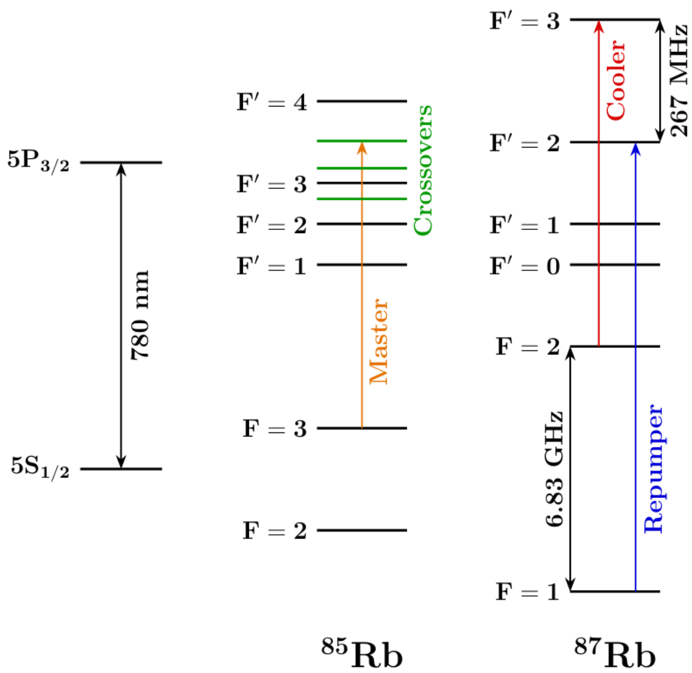


Figure 2.2: *Level scheme of the ^{85}Rb and ^{87}Rb D2 line.*

frequency located between the ^{87}Rb cooling and repumping transitions. REFERENCE

2.1.2.2 Cooler and repumper lasers

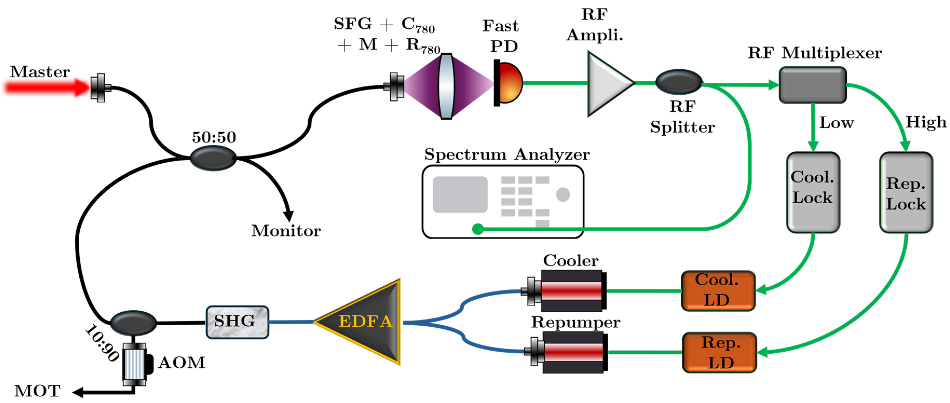


Figure 2.3: *Beat-note locking setup scheme.*

The cooling and repumping light is generated by two 1560 nm DFB diodes. Their outputs are fiber-combined, amplified to ~ 500 mW by an Erbium-Doped Fiber Amplifier (EDFA), and frequency-doubled to 780 nm via a Second Harmonic Generation (SHG) crystal (Fig. 2.3).

Most of this light (90%) is distributed to the MOT optics via a fast-switching AOM. The remaining 10% is mixed with the master laser on a fast (~ 10 GHz) photodiode to generate beat-note signals for frequency offset locking.

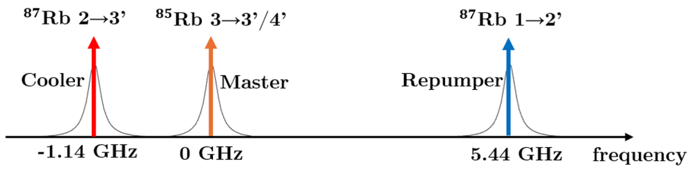


Figure 2.4: *Example of beat-note signals on a spectrum analyzer, used for frequency offset locking of the cooler and repumper lasers relative to the master laser.*

The master-cooler and master-repumper beat notes are isolated using RF filters and used as reference signals to stabilize the frequency offsets of the cooler and repumper lasers with respect to the master laser [18].

2.2 Experimental Procedures and Data Analysis

2.2.1 Cold-Atom Preparation: MOT and Optical Molasses

Each experimental cycle begins with loading atoms into the magneto-optical trap (MOT), using the operating conditions previously identified as optimal for this apparatus. The MOT is run with an axial magnetic-field gradient of $B' = 9.32\,\mathrm{G/cm}$, corresponding to a current of 8 A in the quadrupole coils, a cooling-light detuning of $\Delta = -13\,\mathrm{MHz}$, and all compensation-coil currents set to zero. The loading stage lasts 5 s.

At the end of the MOT phase, the quadrupole magnetic field is switched off and the cloud enters the optical molasses stage. In the absence of the trapping field, the atoms are no longer confined and the cloud starts to expand. At the same time, the atomic temperature decreases further, while the cloud falls under gravity and remains subject to a velocity-dependent damping force generated by the cooling light.

The temperature reached in optical molasses scales approximately as $I/|\Delta|$, where I is the cooling-beam intensity and Δ is the laser detuning (see Sec. 1.2.3). This scaling is valid only within an appropriate detuning range. If $|\Delta|$ is too large, the photon scattering rate, $\Gamma_p \propto 1/\Delta^2$, becomes too small to sustain efficient cooling. On the other hand, if the detuning is too small, excitation of the neighbouring $F = 2 \to F' = 2$ transition is no longer negligible.

A second requirement for efficient molasses cooling is the cancellation of residual magnetic fields. For this reason, 5 ms after the MOT field is switched off, the shim-coil currents are adjusted in order to compensate for the Earth's magnetic field and for other stray fields present in the laboratory. The cooling-light detuning is then changed from the MOT value to the molasses value through a 20-step ramp with total duration 5 ms. This ramp starts 50 μs after the shim currents are set to their molasses values.

2.2.2 Atom Imaging

The atomic cloud is observed inside the vacuum chamber by means of two monochrome CMOS cameras:

- a Teledyne FLIR BFS-U3-04S2M-CS camera, based on the Sony IMX287 sensor, with a resolution of 720×540 pixels and a pixel size of $6.9\,\mu\mathrm{m} \times 6.9\,\mu\mathrm{m}$;
- an Allied Vision Manta G-419 NIR camera, based on the CMOSIS/ams CMV4000 sensor, with a resolution of 2048×2048 pixels and a pixel size of $5.5\,\mu\mathrm{m} \times 5.5\,\mu\mathrm{m}$.

The FLIR camera is used mainly during the optimization of the molasses stage. The Manta camera, owing to its higher sensitivity at 780 nm and its smaller pixel pitch, is used for ODT loading measurements, where resolving the small trapped cloud is more demanding.
Both cameras are equipped with a Thorlabs MVL50M23 adjustable objective, with focal length 50 mm, minimum working distance 200 mm, and minimum magnification of about 0.33. The imaging system is focused so that the HCPCF inside the chamber, and therefore also the atomic cloud located at nearly the same distance, is sharply imaged on the sensor. In both cases the acquisition is triggered externally by the FPGA control system.

2.2.3 FLIR Camera Calibration

A calibration of the imaging system is required in order to convert distances measured on the sensor into physical lengths in the object plane. This is needed in particular for the temperature measurements, where the cloud width extracted from the images must be expressed in absolute units. Starting from the physical pixel pitch of the FLIR sensor, $6.9\,\mu$m, the optical magnification M and the corresponding object-plane pixel size C were estimated through three independent methods.

1. **Direct imaging of a known microstructure**

   The first method uses the hollow-core photonic crystal fiber (HCPCF) as a reference object, exploiting its known physical diameter of $316\,\mu$m. The fiber is imaged directly and its apparent diameter on the sensor is compared with the real one. Since the image is affected by optical blur and the fiber edges cannot be located accurately by eye, the diameter is extracted from a spatial fit.

   The fitted image corresponds to an apparent diameter of 8 pixels. The resulting object-plane calibration is therefore

   $$C = \frac{316\,\mu\text{m}}{8\,\text{px}} = 39.5\,\mu\text{m/px}, \tag{2.1}$$

   which gives a magnification

   $$M = \frac{6.9}{39.5} \approx 0.174. \tag{2.2}$$

2. **Optical calibration by reflected ruler imaging**

   The second method relies on imaging a macroscopic reference target placed at the same optical distance as the atomic cloud. A ruler is observed through a mirror after the camera has already been focused on the atoms. The ruler is translated until its reflected image becomes sharp, ensuring that the target lies on the same focal plane as the cloud.

Several ruler intervals are measured in order to reduce the uncertainty and average over possible local imperfections. This procedure yields an effective pixel size of about 43.3 μm/px, corresponding to

$$M \approx 0.160. \tag{2.3}$$

3. **Calibration from MOT displacement in a bias magnetic field**

As a further cross-check, the imaging scale can be estimated from the displacement of the MOT induced by a controlled bias magnetic field. When a uniform field is applied along the vertical direction by means of the y-axis shim coils, the zero of the quadrupole field is shifted, and the cloud follows the new equilibrium position.

Experimentally, the cloud displacement varies linearly with the current applied to the y-axis shim coils, with measured slope

$$y_{\text{shift}} = 58\,\text{px/A}. \tag{2.4}$$

To convert this result into a physical displacement, one needs the bias field generated per unit shim current and the restoring gradient of the quadrupole field. Modeling the shim coils as a cubic structure of side $L = 645\,\text{mm}$ with $N = 100$ windings gives

$$b = 1.43\,\text{G/A}. \tag{2.5}$$

The quadrupole gradient is estimated from the coil geometry. For a current of 8.4 A, 128 windings, effective radius $R_{\text{eff}} = 62\,\text{mm}$, and effective separation $a_{\text{eff}} = 162\,\text{mm}$, the vertical gradient is

$$G_{y,\text{tot}} = 0.571\,\text{G/mm}. \tag{2.6}$$

The equilibrium displacement per unit current is given by the ratio between the applied bias field and the magnetic-field gradient. Dividing this physical shift by the measured displacement in pixels gives the calibration factor:

$$C = \frac{b}{G_{y,\text{tot}}\, y_{\text{shift}}} = \frac{1.43\,\text{G/A}}{0.571\,\text{G/mm} \times 58\,\text{px/A}} \approx 0.043\,\text{mm/px} = 43\,\mu\text{m/px}. \tag{2.7}$$

If, instead, the empirically measured values previously obtained for this setup are used, namely $b = 1.5\,\text{G/A}$ and $G = 5.6\,\text{G/cm}$ at 8.4 A, one obtains

$$C = \frac{1.5}{5.6 \times 58}\,\text{cm/px} \approx 46\,\mu\text{m/px}. \tag{2.8}$$

Both estimates are consistent with the result obtained from the ruler calibration. In the following analysis, the value provided by the optical method is adopted.

A summary of the three calibration methods is shown in Fig. 2.5.

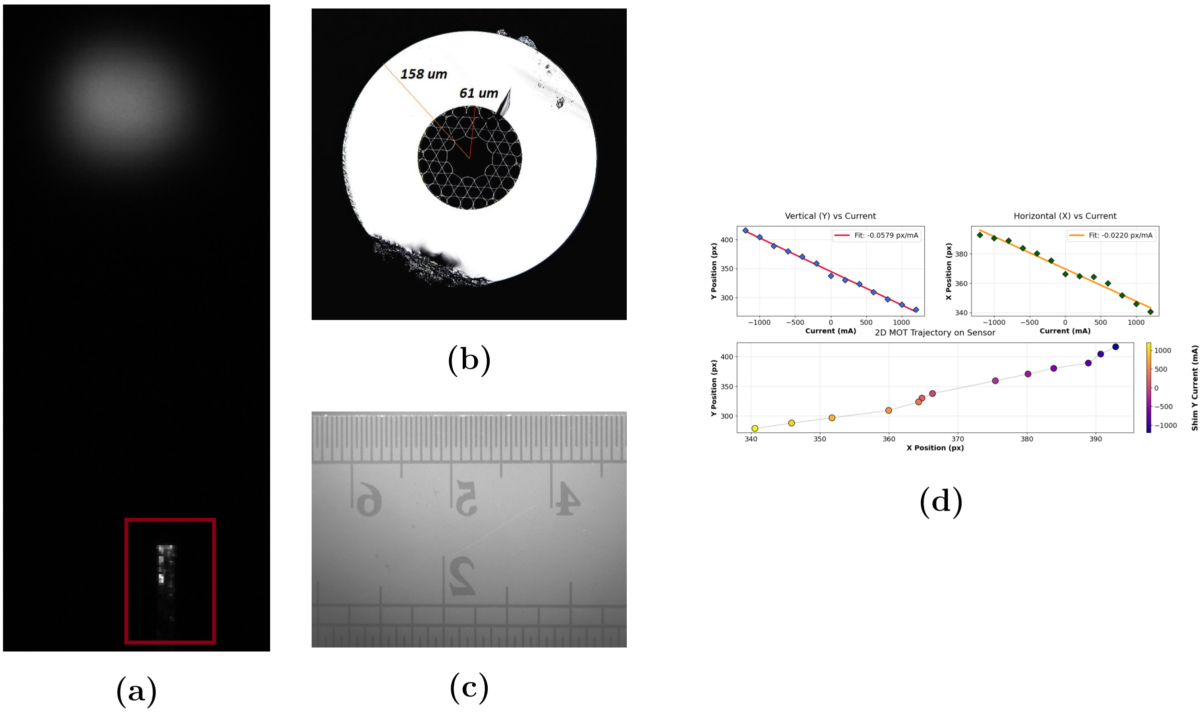


Figure 2.5: *Summary of the spatial calibration methods used for the imaging system.* ***(a)*** *Side-view image of the HCPCF used as the reference object in Method 1.* ***(b)*** *Cross-sectional view of the same fiber, with the nominal diameter of* $316\,\mu\mathrm{m}$*.* ***(c)*** *Ruler used for the reflected-image calibration of Method 2.* ***(d)*** *MOT position as a function of the current applied to the y-axis shim coils in Method 3; the linear fit yields a slope of about* $58\,\mathrm{px/A}$*.*

2.2.4 Temperature Measurements

Once the imaging calibration is known, the cloud size extracted from the images can be expressed in physical units and used to determine the temperature from ballistic expansion. These measurements are used to optimize the molasses parameters by identifying the conditions that minimize the final temperature.

The method is based on the free expansion of the cloud after release. For a non-interacting gas, the position of an atom along one direction evolves as

$$x(t) = x_0 + v_x t. \tag{2.9}$$

Assuming that the initial position and velocity are statistically independent, the variance at time t is

$$\sigma_x^2(t) = \sigma_{x,0}^2 + \sigma_{v_x}^2 t^2, \tag{2.10}$$

where $\sigma_{x,0}$ is the initial spatial width. In the center-of-mass frame, the equipartition theorem gives

$$\sigma_{v_x}^2 = \langle v_x^2 \rangle = \frac{k_B T_x}{m}. \tag{2.11}$$

The fitting law then becomes

$$\sigma_x^2(t) = \sigma_{x,0}^2 + \frac{k_B T_x}{m} t^2, \tag{2.12}$$

with m the mass of ^{87}Rb, $m \approx 1.443 \times 10^{-25}$ kg.
The experimental procedure is as follows:

1. A MOT is loaded and followed by the molasses stage. The molasses parameters and duration are the quantities under optimization.

2. The cooling light is switched off, allowing the cloud to expand freely and fall under gravity. At each selected time of flight, the FLIR camera acquires a fluorescence image of the cloud.

3. A background frame is then acquired after a sufficiently long delay, so that the cloud has completely left the observation region.

4. For each time of flight, the corrected image is obtained as

$$I(t) = I_{\text{raw}}(t) - I_{\text{bg}}(t). \tag{2.13}$$

5. Each corrected image is fitted with a two-dimensional Gaussian, yielding the widths σ_x and σ_y.

6. Equation 2.12 is fitted independently along the two axes to extract T_x and T_y. The temperature quoted in the following is the average of the two values.

Since the imaging pulse destroys the sample, each time-of-flight point must be acquired in a separate experimental cycle.

2.2.5 ODT Loading Measurements

The loading performance of the optical dipole trap is characterized using the Manta camera. Two quantities are extracted from these measurements: the loading efficiency η and the integrated fluorescence signal associated with the trapped cloud, which is proportional to the number of captured atoms.

Because only a small fraction of the molasses atoms, of the order of 10^{-3}, is transferred into the ODT, the trapped-atom signal is much weaker than the fluorescence produced by the untrapped cloud. For this reason, the analysis is based on differential imaging. The sequence for loading the atoms and their subsequent free-fall is illustrated in Figure 2.6.

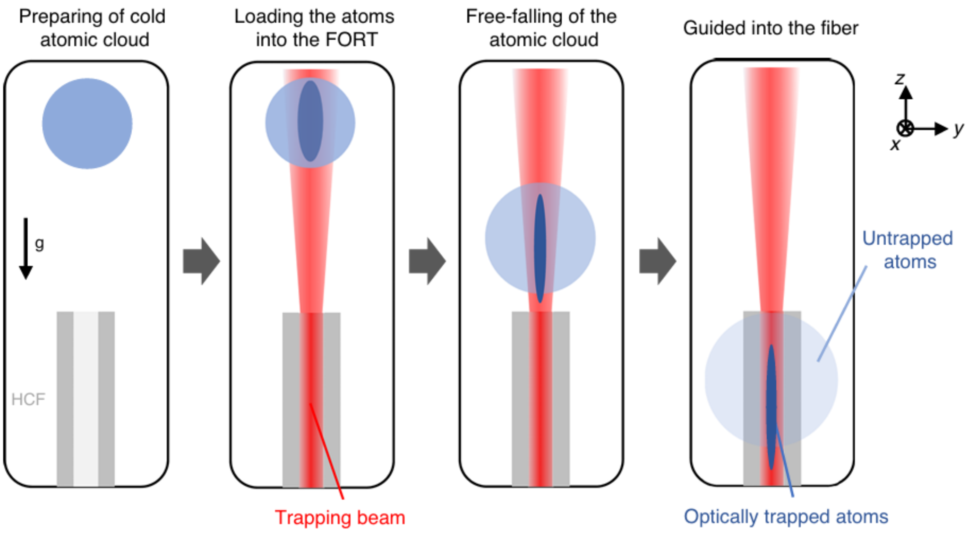


Figure 2.6: *Schematic representation of the experimental sequence for loading and guiding atoms. The process consists of four stages: (1) preparation of the cold atomic cloud, (2) loading the atoms into the far-off-resonance trap (FORT) using a trapping beam, (3) free-falling of the atomic cloud under gravity, and (4) guiding of the optically trapped atoms into the hollow-core fiber (HCF), while the untrapped atoms fall and disperse outside the trap. Credits [21].*

The measurement sequence is the following:

1. Two otherwise identical experimental cycles are performed in succession, one with the ODT on from the MOT loading stage onward and one with the ODT off.

2. After a time of flight of 30 ms, long enough for the untrapped atoms to expand away from the ODT region, fluorescence images I_{on} and I_{off} are acquired with the Manta camera.

3. Each on/off pair is repeated five times.

4. The images are cropped to a pixel region of interest centered on the cloud. This excludes both the visible HCPCF structure and the peripheral parts of the image containing only background.

5. For each condition, the cropped images are averaged:

$$\bar{I}_{\text{on}} = \langle I_{\text{on}} \rangle, \qquad \bar{I}_{\text{off}} = \langle I_{\text{off}} \rangle. \tag{2.14}$$

6. The averaged images are integrated along one direction to obtain one-dimensional profiles with improved signal-to-noise ratio.

7. The total number of atoms in the molasses drifts slowly over time because of variations in vacuum pressure and thermal changes in the optical components. As a consequence, the on and off profiles cannot be subtracted directly: even a small mismatch in the broad molasses background would produce a residual comparable to the ODT signal itself.

   To correct for this effect, the off profile is rescaled by a factor k chosen so that the difference between the two profiles vanishes outside the ODT region. An exclusion window centered at the ODT position, $x_{\mathrm{ODT}} \pm R_{\mathrm{window}}$, is defined. Outside this interval, a linear baseline $L_i(x)$ is estimated for each profile and the residual integrated signal is computed as

$$A_i^{\mathrm{out}} = \int_{\mathrm{outside}} [P_i(x) - L_i(x)]\, dx, \qquad i \in \{\mathrm{on}, \mathrm{off}\}. \tag{2.15}$$

   The normalization factor is then

$$k = \frac{A_{\mathrm{on}}^{\mathrm{out}}}{A_{\mathrm{off}}^{\mathrm{out}}}. \tag{2.16}$$

8. The differential profile attributed to the trapped atoms is therefore

$$\Delta\bar{I}(x) = \bar{I}_{\mathrm{on}}(x) - k\,\bar{I}_{\mathrm{off}}(x). \tag{2.17}$$

9. A Gaussian plus linear background is fitted both to the off profile and to the differential profile. If $A_{\mathrm{off}}^{\mathrm{G}}$ and $A_{\mathrm{ODT}}^{\mathrm{G}}$ denote the Gaussian areas extracted from the two fits, the loading efficiency is defined as

$$\eta = \frac{A_{\mathrm{ODT}}^{\mathrm{G}}}{k\,A_{\mathrm{off}}^{\mathrm{G}}}. \tag{2.18}$$

2.2.6 Measurement of the Number of Atoms

The loading efficiency can be obtained from relative fluorescence signals and therefore does not require an absolute calibration of the camera response. By contrast, estimating the absolute number of trapped atoms requires converting the recorded counts into a known number of detected photons.

To this end, the camera is calibrated against a beam of known optical power derived from the MOT light. The beam is directed onto the sensor after passing through a series of attenuators and through the vacuum-chamber viewports. The power measured along the optical path includes a constant background contribution of $P_{\mathrm{bg}} = 1.5\,\mu\mathrm{W}$, which is subtracted from all radiometric readings.

The total transmission of the optical path is written as the product of the individual transmission factors,

$$T_{\mathrm{tot}} = T_1\,T_{\mathrm{vac}}\,T_2\,T_3 \approx 0.01272. \tag{2.19}$$

During the calibration, the AOM amplitude is set to 50, corresponding to an input power of $22\,\mu\mathrm{W}$ after background subtraction. The optical power reaching the camera is therefore

$$P_{\mathrm{cam}} = 22\,\mu\mathrm{W} \times T_{\mathrm{tot}} \approx 279.8\,\mathrm{nW}. \tag{2.20}$$

The camera response is determined by varying the exposure time t_{exp} while keeping P_{cam} fixed. As shown in Figure **??**, for each image, the beam profile is fitted in two dimensions and the integrated counts are extracted. Denoting by C the total number of recorded counts, the response is assumed to be linear:

$$C = m\,t_{\mathrm{exp}} + q, \tag{2.21}$$

where m is the count rate. Since the incident photon flux is given by $P_{\mathrm{cam}}/(hc/\lambda)$, the conversion factor from counts to detected photons is

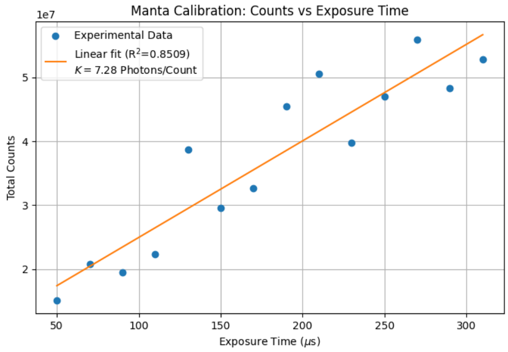


Figure 2.7: *Linear fit of the total camera counts as a function of the exposure time* t_{exp}*. The slope* m *extracted from the fit is used to calculate the photon conversion factor* K*.*

$$K = \frac{P_{\mathrm{cam}}\lambda}{mhc}, \tag{2.22}$$

where h is Planck's constant, c the speed of light, and λ the laser wavelength.

For an ODT fluorescence image, the integrated residual signal C_{tot} can then be converted into the number of detected photons:

$$N_{\mathrm{det}} = K\,C_{\mathrm{tot}}. \tag{2.23}$$

In order to infer the number of trapped atoms, the fluorescence contribution of a single atom during the imaging pulse must be estimated. The scattering rate is described by the standard two-level expression

$$R_{\mathrm{sc}} = \frac{\Gamma}{2}\frac{I/I_{\mathrm{sat}}}{1 + I/I_{\mathrm{sat}} + 4(\Delta/\Gamma)^2}, \tag{2.24}$$

where Γ is the natural linewidth, I is the total imaging intensity, I_{sat} is the saturation intensity, and Δ is the laser detuning.

For an imaging pulse of duration t_{img}, each atom scatters on average $R_{\mathrm{sc}} t_{\mathrm{img}}$ photons. Only a fraction of this fluorescence reaches the sensor. Approximating the collection efficiency by the fractional solid angle subtended by the imaging optics,

$$\eta_{\mathrm{coll}} = \frac{r^2}{4d^2}, \tag{2.25}$$

with r the effective lens radius and d the distance between the cloud and the collection optics, the number of detected photons per atom is

$$N_{\mathrm{ph},1} = R_{\mathrm{sc}}\, t_{\mathrm{img}}\, \eta_{\mathrm{coll}}\, T_{\mathrm{optics}}, \tag{2.26}$$

where T_{optics} accounts for the transmission of the imaging system.

The absolute number of trapped atoms is finally obtained from

$$N = \frac{N_{\mathrm{det}}}{N_{\mathrm{ph},1}} = \frac{K\, C_{\mathrm{tot}}}{R_{\mathrm{sc}}\, t_{\mathrm{img}}\, \eta_{\mathrm{coll}}\, T_{\mathrm{optics}}}. \tag{2.27}$$

2.3 Expected Number of Trapped Atoms

A simple estimate of the number of atoms loaded into the ODT can be obtained by combining the spatial density distribution of the MOT with the thermal velocity distribution of the atoms. An atom at position $\vec{r}$ is assumed to be captured if its transverse kinetic energy is smaller than the local trap depth:

$$\frac{1}{2} m \left(v_x^2 + v_y^2\right) < \left|U_{\mathrm{ODT}}(\vec{r})\right|. \tag{2.28}$$

The spatial density of the initial cloud is denoted by $n_{\mathrm{MOT}}(\vec{r})$, while the velocity distribution is taken to be Maxwell-Boltzmann,

$$f(\vec{v}) = \frac{1}{(2\pi\mu^2)^{3/2}} \exp\left(-\frac{v^2}{2\mu^2}\right), \qquad \mu^2 = \frac{k_B T}{m}. \tag{2.29}$$

The number of captured atoms is obtained by integrating over phase space, restricting the result to atoms moving toward the trap region, namely those with $v_z < 0$. Under this assumption,

$$N_{\mathrm{ODT}} = \int d^3 r\, n_{\mathrm{MOT}}(\vec{r})\, \frac{1}{2} \int_{v_x^2 + v_y^2 < \frac{2|U|}{m}} dv_x\, dv_y\, \frac{1}{2\pi\mu^2} \exp\left(-\frac{v_x^2 + v_y^2}{2\mu^2}\right). \tag{2.30}$$

The velocity integral is more conveniently evaluated in polar coordinates in the transverse velocity plane. Using

$$dv_x\, dv_y = \pi\, d(v_r^2), \tag{2.31}$$

and introducing the variable

$$x = \frac{v_r^2}{2\mu^2}, \tag{2.32}$$

one obtains

$$\int_0^{\frac{2|U|}{m}} \pi\, d(v_r^2)\, \frac{1}{2\pi\mu^2} \exp\left(-\frac{v_r^2}{2\mu^2}\right) = \int_0^{\frac{|U|}{m\mu^2}} e^{-x}\, dx = 1 - \exp\left(-\frac{|U_{\mathrm{ODT}}(\vec{r})|}{k_B T}\right). \tag{2.33}$$

The expression for the number of trapped atoms therefore becomes

$$N_{\mathrm{ODT}} = \int d^3r\, n_{\mathrm{MOT}}(\vec{r})\, \frac{1}{2} \left[1 - \exp\left(-\frac{|U_{\mathrm{ODT}}(\vec{r})|}{k_B T}\right)\right]. \tag{2.34}$$

In the shallow-trap limit, $|U_{\mathrm{ODT}}| \ll k_B T$, the exponential can be expanded to first order one then finds:

$$N_{\mathrm{ODT}} \simeq \int d^3r\, n_{\mathrm{MOT}}(\vec{r})\, \frac{|U_{\mathrm{ODT}}(\vec{r})|}{2k_B T}. \tag{2.35}$$

For a Gaussian dipole trap,

$$U_{\mathrm{ODT}}(\vec{r}) = -U_0\, \frac{W_0^2}{W^2(z)} \exp\left(-\frac{2(x^2+y^2)}{W^2(z)}\right), \tag{2.36}$$

so that

$$N_{\mathrm{ODT}} \simeq \frac{U_0}{2k_B T} \int d^3r\, n_{\mathrm{MOT}}(\vec{r})\, \frac{W_0^2}{W^2(z)} \exp\left(-\frac{2(x^2+y^2)}{W^2(z)}\right). \tag{2.37}$$

If the ODT waist is much smaller than the transverse size of the MOT, namely if $W(z) \ll R_x, R_y$, the transverse integrals can be extended to infinity:

$$\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \exp\left(-\frac{2(x^2+y^2)}{W^2(z)}\right) dx\, dy = \frac{\pi W^2(z)}{2}. \tag{2.38}$$

Substituting this result into the previous expression leads to the cancellation of the $W^2(z)$ term:

$$N_{\mathrm{ODT}} = \frac{U_0}{2k_B T} \frac{\pi W_0^2}{2} \int_{-R_z}^{R_z} n_{\mathrm{MOT}}(0,0,z)\, dz. \tag{2.39}$$

To obtain a compact analytical estimate, the MOT is modeled as a uniform ellipsoid with radii R_x, R_y, and R_z, and constant density

$$n_{\mathrm{MOT}} = \frac{3N_{\mathrm{MOT}}}{4\pi R_x R_y R_z}. \tag{2.40}$$

Under this approximation,

$$N_{\mathrm{ODT}} = \frac{U_0 \pi W_0^2}{4 k_B T} \left(\frac{3 N_{\mathrm{MOT}}}{4\pi R_x R_y R_z} \cdot 2R_z \right) = \frac{3}{8} \frac{U_0 W_0^2}{k_B T R_x R_y} N_{\mathrm{MOT}}. \tag{2.41}$$

2.4 Injection into the black fiber

The second optical field required for the conveyor-belt configuration is delivered by a conventional polarization-maintaining single-mode fiber, referred to throughout this work as the "black fiber". It counterpropagates with respect to the 1064, *mathrmnm* field guided through the Hollow-Core Photonic Crystal Fiber (HCPCF). Their interference generates the moving optical lattice described in Sec. 1.3. Efficient coupling, a well-defined spatial mode, and stable polarization are therefore required to obtain a deep and reproducible conveyor-belt potential.

The black-fiber branch was constructed as an extension of the Optical Dipole Trap (ODT) system described already existing. The injection strategy was first validated on a dedicated test bench and subsequently transferred to the definitive optical branch.

The black fiber is operated at 1064 nm and has a nominal mode-field radius of approximately 5.2 μm. Its input tip includes a glass endcap approximately 300 μm long, which improves power handling but modifies the effective optical distance between the focusing element and the guided mode. Specifically, the fiber is a polarization-maintaining, single-mode cable manufactured by Schäfter+Kirchhoff (model PMC-780-1-18-xxx / PMC-780-5.3-NA012-1-APC-xxx-P, housed in a 900 μm buffer). It features an effective numerical aperture (NA_{e^2}) of 0.08 ± 0.005 at 780 nm, low-stress FC/APC (8°) connectors on both ends with the polarization slow axis aligned to the connector index key, and a specified operating wavelength range from the 770 nm cut-off up to a maximum of 1100 nm.

2.4.1 Gaussian-beam characterization procedure

Throughout this work, the size and divergence of the optical beam are characterized using a common beam-profiling procedure. The same method is applied before and after the fiber whenever the injection geometry, input beam parameters, or output wavefront have to be determined. The beam under test is first aligned horizontally at a fixed height above the optical table. This allows its transverse intensity distribution to be sampled at a sequence of longitudinal positions without introducing additional steering optics between measurements. A camera mounted on a translation stage is then moved along the propagation direction, and an image of the beam cross-section is recorded at each position.

Each recorded image is cropped to isolate the beam spot from background light and stray reflections. The resulting intensity distribution is fitted with a two-dimensional Gaussian function, from which the beam radii along the two transverse axes are extracted. Repeating this analysis at several positions provided the beam radius as a function of the longitudinal coordinate z. For each transverse direction, the measured beam radius is fitted to the standard Gaussian-beam propagation law

$$w(z) = w_0\sqrt{1 + \left(\frac{z - z_0}{z_R}\right)^2}, \tag{2.42}$$

where w_0 is the waist radius, z_0 is the waist position, and z_R is the associated Rayleigh range. This fit provides the size and position of the beam waist and allowed the wavefront curvature at any relevant plane to be determined. The reconstructed beam parameters could then be compared with the target fiber-mode parameters or used to select and position the injection optics.

2.4.2 Injection setup

The injection setup for the black fiber comprises the pre-existing ODT injection setup, shared with the HCPCF branch, and a branch built specifically for the black fiber downstream of PBS2. The HCPCF branch was previously built, but a HWP was added before the focusing lens for the polarization control required to implement the conveyor belt. The pre-existing setup, shown in Figure 2.8, consists of:

- HWP1 and the optical isolator, protecting the laser from back-reflections;
- HWP2 and PBS1, regulating the optical power delivered to the rest of the setup;
- HWP3 and PBS2, splitting the beam between the HCPCF branch and the black-fiber branch;
- on the HCPCF branch, fed by the first output of PBS2: an AOM operating at 110 MHz, used as a fast switch, with the first diffraction order used downstream and the zeroth order blocked by a beam dump
- a $\sim 3\times$ beam expander, collimating and enlarging the beam
- a mirror (M1) and a dichroic mirror, both on tilt stages, used for the fiber-injection alignment, with the dichroic mirror also combining the 1064 nm beam with a 780 nm beam used for atom characterization inside the HCPCF core
- a focusing lens with $f = 50$ mm, on a horizontal micrometric translation stage
- a HWP, for the polarization adjustment before the HCPCF, mounted on a rotation stage
- the HCPCF on an x-y translation stage terminated with an FC/APC connector

Downstream of PBS2, the branch built for the black fiber, shown in Figure 2.9, consists of:

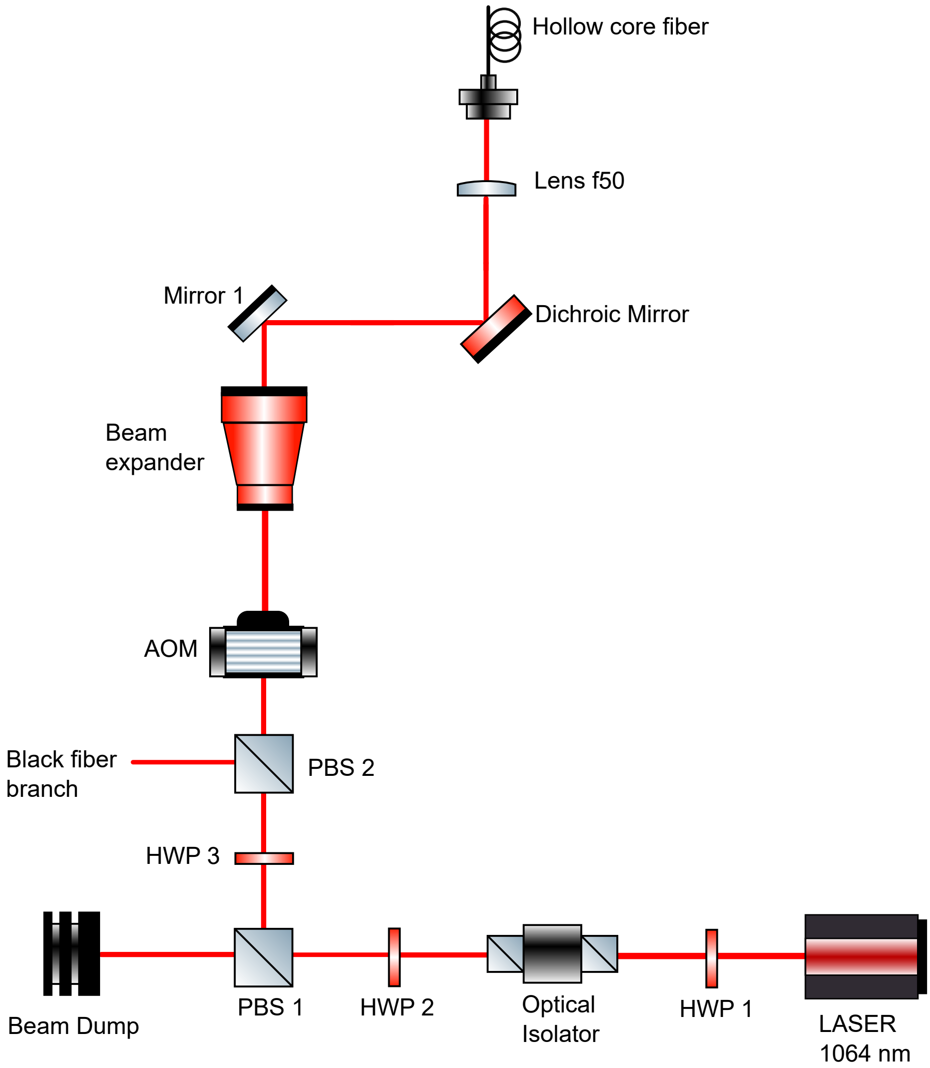


Figure 2.8: *Optical scheme of the pre-existing ODT injection setup, from HWP1 to the HCPCF, with the second output of PBS2 leading to the black-fiber branch described below.* ***[TODO: verify AOM orientation in the schematic]***

- an acousto-optic modulator (AOM), providing fast control of the beam intensity and frequency ([**TODO: manufacturer, model, diffraction order used, drive frequency**]);
- a steering mirror, directing the beam into the telescope;
- a telescope formed by an $f = -30\,\mathrm{mm}$ lens and an $f = 150\,\mathrm{mm}$ lens ([**TODO: lens separation**]), re-collimating the beam to a radius of approximately 1.6 mm, the target value derived in Section 2.4.4;
- two steering mirrors, directing the beam toward the fiber-injection assembly;
- a half-wave plate, setting the incident linear polarization along the slow axis of the fiber ([**TODO: HWP model and alignment procedure**], Figure 2.11);

- an $f = 25\,\text{mm}$ achromatic doublet (Thorlabs AC127-025-C), mounted on a micrometric translation stage along the propagation axis, focusing the beam into the fiber. The axial position of the lens is used to compensate for the displacement of the fiber-mode plane caused by the fiber endcap, discussed in Section 2.4.4;

- the black fiber, mounted on the branch through its output connector.

[**TODO: manufacturer and model; standard telecom-fiber designation; total fiber length; nominal mode-field diameter and numerical aperture at** $1064\,\text{nm}$**; orientation of the slow axis relative to the connector key; input and output connector types and angle; endcap geometry; spectral operating range; distance from the AOM to the telescope; lens mount; translation-stage model, resolution and travel; nominal lens-to-fiber distance; fiber connector and mount; diagnostic pick-offs or power-measurement points along the branch**]

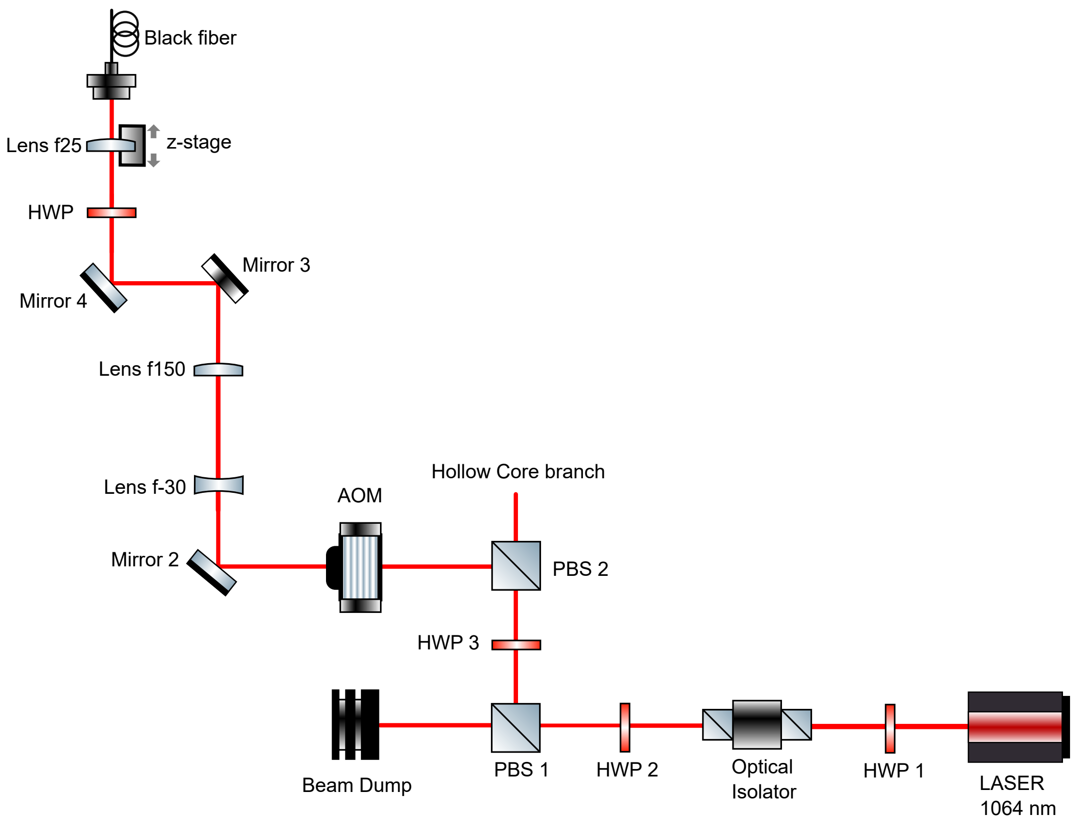


Figure 2.9: *Optical scheme of the black-fiber branch downstream of PBS2.*

Figure 2.10 shows a photograph of the complete injection setup.

2.4.3 Beam-profile characterization method

The size and divergence of the beams involved in the black-fiber injection were characterized with the same beam-profiling approach used in Section 2.4.1.

Figure 2.10: *Photograph of the complete injection branch. The black-fiber branch is on the left, while the pre-existing ODT injection setup is on the right.*

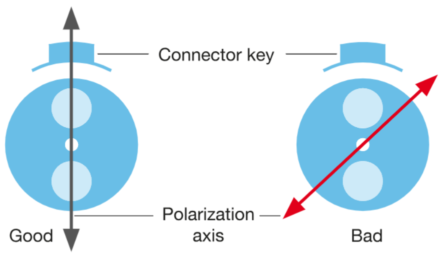


Figure 2.11: *Orientation of the input linear polarization relative to the slow and fast axes of the polarization-maintaining fiber.*

For each transverse direction, the measured radius was fitted to the Gaussian-beam propagation law, this fit gave the size and position of the beam waist, which could then be compared with the target fiber-mode parameters or used to select and position the injection optics. The procedure was repeated both upstream and downstream of the fiber, after each modification or addition of the injection optics, to obtain the beam parameters used in the mode-matching calculations of Section 2.4.4.

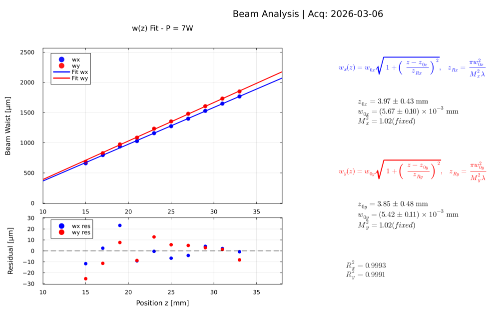
2.4.4 Mode matching and the fiber endcap

Coupling a Gaussian beam into a single-mode fiber requires matching the focused field to the fundamental fiber mode in waist size, waist position and transverse alignment. The standard approach uses a collimated input beam and a fixed collimating lens that focuses it directly onto the fiber facet, and this was the strategy first used for the black fiber, then substituted by a f25 shifting lens. The mode-matching parameters used in the definitive branch (Section 2.4.2) were derived and validated on a dedicated test bench (Fig. 2.14), independent of the complete ODT apparatus and not including the AOM or the telescope. The test bench consisted of:

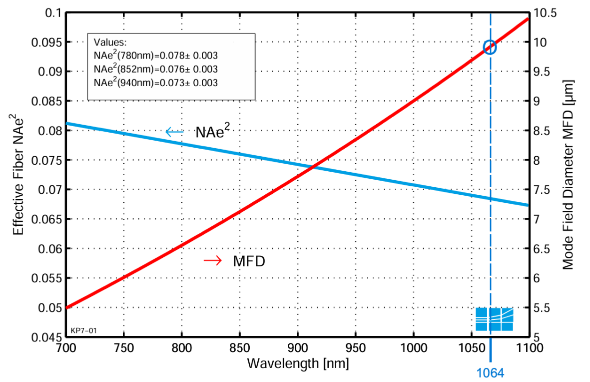
- HWP1 and the optical isolator, as in Section 2.4.2;
- HWP2 and PBS1 (with a beam dump on the rejected port), regulating the optical power delivered to the rest of the setup;
- HWP3 and PBS2, splitting the beam between the HCPCF and black-fiber branches;
- Mirror 1, directing the beam reflected by PBS2 towards the beam expander;
- an adjustable beam expander (ZBE3C2X-8X Achromatic Zoom Beam Expander, AR Coated: 1050 nm-1650 nm), setting the diameter and residual divergence of the beam sent to the black fiber;
- HWP4 and PBS3 (with a beam dump on the rejected port), for additional power control and polarization cleaning;
- Mirror 2 and Mirror 3, routing the beam towards the fiber injection stage;
- HWP5, adjusting the polarization axis before injection into the polarization-maintaining fiber;
- a focusing element coupling the beam into the black fiber: initially a fixed $f = 18\,\mathrm{mm}$ fiber collimator (Schäfter+Kirchhoff, 60FC-4-A18-03), later replaced by the $f = 25\,\mathrm{mm}$ achromatic doublet (Thorlabs AC127-025-C) on a micrometric translation stage (z-stage) also used in the definitive branch;

Injecting a nominally collimated beam directly through the fixed $f = 18\,\mathrm{mm}$ collimator gave an injection efficiency of approximately 20%. Introducing additional divergence upstream, equivalent to inserting a weak $f = -1000\,\mathrm{mm}$ CHI LHA DETTO lens before the collimator, raised the efficiency above 50%. This indicates that the plane of the fiber mode does not coincide with the nominal focal plane of the collimator.
The beam profile after the fiber was measured (Figure 2.12a) and its reconstructed waist compared with the mode-field diameter (MFD) specified by the manufacturer at 1064 nm

(Figure 2.12b). The measured waist agreed with the specified MFD, confirming that the transmitted light corresponds to the fundamental fiber mode.



(a) *Beam-radius measurements after the black fiber and fit to the Gaussian-beam propagation law of Equation* (**??**). ***[TODO: measurement axis, fit parameters and uncertainties, optical configuration]***



(b) *Manufacturer-specified mode-field diameter of the black fiber at* 1064 nm, *used as the target value in the mode-matching calculation below.*

Figure 2.12: *Black fiber beam characterization and manufacturer specifications.*

The beam profile analysis shown in Fig.2.12a uses the standard Gaussian beam propagation model. The beam radius evolution $w(z)$ along the propagation axis z for both the x and y transverse directions is described by:

$$w_x(z) = w_{0x}\sqrt{1+\left(\frac{z-z_{0x}}{z_{Rx}}\right)^2}, \quad \text{with} \quad z_{Rx} = \frac{\pi w_{0x}^2}{M_x^2\lambda} \tag{2.43}$$

$$w_y(z) = w_{0y}\sqrt{1+\left(\frac{z-z_{0y}}{z_{Ry}}\right)^2}, \quad \text{with} \quad z_{Ry} = \frac{\pi w_{0y}^2}{M_y^2\lambda} \tag{2.44}$$

Based on the fit of the experimental data, the extracted parameters for the x-axis are:

$$\begin{aligned} z_{0x} &= 3.97 \pm 0.43\,\text{mm} \\ w_{0x} &= (5.67 \pm 0.10) \times 10^{-3}\,\text{mm} \\ M_x^2 &= 1.02\,(\text{fixed}) \end{aligned}$$

For the y-axis, the extracted fit parameters are:

$$\begin{aligned} z_{0y} &= 3.85 \pm 0.48\,\text{mm} \\ w_{0y} &= (5.42 \pm 0.11) \times 10^{-3}\,\text{mm} \\ M_y^2 &= 1.02\,(\text{fixed}) \end{aligned}$$

The goodness of the fit is confirmed by the high coefficient of determination for both axes:

$$R_x^2 = 0.9993$$
$$R_y^2 = 0.9991$$

The mismatch between the nominal collimator focal plane and the fiber-mode plane is attributed to the endcap spliced onto the fiber input, a glass segment approximately 300 μm long used to improve power handling (Figure 2.13). The endcap modifies the effective optical path between the external focusing lens and the fiber facet, displacing the focused waist relative to the position expected for a bare fiber tip. Standard fiber collimators are designed assuming the fiber mode lies at their nominal focal plane, so this displacement is not compensated automatically and must be corrected in the injection design. Reproducing the correction with a fixed, long-focal-length diverging lens would be impractical to position and would add aberrations to the injected beam. A focusing lens mounted on a micrometric translation stage, giving direct control of the axial position of the focused waist, was used instead.

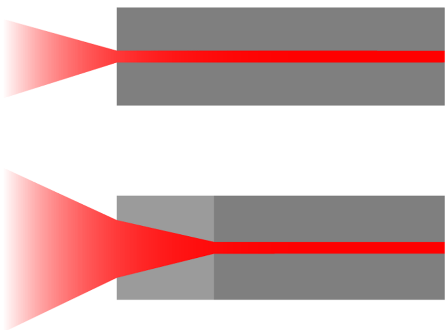


Figure 2.13: *Geometry of the endcap spliced onto the black-fiber input.* ***[TODO: connector orientation, propagation direction]***

For a collimated beam of radius w_{in} focused by a lens of focal length f, Equation (??) gives the resulting waist radius w_{f}. Solving for the input radius required to obtain a target w_{f},

$$w_{\text{in}} = \frac{\lambda f}{\pi w_{\text{f}}}. \tag{2.45}$$

Using $\lambda = 1.064\,\mu$m, $f = 25$ mm, and $w_{\text{f}} = 5.2\,\mu$m, taken as half of the manufacturer-specified MFD (Figure 2.12b), gives

$$w_{\text{in}} = \frac{(1.064 \times 10^{-6}\,\text{m})(25 \times 10^{-3}\,\text{m})}{\pi(5.2 \times 10^{-6}\,\text{m})} \approx 1.6\,\text{mm}. \tag{2.46}$$

The beam expander was set to produce this radius at the injection lens, and the resulting beam was profiled before injection to verify both its size and its residual divergence.

The axial position of the $f = 25\,\mathrm{mm}$ lens was calibrated by propagating light in the reverse direction: with 7 W injected through the fixed $f = 18\,\mathrm{mm}$ collimator and collected at the output by the $f = 25\,\mathrm{mm}$ lens, the lens position was adjusted until the output beam was collimated. This confirmed that the lens had been placed at the axial position corresponding to the effective fiber-mode plane, and fixed the lens position later reproduced in the definitive branch.

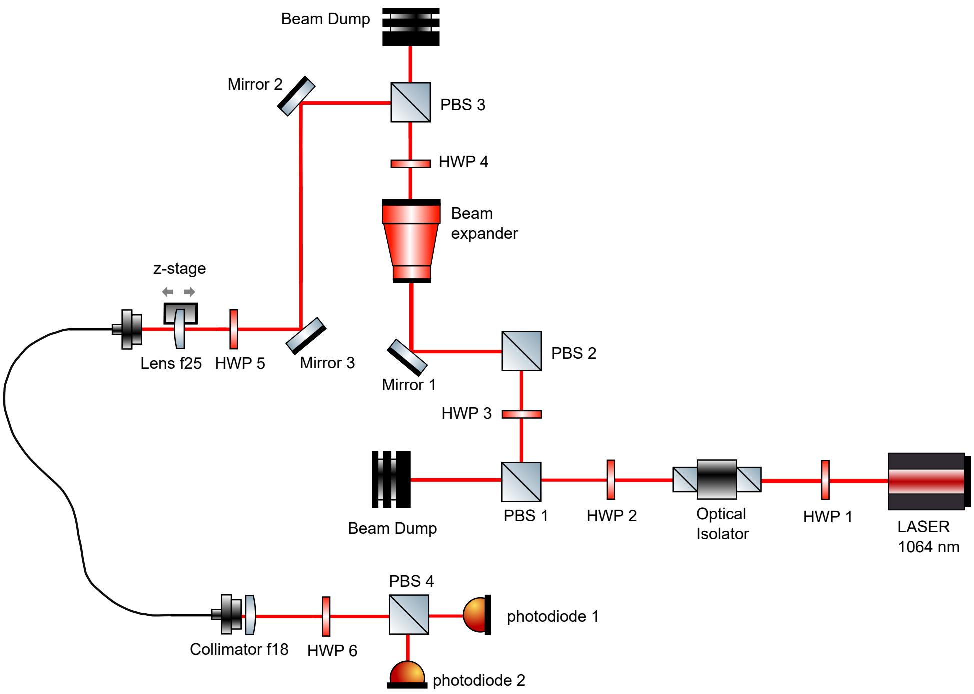


Figure 2.14: *Setup of the black-fiber injection test bench, used to validate the mode-matching procedure and calibrate the axial position of the $f = 25\,\mathrm{mm}$ lens. At the fiber output the polarization stability was characterized with the setup of Section 2.5.*

On the test bench, with the $f = 25\,\mathrm{mm}$ lens at the calibrated position and forward propagation, the power measured after the black fiber and the $f = 18\,\mathrm{mm}$ output collimator reached approximately 70% of the power measured before the fiber, directly after the $f = 25\,\mathrm{mm}$ lens.

With the definitive branch of Section 2.4.2, after adding the AOM and the telescope, the measured injection efficiency reached approximately 75%,confirming that the mode-matching procedure transfers to the complete optical layout.

2.5 Polarization stability characterization

After optimizing the injection efficiency, the polarization stability of the light transmitted through the black fiber was characterized. This quantity is relevant to the optical conveyor belt because polarization fluctuations are converted into optical-power fluctuations by the polarization-sensitive elements of the final setup, affecting the stability of the resulting optical potential. The two orthogonal linear-polarization components at the fiber output were measured simultaneously; their time-domain stability was analyzed with the Allan deviation, and their frequency content with the power spectral density, obtained from the Fourier transform of the recorded signals and compared with the expected shot-noise level.

2.5.1 Detection setup and polarization alignment

The polarization-analysis setup is shown in Figure 2.14. The beam emerging from the black fiber was collimated and sent through a half-wave plate (HWP) followed by a polarizing beam splitter (PBS), whose transmitted and reflected components were detected simultaneously by two photodiodes.

During the initial alignment, the transmitted component was measured with a home-made photodiode connected to channel 1, and the reflected component with a Thorlabs PDA100A2 connected to channel 2, operated at 0 dB internal gain with 6 dB of additional electrical attenuation. The total optical power incident on the PBS was 4.2 mW.

The HWP was adjusted to obtain approximately equal power at the two PBS outputs, corresponding to an input polarization close to 45° relative to the PBS axes. At this operating point, a small polarization rotation transfers power between the two outputs, producing anticorrelated variations in the two detector signals, which gives high sensitivity to polarization fluctuations.

Before the stability measurements, the PBS orientation was optimized to ensure that its finite extinction ratio did not limit the measurement. Rotating the incident linear polarization and finely adjusting the PBS orientation gave polarization extinction ratios of 31 dB in reflection and 33 dB in transmission.

[**TODO: output-collimator focal length; HWP and PBS models; acquisition-device model; photodiode connections and termination impedances; acquisition bandwidth and voltage ranges**]

2.5.2 Alignment to the fiber principal axes

The polarization analyzer was referenced to the principal axes of the polarization-maintaining fiber, using an input HWP before the fiber and, after the fiber, an output collimator, a steering mirror, a second HWP, the PBS, and a power meter.

The two HWPs were adjusted to maximize the power transmitted through the PBS while minimizing the reflected power and its fluctuations. This condition corresponds to exciting one principal axis of the fiber and aligning the emerging linear polarization with one PBS axis; the optimal input-HWP angle was 131°.

The output HWP was adjusted to obtain a maximum transmitted power of 2.68 mW.

With the power meter moved to the reflection port, the reflected signal was minimized by jointly tuning the PBS orientation and the two HWPs, giving a minimum reflected power of $4\,\mu$W, including a background of $1.5\,\mu$W, against a maximum reflected power of approximately 2 mW. Without background subtraction, the corresponding polarization extinction ratio is

$$\mathrm{PER} = 10\log_{10}\left(\frac{P_{\max}}{P_{\min}}\right) = 10\log_{10}\left(\frac{2\,\mathrm{mW}}{4\,\mu\mathrm{W}}\right) \approx 27\,\mathrm{dB}. \tag{2.47}$$

As a further check, the fiber transmission efficiency was measured again, with input and output powers of 4.18 mW and 2.88 mW, respectively:

$$\eta_{\text{fiber}} = \frac{2.88\,\mathrm{mW}}{4.18\,\mathrm{mW}} \approx 69\%, \tag{2.48}$$

consistent with the efficiency measured during injection optimization.

2.5.3 Photodiode calibration

The two detector outputs were converted from voltage to equivalent optical power before comparing the polarization components.

The home-made detector is based on a BPW34 photodiode followed by an OP27G transimpedance amplifier with a gain of $10\,\mathrm{k}\Omega$. Using a photodiode responsivity at 1064 nm of

$$\mathcal{R}_{\text{HM}} \approx 0.7\,\mathrm{A/W} \times 0.3 = 0.21\,\mathrm{A/W}, \tag{2.49}$$

the resulting voltage responsivity is

$$G_{\text{HM}} = \mathcal{R}_{\text{HM}} G_{\text{TIA}} \approx 2.1\,\mathrm{V/mW}. \tag{2.50}$$

The detector output stage has a $1\,\mathrm{k}\Omega$ output impedance; the corresponding loading correction was included where required by the acquisition-input impedance.

For the Thorlabs PDA100A2, a photodiode responsivity of 0.4 A/W at 1064 nm was used, together with the selected internal gain and any additional electrical attenuation, to convert the recorded voltage into optical power.

[TODO: measured offsets and dark signals; acquisition-input impedance; PDA100A2 transimpedance gain at 0 dB and 10 dB; uncertainty of both voltage-to-power conversion factors]

2.5.4 Data acquisition and normalized polarization signal

For the stability measurements, the home-made detector was placed on the reflected port and connected to channel 1, and the Thorlabs PDA100A2 on the transmitted port and connected to channel 2, operated at 10 dB internal gain without external attenuation.
For each measurement, 10^5 samples were acquired simultaneously from both channels. Traces of 600 ms, 60 s, and 600 s total duration were recorded to probe different timescales; the 60 s trace was acquired both in standard sampling mode, preserving faster fluctuations, and in High-Resolution mode, improving the resolution of slower fluctuations.
After background subtraction and detector calibration, the measured voltages were converted into the transmitted and reflected powers $P_{\mathrm{T}}(t)$ and $P_{\mathrm{R}}(t)$. The total detected power is

$$P_{\Sigma}(t) = P_{\mathrm{T}}(t) + P_{\mathrm{R}}(t), \tag{2.51}$$

and the normalized differential signal is

$$D(t) = \frac{P_{\mathrm{T}}(t) - P_{\mathrm{R}}(t)}{P_{\mathrm{T}}(t) + P_{\mathrm{R}}(t)}. \tag{2.52}$$

[TODO: verify that Equation (2.52) **matches the normalization used in the analysis code, e.g.** $D(t) = P_{\mathrm{T}}(t)/\overline{P_{\mathrm{T}}} - P_{\mathrm{R}}(t)/\overline{P_{\mathrm{R}}}$] A fluctuation of the total transmitted power appears mainly in $P_{\Sigma}(t)$, while a polarization fluctuation transfers power between the two PBS outputs and appears predominantly in $D(t)$; the normalization suppresses common-mode power fluctuations to first order.
[TODO: sampling frequency for each trace; Nyquist frequency and frequency resolution; acquisition-device (oscilloscope or DAQ) model; anti-aliasing filter; exact definition of High-Resolution mode]

2.5.5 Allan-deviation analysis

The time-domain stability of the polarization signal was quantified with the overlapping Allan deviation. For a uniformly sampled sequence D_k with sampling interval τ_0, averages were computed over adjacent groups of m samples,

$$\overline{D}_j(\tau) = \frac{1}{m} \sum_{k=jm}^{(j+1)m-1} D_k, \qquad \tau = m\tau_0, \tag{2.53}$$

and the Allan variance from the difference between consecutive averages,

$$\sigma_D^2(\tau) = \frac{1}{2} \left\langle \left[\overline{D}_{j+1}(\tau) - \overline{D}_j(\tau) \right]^2 \right\rangle, \qquad \sigma_D(\tau) = \sqrt{\sigma_D^2(\tau)}. \tag{2.54}$$

The Allan deviation gives the typical change in the average polarization signal between consecutive intervals of duration τ, and identifies the timescales over which averaging improves the polarization stability and those over which slow drift dominates. A decreasing $\sigma_D(\tau)$ indicates that averaging suppresses the dominant fluctuations; a minimum identifies the optimum averaging time; an increase at longer τ indicates the onset of drift or other low-frequency instabilities. The slope of $\sigma_D(\tau)$ on a logarithmic scale distinguishes different classes of noise.

The Allan deviation was evaluated for the individual polarization channels, for the total-power signal $P_\Sigma(t)$, and for the normalized differential signal $D(t)$, to determine whether the observed instability is dominated by common-mode optical-power noise or by a redistribution of power between the two polarization components.

[**TODO: overlapping or non-overlapping estimator used; removal of mean or linear drift; range of averaging times; confidence intervals or number of averages at each τ; treatment of the trace end points**]

2.5.6 Frequency-domain analysis and comparison with the shot-noise limit

The frequency content of the measured power fluctuations was analyzed with the power spectral density (PSD) of the calibrated photodiode signal, computed from its discrete Fourier transform after subtracting the mean value of each trace. The PSD describes how the variance of the measured optical power is distributed over frequency.

[**TODO: window function; direct FFT or Welch method; segment length and overlap; sampling frequency and frequency resolution; single- or double-sided PSD convention; whether a linear trend was removed; units used in the final plots** (W^2/Hz **or** $\mathrm{W}/\sqrt{\mathrm{Hz}}$)]

The measured spectrum was compared with the shot-noise limit expected for a coherent optical field at the same average power. For a photodiode with responsivity $\mathcal{R}$ receiving an average optical power $\overline{P}$, the mean photocurrent is $\overline{I} = \mathcal{R}\,\overline{P}$, and the single-sided shot-noise PSD of the photocurrent is

$$S_{I,\mathrm{SN}} = 2e\overline{I} = 2e\mathcal{R}\,\overline{P}, \tag{2.55}$$

where e is the elementary charge. The equivalent optical-power PSD follows by dividing by the squared responsivity,

$$S_{P,\mathrm{SN}} = \frac{S_{I,\mathrm{SN}}}{\mathcal{R}^2} = \frac{2e\overline{P}}{\mathcal{R}}. \tag{2.56}$$

Using $\mathcal{R} = \mathrm{QE}\, e/(\hbar\omega)$, this can be written as

$$S_{P,\mathrm{SN}} = \frac{2\hbar\omega\,\overline{P}}{\mathrm{QE}}, \tag{2.57}$$

where QE is the photodiode quantum efficiency. For unit quantum efficiency, this reduces to the quantum limit

$$S_{P,\mathrm{QL}} = 2\hbar\omega\,\overline{P}. \tag{2.58}$$

Equation (2.58) was used as the reference shot-noise level for comparison with the measured optical-power PSD, both expressed with the same single-sided spectral convention and in units of W^2/Hz. A measured spectrum above this limit indicates excess technical noise in addition to the fundamental photon shot noise. If the spectra are instead plotted as amplitude spectral densities, the corresponding quantum limit is

$$\sqrt{S_{P,\mathrm{QL}}} = \sqrt{2\hbar\omega\,\overline{P}}, \tag{2.59}$$

with units of $\mathrm{W}/\sqrt{\mathrm{Hz}}$.

2.6 Upper Breadboard Setup and Conveyor Belt Alignment

The implementation of the optical conveyor belt requires the spatial superposition of two counter-propagating optical fields: the bottom-up beam, delivered through the vacuum chamber by the Hollow-Core Photonic Crystal Fiber (HCPCF), and the top-down beam, delivered by the black fiber described in Section 2.4. The combination and alignment of these two optical paths take place on the upper breadboard, located above the main experimental chamber. Achieving a deep and reproducible conveyor-belt potential requires precise mode matching between the two counter-propagating beams and systematic control of their relative alignment along the entire shared optical path.

The experimental work on the upper breadboard was divided into three main phases: the outcoupling and conditioning of the beam emerging from the black fiber; the design and construction of dedicated imaging systems to monitor the spatial modes; and the systematic alignment of the counter-propagating beams to maximize their spatial overlap.

2.6.1 Black-fiber outcoupling and beam conditioning

The light injected into the black fiber on the lower bench (Section 2.4.2) is guided to the upper breadboard, where it is outcoupled, conditioned, and combined with the counter-propagating HCPCF beam. The upper-breadboard branch, shown in Figure 2.15, consists of:

- the black-fiber outcoupling assembly, mounted on a custom-made support shortened by 3 mm compared to its original design to increase mechanical rigidity and pointing stability ([**TODO: support model and mounting details**]);
- an $f = 19\,\mathrm{mm}$ collimating lens, positioned manually on a high-stability mount ([**TODO: lens and mount model**]), collimating the diverging beam emerging from the fiber;
- a half-wave plate (HWP) on a rotation mount, used to adjust the polarization of the collimated beam, and a polarizing beam splitter (PBS);
- a beam expander providing a magnification of approximately 3× ([**TODO: beam-expander model and constituent focal lengths**]), matching the transverse size of the black-fiber beam to the expected size of the beam emerging from the HCPCF – a fundamental prerequisite for achieving a high-contrast interference pattern in the conveyor-belt configuration;

- two steering mirrors, directing the expanded beam toward the dichroic mirror ([**TODO: mirror models**]);
- another set of HWP and PBS, used to adjust the polarization of the top-down beam before it enters the vacuum chamber, ensuring that it matches the polarization of the bottom-up beam emerging from the HCPCF;
- a dichroic mirror, reflecting the 1064 nm light downward, into the vacuum chamber ([**TODO: dichroic mirror model and reflection/transmission specifications**]);
- an $f = 200$ mm lens, mounted along the vertical axis downstream of the dichroic mirror, fulfilling a dual purpose: it focuses the top-down beam from the black fiber onto the upper tip of the HCPCF, while simultaneously collimating the counter-propagating bottom-up beam emerging from the same hollow-core fiber.

Because the optical components on the upper breadboard affect the polarization state of the light, additional waveplates and polarizers are required in the definitive layout to ensure that both counter-propagating beams share the same polarization axis ([**TODO: waveplate models and polarization-alignment procedure**]).

2.6.2 Beam propagation and constraints of the optical elements

The design of the shared optical path requires a careful evaluation of the beam size at each stage. The expected waist radius w_{col} of the bottom-up collimated beam can be estimated from the nominal mode radius within the hollow core, $w_{\mathrm{HCPCF}} \approx 19\,\mu$m, using the relation

$$w_{\mathrm{col}} = \frac{\lambda f}{\pi w_{\mathrm{HCPCF}}}, \tag{2.60}$$

where λ is the wavelength and f is the focal length of the vertical lens. For $\lambda = 1064$ nm and $f = 200$ mm, Equation (2.60) gives $w_{\mathrm{col}} \approx 3.6$ mm, corresponding to a full beam diameter $d_{\mathrm{col}} = 2w_{\mathrm{col}} \approx 7.2$ mm.

This collimated beam is steered by standard 1-inch (25.4 mm) mirrors. At an incidence angle of 45°, the beam footprint on the mirror surface becomes elliptical, with major axis d^* given by

$$d^* = \frac{d_{\mathrm{col}}}{\cos(45°)} \approx 10.2\,\mathrm{mm}. \tag{2.61}$$

Although this footprint is mathematically smaller than the clear aperture of a 1-inch mirror, the available clearance is relatively modest. If the effective mode radius in the

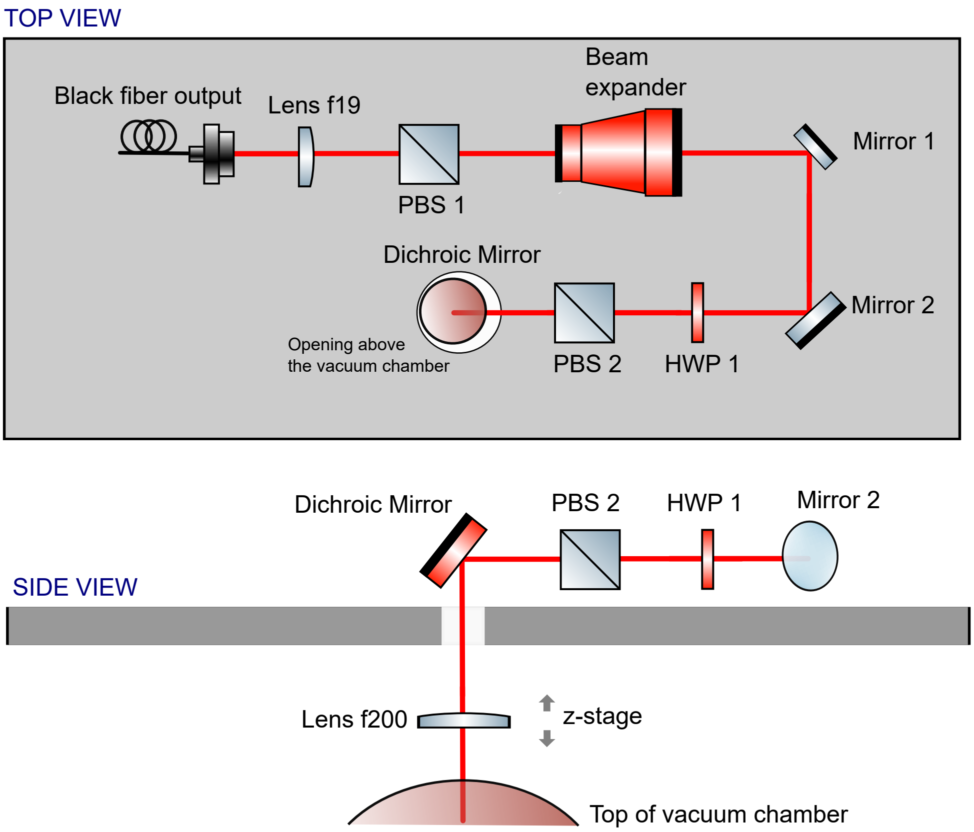


Figure 2.15: *Optical scheme of the upper-breadboard branch, from the black-fiber outcoupling assembly to the shared* $f = 200\,\mathrm{mm}$ *lens where the top-down and bottom-up beams are combined above the vacuum chamber.* ***[TODO: verify AOM orientation in the schematic]***

HCPCF is smaller than the assumed 19 μm – as could be expected given the 48 μm core diameter – the collimated beam divergence and the resulting mirror footprint would be correspondingly larger. Careful alignment, verified with an infrared viewer, is therefore critical to prevent clipping and vignetting at the edges of the steering mirrors.

2.6.3 Mode imaging and alignment strategy

Aligning this counter-propagating configuration and ensuring optimal mode matching requires simultaneous monitoring of the spatial mode of the HCPCF and optimization of the top-down injection. For this purpose, an imaging system (microscope) was assembled on the upper breadboard, exploiting the shared optical path. The imaging branch, shown in Figure 2.16, consists of:

- a magnetic flippable mirror, inserted between the dichroic mirror and the first steering mirror of the black-fiber branch, picking off the bottom-up beam ([**TODO: mirror model**]);
- an $f = 300\,\mathrm{mm}$ lens ([**TODO: lens model**]);

- a Zeiss microscope objective (16.5 mm, Plan Neofluar $10 \times /0.30$);
- a final $f = 500$ mm focusing lens, projecting the image onto a CMOS camera ([**TODO: camera model and pixel size**]).

This multi-stage magnification setup provides the resolution needed to resolve the spatial distribution of the light exiting the microstructured fiber.

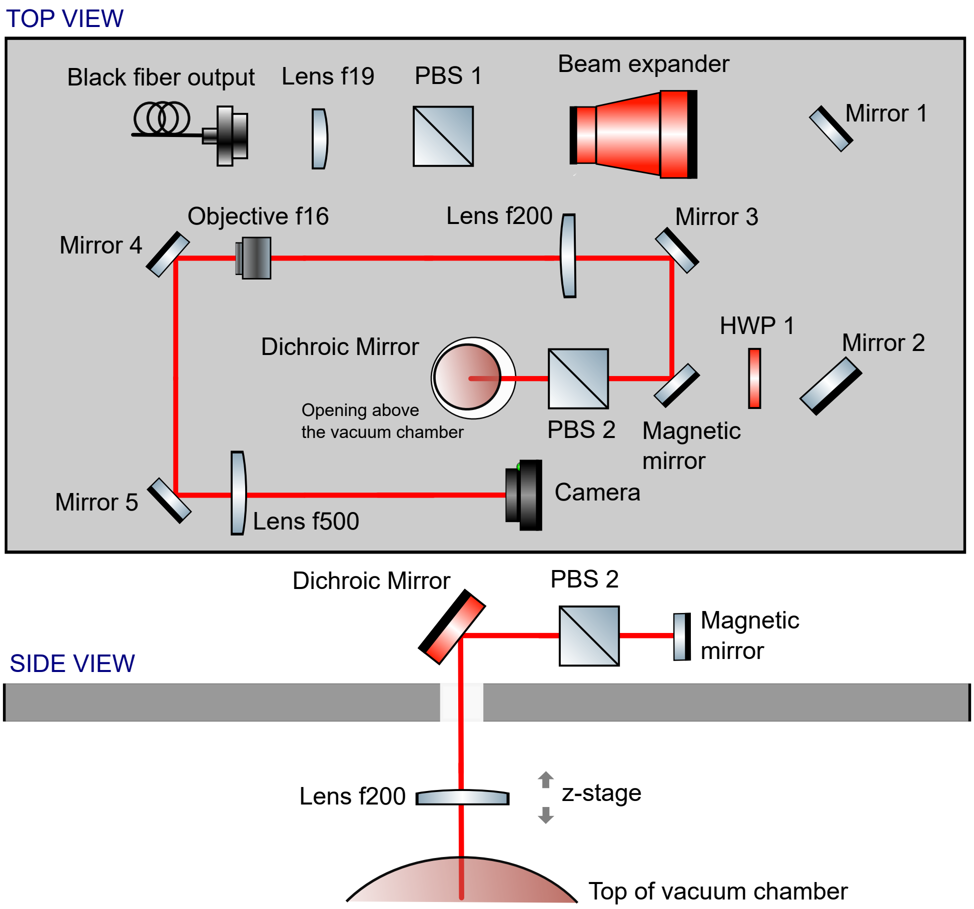


Figure 2.16: *Optical scheme of the imaging (microscope) branch used to monitor the spatial mode of the HCPCF on the upper breadboard.* ***[TODO: verify AOM orientation in the schematic]***

[TODO: insert image of the spatial mode from the HCPCF (bottom-up) acquired with the upper camera; specify in the caption whether the "Pastel" pseudocolor map was used]

The alignment method relies on the assumption that the mode in the HCPCF must be injected symmetrically and propagate accordingly. During the initial observations with the newly constructed microscope, the fundamental mode was found not to propagate symmetrically within the core. By applying gentle mechanical perturbations to the fiber, higher-order propagation modes were intentionally excited; this behavior was systematically exploited to map the spatial boundaries of the hollow core. Once the core region was identified, the fiber position was finely adjusted to ensure that the mode extended symmetrically across the entire available area. This optimization yielded a symmetric fundamental Gaussian mode with an overall bottom-up injection efficiency of approximately 20%–25%.

With the HCPCF mode optimized, the spatial overlap between the top-down and bottom-up beams was verified. A reference mirror, placed after the dichroic mirror, was used to project the beam profiles onto a distant viewing card, checking the collimation of the HCPCF beam; the procedure was repeated for the black-fiber beam, sampling it after the beam expander and at a far-field plane. Both beams were then turned on simultaneously to verify that they were correctly centered on all lenses and mirrors and perfectly overlapped along the entire shared path. Substituting the magnetic mirror with a beam splitter ([**TODO: beam-splitter model and transmission ratio**]) made it possible to operate the microscope continuously, exploiting the small transmission of the dichroic mirror at 1064 nm to simultaneously optimize the injection and monitor the mode in real time.

2.6.4 Figures of merit for the injection efficiency

To quantitatively evaluate the optical setup and the alignment procedure, four figures of merit were established and continuously monitored:

1. **Bottom-up HCPCF mode quality**: the symmetry and spatial distribution of the beam emerging from the vacuum chamber, evaluated through the upper-breadboard microscope (Section 2.6.3);

2. **Bottom-up HCPCF injection efficiency**: the ratio of the power transmitted through the vacuum chamber to the incident power measured before the lower injection lens;

3. **Retro-injection into the black fiber**: the coupling efficiency of the bottom-up HCPCF beam into the black fiber. Since the two beams must be perfectly collinear to form the conveyor belt, optimizing this reverse-coupling signal inherently guarantees the spatial overlap of the two modes on the upper breadboard;

4. **Top-down injection into the HCPCF**: the coupling of the beam delivered by the black fiber into the hollow-core fiber, from above, measured by the power transmitted through the vacuum chamber and collected after the lower collimating lens;

5. **Top-down mode quality**: the symmetry and spatial distribution of the beam entering the vacuum chamber from above.

To evaluate the last figure of merit, a secondary microscope was built on the lower optical bench, exploiting the small but non-zero transmission of the lower 1064 nm dichroic mirror. This diagnostic branch consists of:

- a flippable mirror, picking off the light exiting the bottom tip of the HCPCF ([**TODO: mirror model**]);
- an additional steering mirror, directing the beam toward the focusing optics ([**TODO: mirror model**]);
- an $f = 1000\,\mathrm{mm}$ focusing lens, projecting the beam onto a camera ([**TODO: camera model**]).

This diagnostic tool allows the simultaneous optimization of the top-down injection and the real-time monitoring of the spatial mode traversing the vacuum chamber from top to bottom.
Ultimately, the setup utilized polarization control from both the upper and lower benches. The macroscopic alignment of the beams was successfully achieved by controlling their collimation and utilizing the beam expander, while fine spatial control relied on the upper steering mirrors, the fiber injection optics, and the shared $f = 200\,\mathrm{mm}$ lens. Following this systematic optimization, we achieved a high bottom-up mode quality, as verified through the upper microscope, and a bottom-up injection efficiency of 32%. The retro-injection efficiency into the black fiber reached 60% (compared to a baseline direct injection efficiency of 75%). Furthermore, evaluating the beam with the lower microscope, we achieved a high top-down mode quality and a final top-down injection efficiency of 35%.

2.6.5 RF electronics characterization for intensity control

The depth of the optical dipole trap and the dynamics of the conveyor belt require precise control over the optical power of both beams. The optical power in both the main ODT branch and the black-fiber branch is actively controlled using Acousto-Optic Modulators (AOMs), whose Radio-Frequency (RF) drive signals are generated by a Direct Digital Synthesizer (DDS) board, designated DDS3 ([**TODO: DDS3 board manufacturer and model**]).
Prior to integrating the electronics with the optical setup, the DDS outputs were characterized using a spectrum analyzer ([**TODO: spectrum analyzer model**]). The upper channel of the DDS is dedicated to the primary ODT beam, while the lower channel drives the black-fiber branch (ODT black). For both channels, the emitted RF power was measured as a function of the programmed DDS amplitude parameter.
[**TODO: insert plot of RF power (dBm or W) vs. DDS amplitude for both channels, with fitting curves if applicable**]

2.7 Numerical Model of EIT Cooling

The EIT cooling dynamics were simulated using QuTiP by solving the open-system evolution of an atom coupled to a single quantized motional mode. Two models of increasing complexity were implemented. The first was an ideal three-level Λ system, used to validate the EIT cooling mechanism and the numerical implementation against the analytical predictions introduced in Sec. 1.4. The second model retained the complete hyperfine and Zeeman structure of the ^{87}Rb D_2 line and was used to assess the effects of off-resonant excitation, spontaneous-emission branching, optical pumping, and repumping under experimentally relevant conditions.

Both descriptions were formulated within the same Lindblad master-equation framework. They differed only in the dimension and structure of the internal atomic Hilbert space. The motion was treated in one spatial dimension throughout this work. The simulated direction was selected by assigning to each laser the projection of its wavevector along the corresponding trap axis.

2.7.1 Internal Atomic Basis

In the reduced model, the internal Hilbert space was spanned by two ground states, $|g_1\rangle$ and $|g_2\rangle$, and one excited state, $|e\rangle$. A weak probe field with Rabi frequency Ω_p coupled $|g_1\rangle$ to $|e\rangle$, while a stronger coupling field with Rabi frequency Ω_c coupled $|g_2\rangle$ to $|e\rangle$. At two-photon resonance, this model supports the dark state of Eq. (1.54) and therefore reproduces the ideal carrier suppression discussed in Sec. 1.4. The three-level calculation was used to test the Hamiltonian construction, the motional coupling, the collapse operators, and the time-evolution routines before introducing the complete atomic structure.

The multilevel model included all hyperfine and Zeeman sublevels of the $5S_{1/2}$ and $5P_{3/2}$ manifolds of ^{87}Rb. The internal basis was written as $|F, m_F\rangle$ and comprised the ground manifolds

$$F = 1, \qquad F = 2, \tag{2.62}$$

containing three and five Zeeman states, respectively, and the excited manifolds

$$F' = 0, 1, 2, 3, \tag{2.63}$$

containing a total of sixteen states. The dimension of the internal Hilbert space was therefore

$$\dim(\mathcal{H}_{\mathrm{at}}) = 3 + 5 + 16 = 24. \tag{2.64}$$

All dipole-allowed couplings were constructed explicitly. For a transition between a ground state $|F_g, m_g\rangle$ and an excited state $|F_e, m_e\rangle$, driven by a spherical polarization component $q \in \{-1, 0, +1\}$, the coupling strength was calculated from the corresponding hyperfine dipole matrix element. It was written in the form

$$d_{eg}^{(q)} = \langle F_e, m_e | d_q | F_g, m_g \rangle = \langle F_e \| d \| F_g \rangle (-1)^{F_e - m_e} \begin{pmatrix} F_e & 1 & F_g \\ -m_e & q & m_g \end{pmatrix}, \tag{2.65}$$

where the reduced hyperfine matrix element contains the appropriate Wigner-6j coefficient and the reduced fine-structure matrix element. This construction accounts both for the Zeeman Clebsch–Gordan factors and for the relative strengths of transitions involving different hyperfine manifolds. Transitions that do not satisfy the electric-dipole selection rules were assigned zero coupling.

To reduce optical pumping into states outside the EIT cycle, the target Λ system was built around states near the edge of a Zeeman manifold rather than around $m_F = 0$. In particular, configurations involving $m_F = -2$ were considered because the condition $\Delta m_F = 0, \pm 1$ restricts the number of ground states accessible by spontaneous decay. Two level configurations were implemented. In the first, both EIT ground states belonged to the $F = 2$ manifold, while a repump laser returned population lost to $F = 1$. In the second, the two EIT ground states belonged to different hyperfine manifolds, $F = 1$ and $F = 2$, and the repump field recovered population accumulated in unwanted states outside this pair. The latter configuration was subsequently used for the simulation with the axial trap parameters.

2.7.2 Motional Hilbert Space

The atomic motion along the selected trap axis was approximated by a harmonic oscillator of angular frequency ν. The total Hilbert space was

$$\mathcal{H} = \mathcal{H}_{\text{at}} \otimes \mathcal{H}_{\text{mot}}, \tag{2.66}$$

where

$$\mathcal{H}_{\text{mot}} = \text{span}\{|0\rangle, \dots, |N_{\text{vib}} - 1\rangle\}. \tag{2.67}$$

The motional Hamiltonian was

$$H_{\text{mot}} = \hbar\nu \left(a^\dagger a + \frac{1}{2} \right), \tag{2.68}$$

with a and $a^\dagger$ the annihilation and creation operators. The constant zero-point contribution was omitted in the numerical implementation because it does not affect the dynamics.

The truncation N_{vib} was chosen by balancing numerical cost against convergence of the motional distribution. For each time-dependent calculation, convergence was assessed by verifying that the population of the highest retained Fock states remained negligible and that increasing N_{vib} did not appreciably change the calculated mean occupation,

$$\langle n(t)\rangle = \mathrm{Tr}\Big[\rho(t)\left(\mathbb{I}_{\mathrm{at}} \otimes a^\dagger a\right)\Big]\,. \tag{2.69}$$

2.7.3 Laser Coupling and Lamb–Dicke Expansion

For a laser field l with wavevector $\mathbf{k}_l$, optical phase ϕ_l, and polarization components $\epsilon_{l,q}$, the position-dependent rotating-wave interaction was written as

$$H_l = \frac{\hbar}{2}\sum_{g,e,q} \Omega_l \epsilon_{l,q}\, \mathcal{C}_{eg}^{(q)} |e\rangle\langle g|\, e^{ik_{l,\parallel}x + i\phi_l} + \mathrm{h.c.}, \tag{2.70}$$

where $\mathcal{C}_{eg}^{(q)}$ denotes the normalized transition amplitude obtained from Eq. (2.65), and

$$k_{l,\parallel} = \mathbf{k}_l \cdot \hat{\mathbf{e}}_{\mathrm{mot}} \tag{2.71}$$

is the projection of the laser wavevector along the simulated motional axis.
Using

$$x = x_0(a + a^\dagger), \qquad x_0 = \sqrt{\frac{\hbar}{2m\nu}}, \tag{2.72}$$

the Lamb–Dicke parameter associated with field l was

$$\eta_l = k_{l,\parallel} x_0. \tag{2.73}$$

In the Lamb–Dicke regime, the position-dependent phase was expanded to first order,

$$e^{ik_{l,\parallel}x} \simeq 1 + i\eta_l(a + a^\dagger). \tag{2.74}$$

This expansion retains the carrier and first motional sidebands while neglecting higher-order processes.
For a Raman process involving absorption from one field and stimulated emission into another, the relevant two-photon momentum transfer is determined by

$$\Delta\mathbf{k} = \mathbf{k}_p - \mathbf{k}_c, \tag{2.75}$$

in agreement with the discussion in Sec. 1.4. The corresponding effective Lamb–Dicke parameter along the simulated direction is

$$\eta_{\mathrm{eff}} = (\mathbf{k}_p - \mathbf{k}_c)\cdot \hat{\mathbf{e}}_{\mathrm{mot}}\, x_0. \tag{2.76}$$

In the three-level benchmark, the coupling beam was taken to propagate orthogonally to the motional axis, so that $\eta_c = 0$, while the probe field carried the motional coupling. In the multilevel model, an independent value of η_l could be assigned to each beam according to the experimental geometry.

The first-order expansion is valid when the occupied motional states satisfy

$$\eta_l \sqrt{2\langle n \rangle + 1} \ll 1. \tag{2.77}$$

Calculations initialized outside this regime were treated only as qualitative descriptions of the initial transient. In such cases, the linearized model cannot be regarded as quantitatively exact until the motional occupation has decreased sufficiently.

2.7.4 Hamiltonian in the Rotating Frame

After transformation to a rotating frame and application of the rotating-wave approximation, the Hamiltonian was expressed as

$$H = H_{\mathrm{at}} + H_{\mathrm{mot}} + H_{\mathrm{Z}} + \sum_l H_l. \tag{2.78}$$

Here, H_{at} contains the hyperfine energies and laser detunings in the selected rotating frame, and

$$H_{\mathrm{Z}} = \sum_i g_{F_i} \mu_B B m_{F_i} |i\rangle\langle i| \otimes \mathbb{I}_{\mathrm{mot}} \tag{2.79}$$

describes the linear Zeeman shift produced by the quantization field B. The field was assumed sufficiently weak that the hyperfine quantum numbers F and m_F remained appropriate labels.

For the three-level system, the implemented Hamiltonian was

$$\begin{aligned} H_\Lambda = {} & \hbar\Delta_p |g_1\rangle\langle g_1| + \hbar\Delta_c |g_2\rangle\langle g_2| + \hbar\nu a^\dagger a \\ & + \frac{\hbar\Omega_c}{2} \left(|e\rangle\langle g_2| + |g_2\rangle\langle e| \right) \\ & + \frac{\hbar\Omega_p}{2} \left[|e\rangle\langle g_1| \left(1 + i\eta_p (a + a^\dagger) \right) + \mathrm{h.c.} \right]. \end{aligned} \tag{2.80}$$

In the multilevel model, Eq. (2.70) was constructed separately for the probe, coupling, and repump fields. Each laser was assigned its own Rabi frequency, detuning, polarization, and projected wavevector. The repump field was introduced to return population from ground states outside the intended Λ subsystem. Its motional factor was set to the identity in the simulations reported here,

$$e^{ik_{\mathrm{rep},\parallel} x} \to \mathbb{I}_{\mathrm{mot}}, \tag{2.81}$$

so that repump-induced motional sidebands were neglected. This approximation reduces the numerical complexity and isolates the repumper's internal-state recycling function. It also neglects recoil heating associated with repump absorption and must therefore be regarded as an optimistic approximation when repump scattering is appreciable.

2.7.5 Spontaneous Emission and Master Equation

The density operator evolved according to

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H,\rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2} L_k^\dagger L_k \rho - \frac{1}{2} \rho L_k^\dagger L_k \right). \tag{2.82}$$

For the three-level system, symmetric branching into the two ground states was assumed:

$$L_1 = \sqrt{\frac{\gamma}{2}}\, |g_1\rangle\langle e| \otimes \mathbb{I}_{\mathrm{mot}}, \qquad L_2 = \sqrt{\frac{\gamma}{2}}\, |g_2\rangle\langle e| \otimes \mathbb{I}_{\mathrm{mot}}. \tag{2.83}$$

For the multilevel system, the collapse operator for a decay channel $e \to g$ was written as

$$L_{e\to g} = \sqrt{\gamma_e b_{eg}}\, |g\rangle\langle e| \otimes \mathbb{I}_{\mathrm{mot}}, \tag{2.84}$$

where γ_e is the total decay rate of the excited state and b_{eg} is the normalized branching fraction,

$$b_{eg} = \frac{\sum_q \left| d_{eg}^{(q)} \right|^2}{\sum_{g',q} \left| d_{eg'}^{(q)} \right|^2}, \qquad \sum_g b_{eg} = 1. \tag{2.85}$$

This normalization ensures that

$$\sum_g L_{e\to g}^\dagger L_{e\to g} = \gamma_e |e\rangle\langle e|. \tag{2.86}$$

In the simulations described here, the spontaneous-emission operators acted trivially on the motional subspace. Consequently, the recoil associated with the emitted photon and the corresponding diffusion in phonon number were not explicitly included. The model therefore accounts for motional changes driven by laser absorption and stimulated coupling, but not for recoil heating during spontaneous emission. Within the Lamb–Dicke regime this correction enters at order η^2, but it can affect the final occupation and should be included when a quantitative comparison with the theoretical limit of Eq. (1.68) or with experimental temperatures is required. A recoil-resolved extension can be obtained by replacing the identity in Eq. (2.84) with an angularly dependent operator $e^{-i\mathbf{k}_{\mathrm{em}}\cdot\mathbf{x}}$ and averaging over the dipole emission pattern.

2.7.6 Steady-State Spectroscopy

The EIT spectrum was calculated by scanning the probe detuning while holding the remaining laser parameters fixed. At each detuning, the stationary density operator was obtained from

$$\mathcal{L}_{\mathrm{tot}}(\rho_{\mathrm{ss}}) = 0, \qquad \mathrm{Tr}(\rho_{\mathrm{ss}}) = 1, \tag{2.87}$$

where $\mathcal{L}_{\mathrm{tot}}$ denotes the full Liouvillian superoperator.

The principal spectral observable was the total excited-state population,

$$P_e(\Delta_p) = \mathrm{Tr}\left[\rho_{\mathrm{ss}}(\Delta_p)\sum_e |e\rangle\langle e|\right]. \tag{2.88}$$

In the weak-probe regime, this quantity is proportional to the total photon-scattering rate and was therefore used as an absorption proxy. It should not, however, be identified with the imaginary part of the probe susceptibility when the probe is strong or when several driven transitions contribute simultaneously.

For the multilevel calculation, additional projectors were evaluated to distinguish excitation and population leakage:

$$P_{F'=2} = \sum_{e\in F'=2} |e\rangle\langle e|, \tag{2.89}$$

$$P_{F'=3} = \sum_{e\in F'=3} |e\rangle\langle e|, \tag{2.90}$$

$$P_{\mathrm{out}} = \sum_{g\notin\mathcal{G}_\Lambda} |g\rangle\langle g|, \tag{2.91}$$

where $\mathcal{G}_\Lambda$ is the pair of ground states defining the target Λ system. These observables were used to identify the dark resonance, quantify off-resonant excitation, and determine whether population accumulated in ground states not directly addressed by the EIT beams.

The carrier and first motional sidebands were located relative to the two-photon resonance using the detuning convention of the numerical Hamiltonian. Their assignment was checked directly from the matrix elements connecting $|g,n\rangle$ to $|e,n\rangle$ and $|e,n\pm 1\rangle$, rather than inferred only from the sign of the plotted detuning. This step was necessary because the labels "red" and "blue" sideband reverse under a change in detuning convention.

2.7.7 Time-Dependent Cooling Dynamics

Cooling was quantified by propagating an initial atom–motion state and evaluating Eq. (2.69). The initial internal state was selected within the active ground-state manifold. The motional state was specified either as a Fock state or as a thermal mixture,

$$\rho_{\text{th}} = \sum_{n=0}^{N_{\text{vib}}-1} p_n |n\rangle\langle n|, \qquad p_n = \frac{\bar{n}_0^n}{(1+\bar{n}_0)^{n+1}}, \tag{2.92}$$

with the probabilities renormalized after truncation when necessary.

The three-level model was propagated using QuTiP's deterministic master-equation solver `mesolve`. This method evolves the density operator and gives noise-free ensemble expectation values. For the full multilevel model, the Monte Carlo wavefunction solver `mcsolve` was used. It propagates stochastic pure-state trajectories interrupted by quantum jumps and estimates an observable as

$$\langle O(t)\rangle \simeq \frac{1}{N_{\text{traj}}} \sum_{j=1}^{N_{\text{traj}}} \langle \psi_j(t)|O|\psi_j(t)\rangle. \tag{2.93}$$

The statistical uncertainty decreases approximately as $N_{\text{traj}}^{-1/2}$ for independent trajectories. For an internal dimension N_{at} and a motional cutoff N_{vib}, the state-vector dimension is

$$N = N_{\text{at}} N_{\text{vib}}. \tag{2.94}$$

A density matrix contains N^2 complex entries, whereas a Monte Carlo trajectory stores a state vector with N entries. The trajectory method therefore substantially reduced memory use for the 24-level calculation. The computational time of either solver also depends on the sparsity of the Hamiltonian, the number of collapse channels, the stiffness of the differential equations, and the number of trajectories; it is therefore not described by memory scaling alone.

For Monte Carlo calculations, full state storage was disabled and only the required expectation values were retained. Independent trajectories were distributed across the available processor cores. Numerical convergence was assessed by increasing the trajectory number and comparing the resulting cooling curves within their statistical fluctuations. The output time grid was chosen independently of the adaptive internal integration steps used by the solver.

2.7.8 Units and Conversion to Physical Time

The numerical calculations were performed in units of the natural angular linewidth,

$$\gamma = 1, \tag{2.95}$$

and $\hbar = 1$. All detunings, Rabi frequencies, hyperfine splittings, Zeeman shifts, and trap frequencies supplied to the Hamiltonian were therefore divided by γ. Time was expressed

in units of γ^{-1}.
For the ^{87}Rb D_2 transition, the linewidth was taken as

$$\frac{\gamma}{2\pi} = 6.067 \text{ MHz}, \tag{2.96}$$

so that one simulation time unit corresponds to

$$\gamma^{-1} = \frac{1}{2\pi \times 6.067 \text{ MHz}} \simeq 26.2 \text{ ns}. \tag{2.97}$$

Accordingly, a dimensionless interval t_{sim} was converted to physical time through

$$t_{\text{phys}} = \frac{t_{\text{sim}}}{\gamma}. \tag{2.98}$$

The same angular-frequency convention was used for ν, Ω_l, Δ_l, and Zeeman shifts to avoid factors of 2π within the Hamiltonian.

2.7.9 Validation of the Three-Level Model

The reduced model was validated in two stages. First, the motional degree of freedom was omitted and the probe detuning was scanned to verify the formation of the EIT transparency minimum at two-photon resonance and the narrow dressed-state resonance displaced by the coupling-induced AC Stark shift of Eq. (1.60). The numerically obtained peak position was compared with

$$\delta = \frac{\sqrt{\Delta_c^2 + \Omega_c^2} - |\Delta_c|}{2}. \tag{2.99}$$

Second, the motional basis was restored and the coupling intensity was selected to satisfy the EIT cooling condition. For the sign convention and blue-detuned configuration considered in Sec. 1.4, the exact relation is given by Eq. (1.62),

$$\Omega_c^2 = 4\nu(\nu + \Delta_c), \tag{2.100}$$

which reduces to

$$\Omega_c^2 \simeq 4\Delta_c\nu \tag{2.101}$$

when $\nu \ll \Delta_c$. Under this condition, the narrow absorption resonance is aligned with the phonon-removing sideband, while carrier excitation is suppressed by the dark resonance. The simulated late-time occupation and cooling rate were compared with the analytical rate-equation results of Eqs. (1.65)–(1.68). Agreement was expected only in the weak-probe, unsaturated, Lamb–Dicke, and sufficiently large motional-basis limits. The omission of spontaneous-emission recoil from the numerical collapse operators was also taken into account when interpreting deviations from the analytical steady-state limit.

2.7.10 Optimization of the Multilevel Configuration

The multilevel laser parameters were optimized using steady-state spectra before performing the more expensive time-dependent simulations. For each parameter set, the probe detuning was scanned and the following quantities were recorded: the total excited-state population, the populations of the target and off-resonant excited manifolds, the population outside the active Λ subsystem, the detuning of the transparency minimum, and the detuning of the nearest absorption maximum.
The scans were performed sequentially:

1. The repump Rabi frequency was varied to determine the minimum intensity required to prevent long-lived population accumulation outside the EIT cycle. Excessive repump power was avoided because off-resonant coupling can shift and broaden the EIT spectrum.

2. The probe Rabi frequency was varied within the weak-probe regime. Increasing Ω_p increases the scattering signal and may increase the cooling rate, but it also saturates and power-broadens the dark resonance, increasing carrier scattering.

3. The coupling detuning and coupling Rabi frequency were adjusted to align the narrow absorption maximum with the phonon-removing sideband after accounting for the light shifts introduced by the complete multilevel structure and by the repump field.

The optimization criterion was not the height of the absorption peak alone. Parameter sets were selected to provide low carrier excitation, enhanced scattering at the cooling sideband, reduced scattering at the heating sideband, and limited population leakage. The selected parameters were then tested through a time-dependent `mcsolve` calculation of $\langle n(t)\rangle$. Because the scans were sequential rather than multidimensional, the resulting set should be interpreted as an operational optimum within the explored ranges, rather than as a unique global optimum.

2.7.11 Simulation with Axial Experimental Parameters

The configuration in which the two EIT ground states belong to different hyperfine manifolds was subsequently evaluated using the axial trap parameters of the experimental system. The axial trap frequency was inserted directly into the motional Hamiltonian, while the Lamb–Dicke parameters of the probe and coupling beams were calculated from their wavevector projections according to Eq. (2.73). The magnetic field, laser polarizations, hyperfine splittings, detunings, and Rabi frequencies were retained explicitly in the 24-level Hamiltonian.

The steady-state probe scan was repeated for the axial parameters because changing ν changes the position of the target motional sideband and therefore the required dressed-state shift. The probe, coupling, and repump parameters were adjusted until the absorption maximum was aligned with the axial phonon-removing sideband while the carrier remained close to the transparency minimum. The resulting parameter set was then used in the Monte Carlo time evolution.

For this calculation, convergence was checked with respect to the motional cutoff, the number of trajectories, and the output time interval. The cooling time was extracted by fitting the ensemble-averaged occupation to

$$\langle n(t)\rangle = n_\infty + (n_0 - n_\infty)\, e^{-Wt}, \tag{2.102}$$

over the interval in which a single-exponential description was adequate. Here, W is the fitted cooling rate and n_∞ is the asymptotic mean occupation. When the curve exhibited multiple timescales due to optical pumping among internal states, Eq. (2.102) was used only as an effective characterization and the fitted interval was reported together with the result.

2.7.12 Adaptation to In-Fiber Cooling

To extend the model to cooling inside the hollow-core photonic crystal fiber, the spatially dependent differential AC Stark shift derived in Sec. 1.4 must be included in the internal Hamiltonian. For a transverse atomic position $\mathbf{r}$, the trap-induced shift of each internal state is

$$H_{\mathrm{trap,AC}}(\mathbf{r}) = \sum_i U_i(\mathbf{r})|i\rangle\langle i|, \tag{2.103}$$

with $U_i(\mathbf{r})$ determined by the dynamic polarizability at the trapping wavelength. The local optical detuning then becomes

$$\Delta_l(\mathbf{r}) = \Delta_l^{(0)} - \frac{U_e(\mathbf{r}) - U_g(\mathbf{r})}{\hbar}. \tag{2.104}$$

A direct three-dimensional treatment would require simultaneous quantization of the axial and two transverse motional modes and would substantially increase the Hilbert-space dimension. In the present one-dimensional framework, the leading effect of the transverse motion can instead be incorporated by averaging the local steady-state or sideband rates over the transverse spatial distribution:

$$\overline{A}_{\pm} = \int d^2r\, p_T(\mathbf{r})\, A_{\pm}[\Delta_l(\mathbf{r}), \Omega_l(\mathbf{r}), \nu(\mathbf{r})]\,, \tag{2.105}$$

where $p_T(\mathbf{r})$ is the thermal transverse probability density. In the harmonic approximation,

$$p_T(r) = \frac{m\nu_r^2}{2\pi k_B T} \exp\left(-\frac{m\nu_r^2 r^2}{2k_B T}\right). \tag{2.106}$$

This procedure incorporates the inhomogeneous broadening estimated in Eq. (1.77) without explicitly adding two further harmonic-oscillator Hilbert spaces.
For each sampled position, the local probe and coupling detunings, Rabi frequencies, and trap frequencies must be recalculated from the fiber-mode intensity. The on-axis laser parameters can then be chosen to satisfy the modified resonance condition discussed in Sec. 1.4, while Eq. (2.105) quantifies the degradation caused by atoms sampling different local light shifts. This extension provides the connection between the free-space multilevel simulation and the in-fiber cooling geometry.

2.7.13 Scope and Approximations

The numerical model includes the full internal hyperfine and Zeeman structure, coherent coupling by the applied laser fields, first-order motional sidebands for the EIT beams, and all dipole-allowed spontaneous-emission branches. The main approximations are the restriction to one motional dimension, harmonic confinement, the first-order Lamb–Dicke expansion, omission of spontaneous-emission recoil, neglect of repump motional coupling, and sequential rather than global parameter optimization. The idealized three-level results are therefore used primarily as a validation of the EIT mechanism, while the 24-level calculations quantify the internal-state effects relevant to the experimental implementation. Quantitative predictions for the final in-fiber temperature require the spatial averaging of Eq. (2.105), together with recoil-resolved collapse operators and experimentally calibrated position-dependent light shifts.

Chapter 3

Results and Discussion

Following the experimental and analysis procedures explained in Chapter 2, the ODT capture efficiency was characterized and optimized. This chapter presents the corresponding experimental results.

The first stage of the optimization process consists of minimizing the molasses temperature through the adjustment of the cooling laser detuning, the cooling laser intensity, and the compensation magnetic field.

Subsequently, the ODT loading is optimized by exploiting the compensation coils to displace the MOT after loading. This optimization aims to maximize the spatial overlap between the atomic cloud and the ODT beam, thereby enhancing the capture efficiency.

3.1 Molasses Temperature Optimization

Before starting the optimization, the six MOT beams were realigned to remove any residual overlap mismatch at the trap position — a necessary precaution, since thermal drifts in the optical components tend to slowly spoil this alignment over time. The compensation coil currents were initially set to the values found in earlier characterizations of the apparatus,

$$I_x = 80\,\mathrm{mA}, \qquad I_y = -200\,\mathrm{mA}, \qquad I_z = -400\,\mathrm{mA}, \tag{3.1}$$

which served as the starting point for the systematic optimization described below.

3.1.1 Detuning Frequency Optimization

The optimization began with the cooling-laser detuning, scanned from $-24.4\,\mathrm{MHz}$ ($4.0\,\Gamma/2\pi$) to $-184\,\mathrm{MHz}$ ($30.4\,\Gamma/2\pi$) in steps of $-11.4\,\mathrm{MHz}$, and the molasses temperature measured at each point.

At small detunings the measured temperature closely tracks the $1/|\delta|$ dependence expected for sub-Doppler (Sisyphus) cooling. This trend breaks down as the detuning grows further: the scattering rate falls off as $\Gamma_p \propto 1/\delta^2$, and the detuning approaches the $F = 2 \to F' = 2$ transition, only 267 MHz below the main cooling transition $F = 2 \to F' = 3$. Both effects flatten the temperature curve in the range $|\delta| \approx$ 15–20 and eventually turn it upward again at larger detunings.

The lowest temperature was found at a normalized detuning of $|\delta| = 17.2$, i.e. 104.2 MHz in absolute terms. Fitting the ballistic expansion at this detuning gave $T_x = (14.1 \pm 0.5)\,\mu$K and $T_y = (13.5 \pm 0.7)\,\mu$K along the two axes, whose weighted average yields a final temperature of $T = (13.9 \pm 0.4)\,\mu$K.

The fit at this detuning shows a systematic feature common to all such measurements: the data points at the longest times of flight fall consistently below the linear trend expected from the expansion law. The most likely explanation is that, at sufficiently long expansion times, the cloud radius is no longer small compared to the probe beam waist, so the assumption of uniform illumination breaks down and the outer, dimmer part of the cloud is under-weighted in the fit — an effect that systematically biases the extracted variance downward. Fortunately, this bias is expected to become less severe as the temperature (and hence the expansion rate) decreases further.

3.1.2 AOM Amplitude Optimization

The next optimization step examined the influence of the cooling-beam power itself, controlled via the RF drive amplitude of the switching AOM, together with the duration of the ramp used to transition this amplitude from its MOT value down to the molasses value at the start of the molasses phase. Three amplitude settings were tested — 385, 475, and 550 a.u., corresponding to 5.8, 6.6, and 6.3 mW of cooling power measured after the z-axis MOT collimator (the MOT itself is operated at 550 a.u.) — combined with ramp durations of either 5 ms or 20 ms, and the whole set was repeated across several values of the cooling detuning.

No statistically significant dependence of the molasses temperature on the AOM amplitude emerged from this scan — the small variations observed between the three settings remained within the experimental uncertainty, so no clear optimal amplitude could be identified in the tested range. The ramp duration, on the other hand, had a small but systematic effect: the 20 ms ramp consistently gave temperatures about $0.5\,\mu$K lower than the 5 ms ramp. Given that this improvement was already small and no further gain at the level of a few tenths of a microkelvin was required for the purposes of this work, no additional ramp durations were tested, and the 20 ms ramp was adopted for the subsequent measurements.

It is worth noting, anticipating the ODT characterization described later in this chapter, that this particular choice was later found to introduce extra fluctuations once the ODT loading was being optimized. For that reason, the compensation-coil optimization described next was actually carried out with the AOM amplitude set to 385 a.u. — the setting that gave the lowest temperature at $|\delta| = 17.2\,\Gamma/2\pi$ — together with the 20 ms ramp, even though both parameters were revisited again during the ODT optimization stage. As shown by the measurements in this section, however, their influence on the molasses temperature itself is at most a few tenths of a microkelvin, so this later readjustment does not affect the temperature results reported here.

3.1.3 Compensation Field Optimization

The last step of the molasses optimization concerned the compensation (shim) magnetic field, whose currents are switched from their MOT values to their molasses values 5 ms after the MOT quadrupole field itself is turned off. The three axes were optimized sequentially: first the z-axis current, with the x- and y-axis currents held at their preliminary values; then the x-axis current, using the newly found optimal z value; and finally the y-axis current, with both z and x set to their respective optima. At each step, a parabolic fit to the measured temperature versus current was used to extract the optimal current and the corresponding minimum temperature.

The z-axis scan gave a minimum temperature of $(11.46 \pm 0.03)\,\mu\mathrm{K}$ at $I_z = (-223.2 \pm 0.7)\,\mathrm{mA}$; the subsequent x-axis scan, performed at this z current, gave $(9.2 \pm 0.1)\,\mu\mathrm{K}$ at $I_x = (-17 \pm 2)\,\mathrm{mA}$; and the final y-axis scan gave $(9.6 \pm 0.1)\,\mu\mathrm{K}$ at $I_y = (-184 \pm 3)\,\mathrm{mA}$. The currents actually adopted for the subsequent ODT loading stage were

$$I_x = -15\,\mathrm{mA}, \qquad I_y = -185\,\mathrm{mA}, \qquad I_z = -230\,\mathrm{mA}, \tag{3.2}$$

corresponding, via the coil calibration given in Eq. (**??**), to compensation fields of

$$B_x = -21\,\mathrm{mG}, \qquad B_y = -241\,\mathrm{mG}, \qquad B_z = -345\,\mathrm{mG}. \tag{3.3}$$

With this compensation field, the molasses temperature reached its minimized value of $T = (9.6 \pm 0.1)\,\mu\mathrm{K}$. Comparing the three axes, the z-axis current required the largest correction from its preliminary value, the x-axis current a smaller but still appreciable one, while the y-axis current turned out to already be close to its optimal value beforehand.

The operating current actually used for z, $-230\,\mathrm{mA}$, differs slightly from the true fitted optimum of $-223.2\,\mathrm{mA}$; since both values sit close to the vertex of the fitted parabola, where the temperature is only weakly sensitive to current, this small offset does not introduce any appreciable systematic error in the reported temperatures.

3.2 ODT Loading Optimization

With the molasses parameters established in the previous section, the next stage of the optimization targets the ODT loading itself, primarily by tuning the compensation magnetic field. The idea behind this step is that the MOT forms at the zero of the magneto-optical potential, whose location can be shifted by adjusting the compensation coil currents; displacing the trapped cloud in this way, immediately before the molasses phase, allows its position to be matched to that of the ODT beam. Moving the MOT away from the geometric center where the six cooling beams overlap best inevitably costs some atoms and raises the temperature slightly, but this trade-off is worthwhile if it improves the spatial overlap with the ODT and thereby increases the number of atoms actually transferred into the dipole trap. As long as the displacement stays small, the molasses parameters optimized earlier remain effective at cancelling the residual magnetic field and require no further adjustment.

In practice, once the MOT has loaded, the compensation coil currents are ramped smoothly to their ODT-loading values over 200 ms, discretized into 20 steps, so as to minimize atom loss and heating during the displacement. As with the molasses optimization, preliminary current values — obtained before the beam realignment and molasses optimization described above — were available as a starting point,

$$I_x = -200\,\mathrm{mA}, \qquad I_y = -350\,\mathrm{mA}, \qquad I_z = 50\,\mathrm{mA}. \tag{3.4}$$

Throughout these measurements, the ODT laser power before injection into the fiber was 1.4 W. The transmission efficiency up to just before lens L3 (see the apparatus schematic in the previous chapter) was measured at low power to be about 42%, corresponding to roughly 0.59 W; since several optics sit upstream of this measurement point, the actual power reaching the fiber facet is likely somewhat higher, so a conservative 10% relative uncertainty was assigned, giving an estimated fiber-facet power of $P = 0.59 \pm 0.06\,\mathrm{W}$.

3.2.1 Imaging System Optimization

The first attempts to characterize the ODT capture efficiency used the FLIR camera, with the molasses parameters already optimized above and the preliminary compensation currents quoted for the ODT stage. The analysis followed the procedure described in the previous chapter, but without the background rescaling step.

With the 20 ms AOM ramp used for the molasses optimization, the resulting 2D difference image showed no distinct bright region attributable to the ODT, and the 1D integrated profile only hinted at a weak, localized increase spanning about 4 pixels around the fiber position — later confirmed, from subsequent measurements, to indeed correspond to the trapped atoms, but too small to be reliably distinguished from noise. Since integrating along one dimension is specifically meant to boost the signal-to-noise ratio, this persistent weak signal pointed to a real underlying problem rather than a simple statistical limitation: the residual background pedestal left after subtraction was significantly larger than the ODT signal itself, suggesting oscillatory or shot-to-shot correlated noise that five-image averaging alone could not remove. In addition, the ODT signal spanned only a handful of pixels, which is both consistent with a lower-than-expected initial capture (even accounting for a non-optimal cloud-ODT overlap) and, independently, an obstacle to analysis: too few illuminated pixels to support a reliable 2D Gaussian fit, while direct pixel summation would introduce its own systematic errors.

To address the background problem, the molasses parameters themselves were revisited, and the AOM ramp duration was found to be the key culprit: shortening it from 20 ms to 5 ms markedly improved the background subtraction and thus the signal-to-noise ratio. At the same time, the AOM amplitude was set to 480 a.u. (6.6 mW of cooling power on the z-beam), a value chosen because it sits in the saturated part of the AOM diffraction-efficiency curve, where $dP/dA \approx 0$; operating at this point intrinsically suppresses the sensitivity of the delivered optical power to amplitude noise in the AOM drive.

With this shorter ramp, a distinct bright strip corresponding to the ODT-trapped atoms became visible in the 2D difference image, and comparing the 1D profiles obtained with the two ramp durations shows that the residual pedestal is suppressed to roughly half its previous amplitude and area with the 5 ms ramp — enough for the ODT signal to clearly emerge above the background.

Even with this improved contrast, the FLIR camera's spatial resolution remained insufficient. The subsequent measurements therefore switched to the Manta camera, whose smaller pixel pitch was further complemented by mounting it closer to the chamber, at the minimum working distance of the MVL50M23 objective, to maximize magnification. Comparing the apparent width of the fiber as imaged by the two cameras indicated that this new configuration increased the object size in pixels by a factor of about 2.25 — the combined effect of the higher magnification and the smaller pixel size — which finally provided a sufficiently well-resolved ODT signal for robust quantitative analysis.

3.2.2 Capture Efficiency Optimization

With adequate imaging in place, the compensation coil currents were optimized systematically using the full analysis procedure from the previous chapter, this time with the Manta camera. The three axes were scanned sequentially in the order $x \to z \to y$, with the currents on the not-yet-optimized axes held at the preliminary values quoted above.

Both the ODT signal area — directly proportional to the number of captured atoms, though only measurable in arbitrary units since the camera was not calibrated for absolute atom counting — and the capture efficiency extracted from these scans showed matching trends along all three axes, as expected since they are two ways of looking at the same underlying signal. This first pass yielded the optimal currents

$$I_x = -220\,\mathrm{mA}, \qquad I_y = -290\,\mathrm{mA}, \qquad I_z = -90\,\mathrm{mA}. \tag{3.5}$$

To confirm this result, the z-axis scan was repeated, and this second run revealed an unexpected feature.

Rather than the expected single-peaked, roughly parabolic dependence, both the ODT area and the capture efficiency showed an oscillatory behavior as a function of the z-axis current: starting from $-1000\,\mathrm{mA}$, the signal rose to a first local maximum near $-800\,\mathrm{mA}$, dipped around $-700\,\mathrm{mA}$, and rose again to further local maxima near $-600\,\mathrm{mA}$ and $-400\,\mathrm{mA}$. The same periodicity appeared in the background area — the molasses cloud size measured with the ODT off at the end of the time of flight — with peaks at essentially the same current values, and again in the x-centroid of the cloud on the camera, an oscillation not seen when scanning the other two compensation axes. While a larger initial MOT population naturally leads to a larger captured ODT fraction, the fact that this covariation is periodic in current is not obviously explained; a plausible contributing factor is the complex spatial intensity distribution of the cooling beams near the trap center combined with residual misalignment, but confirming this would require dedicated further measurements, which were beyond the scope of this work.

Taking this behavior into account, the compensation currents and the corresponding generated field components were finally set to

$$I_x = -220\,\mathrm{mA}, \quad I_y = -290\,\mathrm{mA}, \quad I_z = -600\,\mathrm{mA}, \qquad B_x = -308\,\mathrm{mG}, \quad B_y = -377\,\mathrm{mG}, \quad B_z = -900 \tag{3.6}$$

Converting these compensation fields into an equivalent displacement of the MOT position via $t_\alpha = B_\alpha / B'$ for $\alpha = x, y$ and $t_z = -B_z/2B'$, with $2B' = 9.32\,\mathrm{G/cm}$ the axial gradient of the quadrupole field, gives

$$t_x = -661\,\mu\mathrm{m}, \qquad t_y = -809\,\mu\mathrm{m}, \qquad t_z = 966\,\mu\mathrm{m}. \tag{3.7}$$

This configuration maximized the ODT trap area — and hence the number of captured atoms — and corresponds to a capture efficiency of $\eta = (0.160 \pm 0.005)\%$.

This measured efficiency can be compared with the shallow- and non-shallow-trap models introduced in the previous chapter, evaluated at the best molasses configuration found above: $w_0 = 19.0\,\mu\mathrm{m}$, $P = (0.59 \pm 0.06)\,\mathrm{W}$, $R_x R_y \approx (1.043 \pm 0.004)\,\mathrm{mm}^2$, and $T = (9.60 \pm 0.10)\,\mu\mathrm{K}$. These give $\eta = (0.45 \pm 0.05)\%$ for the shallow-trap model and $\eta = (0.41 \pm 0.04)\%$ for the non-shallow-trap model; given $\xi = 0.39 \pm 0.04$, the close agreement between the two models is expected. Note that, unlike in the earlier discussion of these models, no factor of 1/2 for the fraction of atoms with $v_z < 0$ was applied here, since that factor was originally introduced to estimate only the atoms guided into the fiber core, whereas the present measurement targets the total number of captured atoms regardless of their velocity direction.

The measured efficiency is of the same order of magnitude as the theoretical prediction, indicating that the apparatus is operating correctly and close to its optimal parameters, but the theoretical value remains about 2.5 times larger than the measured one. Part of this discrepancy may be a genuine model limitation: the calculation assumes a purely Gaussian ODT beam, an excellent approximation for the fundamental LP_{01} mode, but the beam actually transmitted by the fiber also carries higher-order modes, as discussed in the previous chapter. Before attributing the full discrepancy to the model, however, a procedural effect should also be considered: the imaging sequence requires $600\,\mu$s with the ODT switched off at the end of the time of flight, and atom loss during this dead time could artificially lower the measured trapping efficiency.

3.3 Simulation Results

3.3.1 Validation on the Ideal Three-Level System

Before addressing the full atomic structure, the cooling dynamics were first verified on the idealized three-level Λ system, for which the dark-state interference is exact by construction and the numerics are comparatively cheap to run with the exact master-equation solver.

The resulting animation shows the three expected features side by side: the vibrational population starting from a hot initial distribution ($n = 15$) and progressively piling up on the motional ground state; the steady-state Fano spectrum, with its dark transparency window sitting exactly on the carrier frequency and its narrow absorption peak aligned with the red motional sideband; and the corresponding decay of the mean phonon number $\langle n \rangle$ over time as the atom is cooled. This confirmed that the solver and the underlying EIT cooling mechanism behave as expected before the added complexity of the real atomic structure was introduced.

One caveat applies to this illustrative run: starting from $n = 15$ with $\eta = 0.35$ puts the system only marginally inside the Lamb–Dicke regime at $t = 0$ ($\eta\sqrt{\langle n \rangle + 1} \approx 1.4$), so the linearized motional coupling used in the interaction Hamiltonian mildly underestimates higher-order sideband transitions during the very earliest stage of the evolution. Since the population is rapidly cooled toward $\langle n \rangle \approx 0$, however, the system quickly settles deep into the Lamb–Dicke regime, so the computed steady-state cooling limit itself is unaffected by this initial-transient approximation.

3.3.2 Repumper Optimization

Within the full 24-level model, an accumulation of population in the uncoupled $F = 1$ ground-state manifold was observed during the cooling dynamics — atoms that fall out of the $F = 2$ cooling cycle and stop interacting with the probe and coupling lasers, introducing effective dead time that limits the achievable cooling rate. The repump Rabi frequency Ω_{repump} was therefore scanned, using the steady-state Fano spectrum and ground-state leakage populations as diagnostics.

The scan revealed a direct trade-off: a stronger repump field is more effective at depleting the $F = 1$ population, but it also dresses the atomic states more strongly, producing an AC Stark shift that displaces the entire Fano profile toward higher probe detuning. At low repump power ($\Omega_{\text{repump}} = 0.2\,\gamma$), population trapping in $F = 1$ reached about 35%, substantially isolating a third of the ensemble from the active cooling transition; increasing Ω_{repump} to $0.7\,\gamma$ and beyond was sufficient to deplete this residual population and restore the ensemble to the $F = 2 \leftrightarrow F' = 2$ cycle. At the same time, the induced light shift grows with Ω_{repump}, and at $\Omega_{\text{repump}} = 0.7\,\gamma$ this shift was found to align the Fano absorption maximum precisely with the target red sideband; for higher values (e.g. $1.5\,\gamma$ or $2.0\,\gamma$) the shift overshoots this alignment and the absorption peak moves past the sideband, decoupling the atom from the cooling transition. Based on this trade-off, $\Omega_{\text{repump}} = 0.7\,\gamma$ was identified as the optimal repump strength for this parameter set, balancing dark-state depopulation against excess AC Stark shift.

3.3.3 Probe Optimization

The same diagnostic procedure was then applied to the probe Rabi frequency Ω_p, scanned from $0.05\,\gamma$ to $0.35\,\gamma$.

Here the limiting factor is power broadening: EIT cooling relies on a steep asymmetry between near-zero absorption at the carrier and strong absorption at the red sideband, and at low probe power ($\Omega_p = 0.05$–$0.10\,\gamma$) this contrast is well preserved. As Ω_p increases, the dark-resonance dip fills in and the absorption peak broadens, degrading the achievable contrast and, correspondingly, raising the expected carrier-scattering rate and the resulting steady-state temperature. A probe-induced AC Stark shift was also observed, distinct from the repumper-induced one: for Ω_p between $0.15\,\gamma$ and $0.35\,\gamma$ the absorption peak systematically drifts to higher detuning, eventually detaching from the target sideband at the highest powers tested. Increasing the probe power also degrades the steady-state ground-state preparation, by increasing the overall scattering rate and driving more population into $F = 1$ and into spurious $F = 2$ Zeeman sublevels, even though the fractional leakage into the off-resonant $F' = 3$ state actually decreases with probe power (since it is a relative measure against a growing target-transition signal). On balance, the weak-probe regime was found to be preferable, and $\Omega_p = 0.10\,\gamma$ was adopted as the operating point, preserving a sharp Fano feature aligned with the red sideband while keeping the excitation amplitude adequate.

3.3.4 Coupling Detuning Optimization

With the repumper and probe powers fixed, the coupling-laser detuning Δ_c was scanned from $+6.0\,\gamma$ to $+13.0\,\gamma$.

Because the coupling field itself induces an AC Stark shift, the position of the Fano absorption maximum drifts with Δ_c: at the lower end of the scanned range ($+6.0$–$+8.0\,\gamma$) the peak falls short of the target red sideband, while the absolute height of the peak decreases monotonically with Δ_c (from about 0.0008 at $+6.0\,\gamma$ down to about 0.0004 at $+13.0\,\gamma$), which on its own would favor smaller detunings for a larger excitation amplitude. The two effects were found to balance at $\Delta_c = +9.0\,\gamma$, where the absorption maximum aligns with the red sideband while retaining sufficient amplitude, and where the corresponding minimum in the off-resonant $F' = 3$ leakage spectrum — marking the EIT transparency window — is properly positioned to suppress carrier scattering. This value was therefore adopted as the operating detuning for the subsequent time-dependent simulations.

3.3.5 Cooling Dynamics with the Optimized Parameters

Combining the optimized repumper, probe, and coupling-detuning values ($\Omega_{\text{repump}} = 0.7\,\gamma$, $\Omega_p = 0.10\,\gamma$, $\Delta_c = +9.0\,\gamma$), the full time-dependent cooling dynamics were computed for the 24-level configuration in which both Λ-system ground states lie within $F = 2$.

With this parameter set the simulated system cools in fewer than 10^5 simulation steps, corresponding to a cooling time of approximately 2 ms — a substantial improvement over the unoptimized configuration used during the initial parameter scans, whose cooling time was around 15 ms. Obtaining this Monte Carlo trajectory for the full 24-level system required about 8 hours of computation; for comparison, the exact master-equation solver applied to the same system only reached about 70,000 simulated steps after three days of continuous running, confirming that the Monte Carlo wavefunction method is the only practical option for the complete atomic structure.

3.3.6 Cooling Dynamics with Ground States Split Across $F = 1$ and $F = 2$

The same optimization methodology was then repeated for the alternative level assignment, in which the two Λ-system ground states are split between $F = 2$ and $F = 1$ rather than both lying in $F = 2$, and the repumper instead recovers population left behind in $F = 2$. This configuration is the one that more closely matches the level structure accessible in our experimental setup, and is therefore the one carried forward into the real-parameter simulation described below.

Applying the same procedure — steady-state Fano spectrum evaluation followed by sequential optimization of the laser parameters and a Monte Carlo time evolution — yielded the optimized parameters $\Delta_c = +9\,\gamma$, $\Omega_p = 0.17\,\gamma$, $\Omega_r = 1.27\,\gamma$ for this configuration. As in the previous configuration, the repumper was found to be essential: without it, population is rapidly lost to uncoupled dark states and the cooling dynamics stall.

This behavior was checked explicitly by running the simulation without the repump field, at zero magnetic field, and was found to hold equally in the presence of a magnetic field — in fact, the Fano profile itself is further degraded once Zeeman shifts are included, making the repumper's role even more important in the presence of a realistic bias field.

3.3.7 Axial-Direction Simulation with Real System Parameters

As a final step, the $F = 1/F = 2$-split configuration was re-evaluated using the trap frequency ν and Lamb–Dicke parameter η specific to the axial direction of our experimental trap geometry, in place of the generic illustrative values used in the parameter scans above. The same sequential optimization procedure was applied to this realistic configuration, converging again on parameters close to $\Delta_c = +9\,\gamma$, $\Omega_p = 0.17\,\gamma$, $\Omega_r = 1.27\,\gamma$, and the resulting parameter set was then used to run the full Monte Carlo time evolution for this specific geometry.

The resulting cooling dynamics for the axial direction of our actual trap follow the same qualitative behavior established in the generic $F = 1/F = 2$-split configuration above, confirming that the optimized EIT cooling scheme identified through the generic parameter scans remains effective once the specific trap parameters of our setup are substituted in place of the illustrative values used during the optimization stage.

Conclusions

This thesis work implemented and characterized an experimental apparatus for an Optical Dipole Trap (ODT) designed to trap a sample of ^{87}Rb atoms, which was integrated into a pre-existing setup based on a Magneto-Optical Trap (MOT) and optical molasses cooling. The ODT was realized using 1064 nm laser light coupled into a kagome Hollow-Core Photonic Crystal Fiber (HCPCF). A crucial objective of this work was the optimization and reliable light injection into the HCPCF. Standard injection procedures relying solely on power monitoring frequently resulted in light coupling into the silica jacket or into higher-order modes. To overcome this, a custom-built optical microscope was employed for near-field monitoring. On a test fiber, this approach enabled coupling into a fundamental LP_{01}-like mode with minor deformations, while also achieving $\sim 96\%$ transmission efficiency. The application of the same procedure to the HCPCF installed in the vacuum chamber yielded a final operating efficiency of $\sim 40\%$ with a partially asymmetric mode profile. This reduced performance on the in-chamber fiber is likely due to mechanical stresses applied to the fiber itself by the vacuum sealing system.

Once the ODT was operative, the experimental focus shifted to measurements inside the vacuum chamber. First, the molasses stage temperature was optimized through systematic investigation of the cooling beam detuning, the Acousto-Optic Modulator (AOM) ramp durations and amplitudes, and the cancellation of residual magnetic fields via compensation coils. These adjustments led to a minimum x-y-averaged atomic cloud temperature of $T = (9.6 \pm 0.1)\ \mu$K.

The final characterization focused on the atom loading efficiency from the molasses into the ODT. The spatial overlap between the atomic cloud and the ODT beam was optimized by utilizing the compensation coils during the MOT phase to displace the atomic cloud from its optimal position, thereby increasing the overlap with the ODT. The successful isolation of the trapped atom signal relied on the implementation of a higher-resolution imaging system and, most importantly, on the mitigation of the intensity noise introduced by the AOM. This enhanced the signal-to-noise ratio, enabling the correct optimization of the ODT loading process, leading to a maximum observed capture efficiency of $(0.160 \pm 0.005)\%$.

Although numerically small, the observed capture efficiency is consistent with the order of magnitude predicted by the non-shallow trap theoretical model presented in this thesis. Applying the experimental parameters of the molasses and the trap geometry to the model yields an expected capture efficiency of $(0.41 \pm 0.04)\%$. The agreement on the order of magnitude suggests that the primary limitations to the capture efficiency are given by the intrinsic properties of the apparatus such as the distance between the fiber facet and the MOT center, the beam geometrical properties and the ODT optical power, together with the minimum temperature achieved. However, the experimental value remains approximately one-third of the theoretical expectation. It should be investigated whether this discrepancy comes from procedural artifacts such as the 600 μs delay between the end of the Time-of-Flight (TOF) sequence and the conclusion of the imaging flash.

Beyond addressing this discrepancy, the results obtained in this thesis establish a robust foundation for the future development of the experiment. The identification of the optimal parameters for both the optical molasses and the ODT loading sequence provides a reliable experimental baseline. This ensures that these operational stages are fully characterized and will not require fundamental recalibration in subsequent stages. Furthermore, the developed injection setup and near-field inspection protocol will be instrumental for introducing a second, counter-propagating 1064 nm beam. This configuration is necessary to realize an optical conveyor belt, enabling the efficient loading and deep confinement of atoms directly within the hollow core of the HCPCF.

Bibliography

[1] Rui Wang *et al.* “Optical Trapping of Mesoscale Particles and Atoms in Hollow-Core Optical Fibers: Principle and Applications”. In: *Light: Science & Applications* 14.1 (2025), p. 146. DOI: 10.1038/s41377-025-01801-5.

[2] Mingjie Xin *et al.* “An Atom Interferometer inside a Hollow-Core Photonic Crystal Fiber”. In: *Science Advances* 4.1 (Jan. 2018), e1701723. DOI: 10.1126/sciadv.1701723.

[3] Martin Fasser *et al.* “Subradiance and Superradiant Long-Range Excitation Transport among Quantum Emitter Ensembles in a Waveguide”. In: *Optica Quantum* 2.6 (May 2024), p. 397. DOI: 10.1364/opticaq.531434.

[4] Matej Komanec *et al.* “Hollow-Core Optical Fibers”. In: *Radioengineering* 29 (2020), pp. 417–430. DOI: 10.13164/re.2020.0417.

[5] M. J. F. Digonnet *et al.* “Understanding air-core photonic-bandgap fibers: analogy to conventional fibers”. In: *Journal of Lightwave Technology* 23.12 (2005), pp. 4169–4177. DOI: 10.1109/JLT.2005.859406.

[6] B. Debord *et al.* “Hollow-Core Fiber Technology: The Rising of “Gas Photonics””. In: *Fibers* 7.2 (2019), p. 16. DOI: 10.3390/fib7020016.

[7] R. F. Cregan *et al.* “Single-Mode Photonic Band Gap Guidance of Light in Air”. In: *Science* 285.5433 (1999), pp. 1537–1539. DOI: 10.1126/science.285.5433.1537.

[8] J. von Neumann and E. Wigner. “Über merkwürdige diskrete Eigenwerte”. In: *Physikalische Zeitschrift* 30 (1929), pp. 465–467.

[9] E. L. Raab *et al.* “Trapping of Neutral Sodium Atoms with Radiation Pressure”. In: *Physical Review Letters* 59.23 (1987), pp. 2631–2634. DOI: 10.1103/PhysRevLett.59.2631.

[10] Stefan Kuhr. “A Controlled Quantum System of Individual Neutral Atoms”. PhD thesis. Bonn, Germany: Rheinische Friedrich-Wilhelms-Universität Bonn, 2003.

[11] Harold J. Metcalf and Peter van der Straten. *Laser Cooling and Trapping.* Springer New York, 1999. DOI: 10.1007/978-1-4612-1470-0.

[12] J. Dalibard and C. Cohen-Tannoudji. “Laser cooling below the Doppler limit by polarization gradients: simple theoretical models”. In: *Journal of the Optical Society of America B* 6.11 (1989), pp. 2023–2045. DOI: 10.1364/JOSAB.6.002023.

[13] Rudolf Grimm, Matthias Weidemüller, and Yurii B. Ovchinnikov. “Optical Dipole Traps for Neutral Atoms”. In: *Advances in Atomic, Molecular, and Optical Physics* 42 (2000), pp. 95–170. DOI: 10.1016/S1049-250X(08)60186-X.

[14] L. D. Landau and E. M. Lifshitz. *Mechanics*. 3rd. Vol. 1. Course of Theoretical Physics. Section 50: Adiabatic invariants. Butterworth-Heinemann, 1976. ISBN: 978-0750628969.

[15] Giovanna Morigi, Jürgen Eschner, and Christoph H. Keitel. "Ground State Laser Cooling Using Electromagnetically Induced Transparency". In: *Physical Review Letters* 85.21 (2000), pp. 4458–4461. DOI: 10.1103/PhysRevLett.85.4458.

[16] G. Morigi. "Cooling atomic motion with quantum interference". In: *Physical Review A* 67 (2003), p. 033402. DOI: 10.1103/PhysRevA.67.033402.

[17] Brahim Lounis and Claude Cohen-Tannoudji. "Coherent population trapping and Fano profiles". In: *Journal de Physique II* 2 (1992), pp. 579–592. DOI: 10.1051/jp2:1992153.

[18] Leonardo Razzai. "Setup of a Magneto-Optical Trap of cold 87Rb atoms in proximity to an hollow core photonic crystal fiber". Relatore: Prof. Francesco Minardi; Correlatore: Prof. Marco Prevedelli. MA thesis. Alma Mater Studiorum – Università di Bologna, 2024.

[19] Matteo Marchesini *et al.* "All-fiber, near-infrared, laser system at 780 nm for atom cooling". In: *Opt. Continuum* 3.10 (2024), pp. 1868–1879. DOI: 10.1364/OPTCON.525738.

[20] M. Marchesini. "A new magneto-optical trap for cold rubidium atoms in hollow-core fibres". PhD thesis. University of Bologna, 2025.

[21] R. Wang *et al.* "Optical trapping of mesoscale particles and atoms in hollow-core optical fibers: principle and applications". In: *Light: Science & Applications* 14 (2025), p. 146. DOI: 10.1038/s41377-025-01801-5.